Bayesian Inference for Biologists

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Welcome

These notes accompany **BIOL 7950: Bayesian Inference for Biologists** at Kennesaw State University.

The goal of this course is to introduce Bayesian statistical thinking and modern Bayesian analysis for biological data.

Course organization

The book is organized into three parts:

1. Foundations: introduces the basic ideas behind Bayesian inference, Bayes' rule, and how Bayesian inference differs from frequentist inference. Also introduces some key aspects of working with Bayesian models in R and JAGS.
2. Bayesian analyses in R and JAGS: demonstrates Bayesian versions of common models for biology like linear models, GLMs, and nonlinear models.
3. Special applications: demonstrates some situations where Bayesian methods really shine, like mark-recapture and state space models.

Work in progress warning!

This book is always a work in progress and I welcome constructive criticism and feedback. You can get in touch with me by email, on Bluesky ([at]greenquanteco), and Twitter ([at]GreenQuantEco).

Suggested reading

There is no required textbook for this course, but if you want a book to help you on your Bayesian journey you could do worse than the following:

Albert J. 2009. Bayesian computation with R. Springer Science+Business Media, New York. *Unlike most books on this list, this book focuses on implementing Bayesian methods in R itself. The mathematical exposition is a bit dense but was helpful for understanding exactly how MCMC relates to inference.*

Gelman A, Carlin JB, Stern HS, Rubin DB. 2013. Bayesian data analysis. 3rd edition. Taylor & Francis Group, Boca Raton, Florida, USA. *The all-time classic, written by some of the biggest names in modern*

Bayesian theory.

Kéry M. 2010. Introduction to WinBUGS for Ecologists. Academic Press, Burlington, Massachusetts, USA. *This was my first Bayesian book, and I can't recommend it highly enough. It explains the basics of Bayesian inference, WinBUGS, and how to implement some common statistical models ranging from t-tests to ANCOVA. Even if you are not an ecologist, the exposition and example code in this book are clear and easy to follow.*

Kéry M, Kellner KF. 2024. Applied Statistical Modelling for Ecologists. Academic Press, Burlington, Massachusetts, USA. *I just learned about this (August 2026) and will be buying it soon. If it is as good as the other books in the series, it'll go into my regular rotation.*

McCarthy MA. 2007. Bayesian methods for ecology. Cambridge University Press, New York. *Another WinBUGS-focused book, this is a nice complement to Kéry (2010) with a bit more theory. Even if you are not an ecologist, the exposition and example code in this book are clear and easy to follow.*

Get in touch!

Visit the lab website for more information about Dr. Green's other courses and QuantEco lab research!

Part I

Introduction to Bayesian inference

If it feels backwards, that’s because it is.

Chapter 1

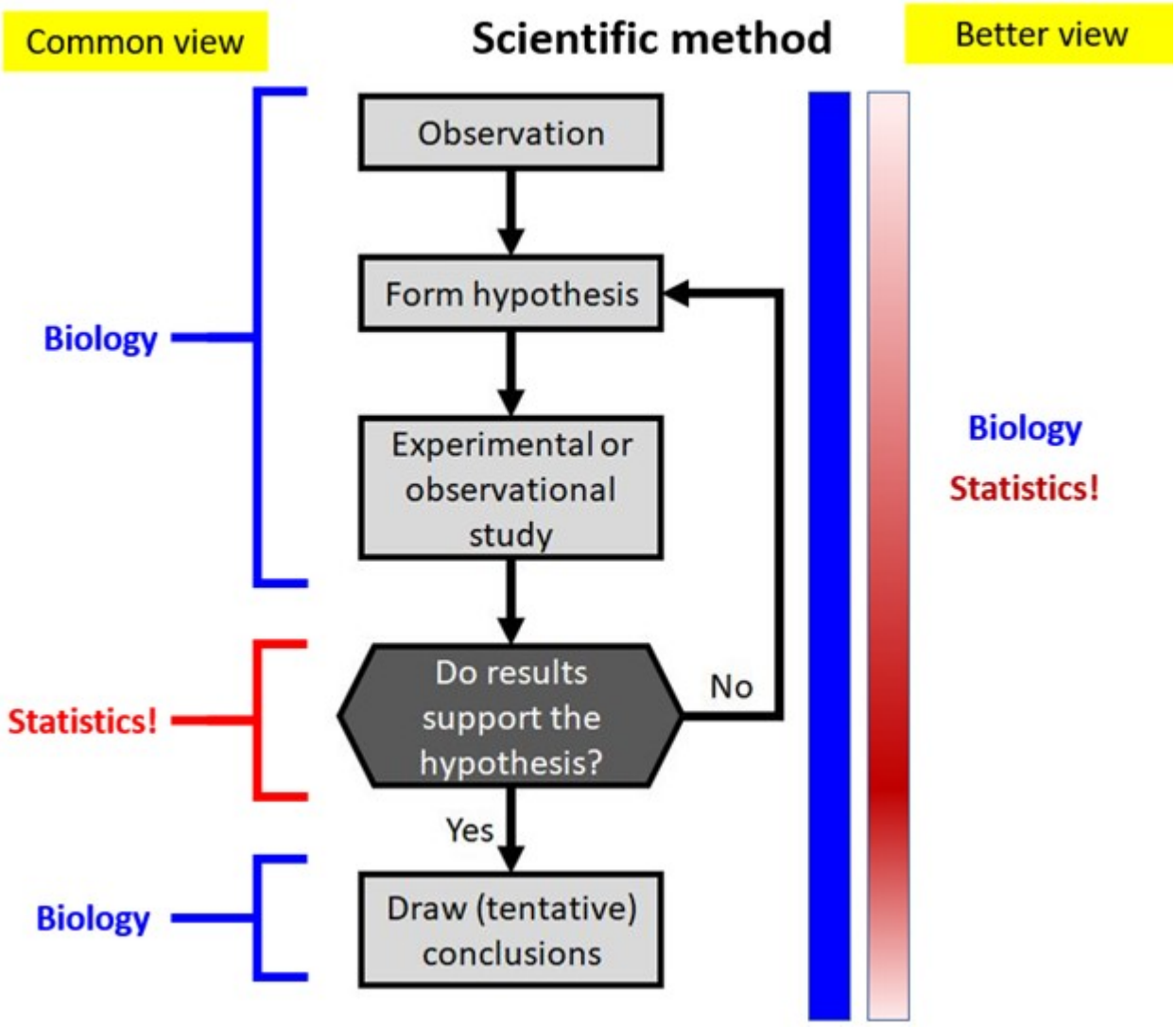
Introduction: Modes of inference

1.1 Motivation: hypothesis testing in biology

Biology, like all sciences, depends on the **scientific method**. This means testing proposed explanations, or **hypotheses**, against the data to see if the data support the hypothesis or not. That critical step, of deciding whether observations support or refute a hypothesis, is the domain of **statistics**: the branch of mathematics concerned with finding and interpreting patterns in data.

Many biologists traditionally think of “biology” and “data analysis” as two completely separate enterprises, with statistics being like a special wrench that a mechanic pulls out of the toolbox so infrequently that its use is unfamiliar. I think this is a mistake. As the figure below shows, every part of the scientific method can benefit from some knowledge of statistics. This includes experimental design, data collection, and interpreting the findings.

Figure 1.1: Statistics and thinking about data is part of the entire scientific process, not just a single stage.



So why do we, as biologists, need statistics? Some common answers might include:

1. "To tell us whether the patterns we see in our data are real"
2. "To tell whether the patterns in our data are due to chance or not."
3. "To prove our hypotheses."

All 3 of these are completely wrong. **1** is wrong because confirming reality requires observation and experiment, not calculating. **2** is a common misconception. *Any* pattern in data could be cause by chance. Statistics can help you estimate that probability, but cannot tell you for sure. Finally, **3** is wrong because that's not how science works. Hypotheses can be supported or not supported. No amount of statistical analysis will ever completely rule out chance.

Strange as it may seem, though, **2** actually has a lot to do with how most statistical analysis in biology is done, and has for most of the last century.

1.2 Null hypothesis significance testing and *p*-values

1.2.1 Introduction

When scientists design experiments, they are testing one or more hypotheses, or proposed explanations for a phenomenon. Contrary to popular belief, the statistical analysis of experimental results usually does not test a researcher's hypotheses directly. Instead, statistical analysis of experimental results focuses on comparing experimental results to the results that *might have occurred* if there was no effect of the factor under investigation.

In other words, traditional statistics sets up the following:

- **Null hypothesis or** H_0: default assumption that some effect (or quantity, or relationship, etc.) is zero and does not affect the experimental results.
- **Alternative hypothesis or** H_a: assumption that the null hypothesis is false. Usually this is the researcher's scientific hypothesis, or proposed explanation for a phenomenon. Also sometimes called H_1.

In (slightly) more concrete terms, if researchers want to test the hypothesis that some factor X has an effect on measurement Y, then they set up an experiment comparing observations of Y across different values of X. They would then have the following set up:

- **Null hypothesis:** Factor X does not affect Y.
- **Alternative hypothesis:** Factor X does affect Y

Notice that the alternative hypothesis is the **logical negation** of the null hypothesis: the alternatives "does not affect" and "does affect" cover all possibilities. There is no "sort of affects".

Figure 1.2: Affect or affect not. There is no try.



The alternative hypothesis is also ***not*** a statement about another variable (e.g., "Factor Z affects Y"). This is an easy mistake to make and one of the weaknesses of NHST: the word "hypothesis" in the phrase "null hypothesis" is not being used in the sense that scientists usually use it. This unfortunate choice of vocabulary

is responsible for generations of confusion among researchers[1].

In order to test an "alternative hypothesis", what need is to *estimate* how likely the data would be if the null hypothesis were true. If a pattern seen in the data is highly unlikely, this is taken as evidence that the null hypothesis should be rejected in favor of the alternative hypothesis. This is not the same as "accepting" or "proving" the alternative hypothesis.

The null hypothesis is usually translated to a prediction like one of the following:

- The mean of Y is the same for all levels of X.
- The mean of Y does not differ between observations exposed to X and observations not exposed to X.
- Outcome Y is equally likely to occur when X occurs as it is when X does not occur.
- Rank order of Y values does not differ between levels of X.

All of these predictions have something in common: Y is **independent** of X. That's what the null hypothesis means. To test the predictions of the null hypothesis, we need a way to calculate what the data *might have looked like if the null hypothesis was correct*. Once we calculate that, we can compare the data to those calculations to see how "unlikely" the actual data were. Data that are sufficiently unlikely under the assumption that the null hypothesis is correct—a pattern that is **statistically significant**—are a kind of evidence that the null hypothesis is false. If this sounds convoluted, that's because it is.

This kind of analysis is referred to as **Null Hypothesis Significance Testing (NHST)** and is one of the most important paradigms in modern science. Whether or not this is a good thing is a subject of fierce debate among scientists, statisticians, and philosophers of science. For better or worse, NHST is all but the *de facto* standard of inference in scientific publishing, so regardless of its merits in your particular case you will need to be able to use it. The purpose of this course module is to encourage you to use it correctly.

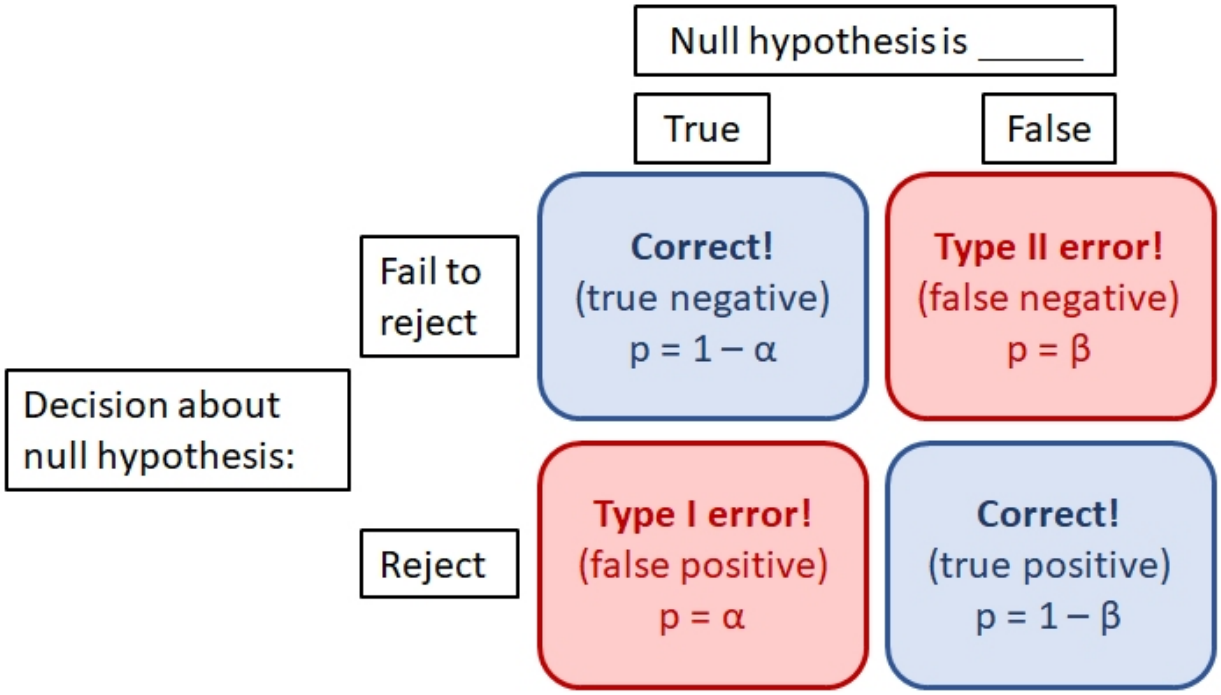
The *p*-value allows us to make binary distinctions, in a systematic way, between situations where a null hypothesis is true and no pattern is present, and situations where the null hypothesis is highly unlikely to be true and thus a pattern might be present. Any binary decision process has four possible outcomes:

1. **True positive**: declare a pattern is present when it is present.
2. **True negative**: declare a pattern is absent, when it is absent.
3. **False positive**: declare a pattern is present, when it is actually absent. Also called a **Type I error**.
4. **False negative**: declare a pattern is absent, when it is actually present. Also called a **Type II error**.

These outcomes can be summarized as shown below. Statisticians are particularly concerned with the probabilities of false positives (α) and false negatives (β). The false positive probability α is the standard against which experimental *p*-values are compared. This is largely because we humans are inordinately prone to finding patterns even when no patterns are present–i.e., committing type I errors.

[1] Not to mention among generations of students taking statistics courses, myself included.

Figure 1.3: Relationship between the null hypothesis and our decision about experimental results. Type I and II errors result when we reject a true null hypothesis and when we fail to reject a false null. The false discovery rate, or Type I error rate, defined by α is a very important value in our approach to science.



Why have we evolved this way? Finding patterns in noisy data has helped us discover virtually everything we know. So, pattern recognition must be a powerful cognitive tool. But there is another interpretation, that in our ancestral lifestyle life false positives were far less costly than false negatives. Imagine you are an Australopithecine or an early *Homo erectus* (or *H. sapiens*, for that matter), wandering through the savanna of what will someday be modern Ethiopia, looking for food. You hear a rustle in the tall grass a few meters ahead. Is that rustle just the wind? Or is it a dangerous animal that will kill and possibly eat you? Here are your choices and their consequences:

Figure 1.4: Illustration of the consequences of Type I and Type II errors, and why human minds may be so prone to Type I errors.

	True source of the noise	
Your choice:	**Just the wind**	**Dangerous animal**
Ignore the noise and keep foraging.	**True negative** No worries! You get to keep foraging a while longer.	**False negative! Type II error!** You are killed and eaten by a lion. And if you are very, very lucky…they'll do it in that order.
Run away from the possible lion!	**False positive! Type I error!** You spend some calories running away, and lose some foraging time, but at least you're alive.	**True positive** You run for your life and live to forage another day. Everyone at the camp totally believes your story about the lion.

Notice that of your choices, only the type II error has immediate evolutionary consequences. In terms of fitness, the difference between a type II error and a true positive (death vs. survival) is much greater than the difference between a true negative and a false positive (a bit more food vs. a mild inconvenience).

What does this have to do with statistics in biology? Our human minds have evolved to find patterns in noisy data, and we are so good at it that we frequently "find" patterns that aren't really there. So, we use the scientific method and statistical analysis as tools to help mitigate our bias toward pattern-finding.

1.2.2 Example

Let's do a quick example of null hypothesis significance testing. We'll use the famous `iris` dataset, which is built into R. It contains measurements of 4 different parts of 150 *Iris* flowers, with 50 from each of three species (*I. setosa*, *I. versicolor*, and *I. virginica*). We are going to use NHST to investigate whether the three species have different petal lengths.

Figure 1.5: Three species of purple iris. From left to right, *Iris setosa*, *I. versicolor*, and *I. virginica*. The petal is one of the flowery parts.



Iris setosa



Iris versicolor



Iris virginica

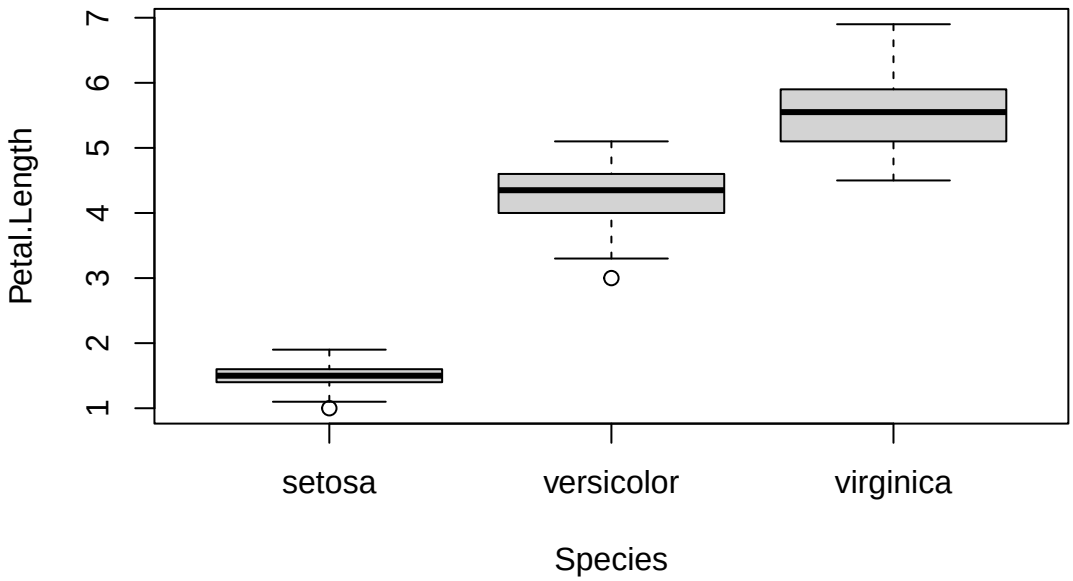
Our research question is:

Do different species of *Iris* differ in their petal lengths?

Since we already have the measurements, one good first step would be a make a plot of the petal sizes in each species. The best tool for this kind of question is a **boxplot**, or **box-and-whiskers plot**.

```
boxplot(Petal.Length~Species, data=iris)
```

Figure 1.6: Variation in petal length among 3 species of *Iris*



The boxplot shows us that petal lengths for *I. setosa* are clustered around 1.5 cm, about 4.5 cm for *I. versicolor*, and about 6 cm for *I. virginica*. Can we conclude that petal lengths differ between species?

No, not yet. Obviously, there is some intraspecific (within-species) variation in petal length, and some interspecific variation. And, it sure looks like *I. setosa* is quite a bit smaller than the other two species. *I. versicolor* and *I. virginica* look like they might be different, but there is some overlap. So no, we can't answer our question just by eyeballing the figure.

If you think about it for a while, you might think of another question: what does it even mean for the species to have different petal lengths? No overlap in size at all? Only 1% overlap? Only 2%? What is the threshold? And even if you could think of a threshold, how would you defend it against other scientists who have other ideas about flower petal lengths?

To answer that question, what does "different" even mean, we have to realize that these 150 flowers are not all of the flowers. They are a sample of millions or billions of flowers. And even if we could measure every single *Iris* flower on Earth, those untold billions are an infinitesimally tiny fraction of all of the *Iris* flowers that there ever could have been. So, the means of our three species of flowers are not the true, underlying, eternal Platonic ideal means of all *Iris*. They are the means of a *sample* of *Iris*: **sample means**, often labeld $\bar{x}$. That unknown, unobservable quantity that underlies the samples is the **population mean**: the mean of all possible flowers. The population mean is usually denoted μ. We can obtain a sample mean, but only ever estimate the population mean. Likewise, each species appears to have a sample SD calculated from the data, s, and a population SD σ that we can only estimate.

```
# estimate the sample mean of petal length
aggregate(Petal.Length~Species, data=iris, mean)
```

```
     Species Petal.Length
1     setosa        1.462
2 versicolor        4.260
3  virginica        5.552
```

```
# calculate sample SD of petal length
aggregate(Petal.Length~Species, data=iris, sd)
```

```
     Species Petal.Length
1     setosa    0.1736640
2 versicolor    0.4699110
3  virginica    0.5518947
```

With that in mind, we now have a way to define a "difference" between species in terms of petal length: could we collect a sample of these 3 species, assuming they had the same μ and σ, and observe a difference between groups as large as the one we saw in Figure 1.6? If the population means are the same, then observing a consistent difference should be very, very unlikely. If it is so unlikely that we reject that possibility, then we can tentatively accept the alternative: that the population means are not the same, and are different.

So, our question of "Do different species of *Iris* differ in their petal lengths?" can be rephrased into a statistical question:

Do different speceies of *Iris* have different population means?

Our research hypothesis is that they do. Thus, the null hypothesis, the hypothesis of no effect, must be:

H_0: The three species of *Iris* have the same population mean.

and for H_1 to cover all possibilities:

H_1: At least one species of iris must have a different population mean than the other species.

That H_1 covers the possibilities of 1 pairwise difference in μ, 2 pairwise differences in μ, and 3 pairwise differences in μ. That is, all cases other than 0 pairwise differences.

The test for this situation is **analysis of variance (ANOVA)**:

```
mod1 <- aov(Petal.Length~Species, data=iris)
summary(mod1)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
Species       2  437.1  218.55    1180 <2e-16 ***
Residuals   147   27.2    0.19
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The result of `summary()` is the **ANOVA table**. the ANOVA table shows us that the factor `Species` uses 2 degrees of freedom (variously abbreviated as DF, Df, or d.f.) to account for 437.10 of the squared errors; the residuals use 147 d.f. to account for 27.22 of the squared errors. The ratio of the mean squared error per d.f. associated with each source, 218.551 / 0.185 = 1180.2, is the test statistic F. This is used to calculate the *p*-value. Like most test statistics such as t or χ^2, F follows a particular distribution defined by DF, which allows $p(F)$ to be calculated.

Under the assumption of no difference in popuation means, we would expect a random dataset to display an F statistics of 1180 very, very rarely: a proportion so small that R rounds it off to 0.00000000000000022 (`<2e-16` or $p < 2.2 \times 10^{-16}$), or 0.0000000000000022% of the time. In biology we can round this off to $p < 0.001$ or $p < 0.01$. How unlikely is $p < 2.2 \times 10^{-16}$?

- An event with probability has about 1 chance in 4.55×10^{15}, or 1 to 4550000000000000 against.
- The odds of winning the Powerball average around 1 in 2.92×10^{8}. This means that under the null hypothesis an F statistic this large is about 15 million times less likely than winning the Powerball.
- If you flipped a coin every second with $p\left(Heads\right) = 2.2 \times 10^{-6}$, you would wait on, on average, about 144 million years until the coin came up heads. Flipping the coin that many times (4.5×10^{15}) would probably wear the coin down to nothing through erosion. That's so long, that by that time the length of a day (Earth's rotation) will have changed. That is so long ago, that you would find yourself surrounded by non-avian dinosaurs, but would be safe from *Tyrannosaurus rex* because it would not evolve for another 70 million years!

1.2.3 Conclusions

The point is, what this *p*-value tells us is that the pattern we see in these data would be absurdly unlikely if the null hypothesis was true, and the populations means of the *Iris* species did not differ. Not impossible, but so astronomically unlikely that we can provisionally *reject* the null hypothesis. Note that we do not automatically *accept* the alternative hypothesis, because the probability of the pattern under the null is not 0. This is close to what Stephen Jay Gould meant when he wrote,

> In science, "fact" can only mean "confirmed to such a degree that it would be perverse to withhold provisional assent." I suppose that apples might start to rise tomorrow, but the possibility does not merit equal time in physics classrooms (Gould 1983).

So at the end of this analysis, we have not established that the hypothesis is true. Instead, we have collected evidence that would have been very unlikely if the hypothesis was not true. In other words, we have quantified that $p\left(E|\neg H\right)$ is very small.

$$\underbrace{p\left(E|\neg H\right)}_{\text{p-value}} \quad \neq \quad \underbrace{p\left(H|E\right)}_{\text{what we actually want to know}}$$

> **Note**
>
> Technically, the E here is not the evidence, but "a pattern at least as extreme as that observed", which is a close enough approximation for the point I'm making.

1.3 Maximum likelihood estimation (MLE)

1.3.1 Introduction

Much of modern statistics is based on fitting models by **maximum likelihood estimation** or **MLE*. This revolves around finding model parameters that maximize the **likelihood function** of the model given the data. The likelihood describes how well different possible parameter values are supported by the observed data under a particular statistical model. A statistical model specifies the mathematical relationships among variables and the probability distribution of the observations; examples include a linear regression model or a Poisson GLM.

In symbols, we can write the likelihood function as:

$$\mathcal{L}\left(\theta|x\right)$$

where x represents the observed data and θ represents one or more unobserved parameters. For the data we actually observed, the likelihood function asks: which possible values of θ would make these data more or less plausible?

Importantly, likelihood is ***not*** the probability that a particular parameter value or model is correct. Although $\mathcal{L}\left(\theta|x\right)$ is written with the parameters to the left of the vertical bar, it is based on the probability of the data given the parameters, $p\left(x|\theta\right)$, not the probability of the parameters given the data, $p\left(\theta|x\right)$. The latter is a Bayesian posterior probability (which we will explore in Section 1.4).

The takeaway is that probability asks about data when the model is known, while likelihood asks about parameter values after the data have been observed.

1.3.2 Example

Here's a quick example: imagine you have measured 5 squirrel femurs and found lengths of 63.6, 66.5, 64.2, 67.6, and 68.3 mm, yielding an observed sample mean and SD of 66.0 ± 2.1 mm. What estimate of the population mean μ that makes this set of measurements most plausible?

```
a <- c(63.6, 66.5, 64.2, 67.6, 68.3)
mean(a)
```

```
[1] 66.04
```

```
sd(a)
```

```
[1] 2.067124
```

Consider 3 candidates for μ: 64, 66, and 68. For any proposed value of μ, `dnorm()` tells us the probability density of observing each femur length under a normal distribution with that mean:

```
samp.sd <- 2.1
dnorm(a, mean=64, sd=samp.sd)
```

```
[1] 0.18655737 0.09352817 0.18911291 0.04370627 0.02334791
```

```
dnorm(a, mean=66, sd=samp.sd)
```

```
[1] 0.09887122 0.18466340 0.13156914 0.14211406 0.10428277
```

```
dnorm(a, mean=68, sd=samp.sd)
```

```
[1] 0.02115483 0.14719781 0.03695464 0.18655737 0.18804388
```

Because we observed more than 1 measurement, and assume that they are independent, the joint likelihood is the product of their densities:

```
prod(dnorm(a, mean=64, sd=samp.sd))
```

```
[1] 3.367191e-06
```

```
prod(dnorm(a, mean=66, sd=samp.sd))
```

```
[1] 3.560036e-05
```

```
prod(dnorm(a, mean=68, sd=samp.sd))
```

```
[1] 4.036932e-06
```

It looks like 66 has the greatest joint likelihood across all observed data. But we can get a fuller picture if we test more values:

```
# series of possible mu to test
test.mu <- seq(60, 72, by=0.01)

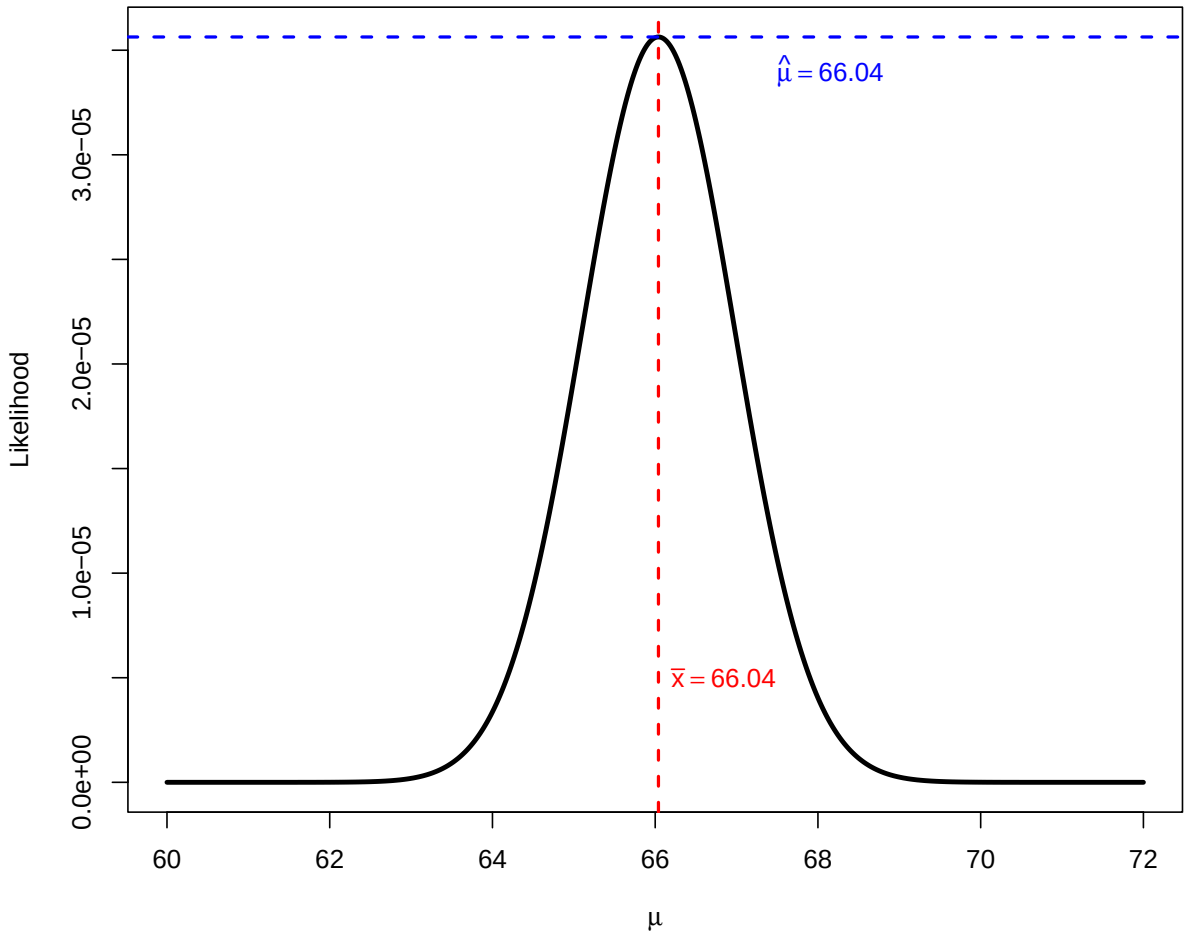
# number of possible mu
n <- length(test.mu)

# set up vector to hold likelihood
# function of each candidate mu
lik <- numeric(n)

# calculate likelihoods of mu's in a loop
for(i in 1:n){lik[i] <- prod(dnorm(a, test.mu[i], sd=samp.sd))}#i
```

```r
# now plot the likelihood function vs. mu
plot(test.mu, lik, type="l", lwd=3,
     xlab=expression(mu),
     ylab="Likelihood")
abline(v=mean(a), lwd=2, lty=2, col="red")
text(66.2, 0.5e-5, expression(bar(x)==66.04), adj=0, col="red")
abline(h=lik[which.max(lik)], lty=2, lwd=2, col="blue")
text(67.5, 3.4e-5, expression(hat(mu)==66.04), adj=0, col="blue")
```

Figure 1.7



In this case, the sample mean $\bar{x} = 66.04$ (`mean(a)`) is the same as the μ value that maximized the joint likelihood of the 5 observed values, $\hat{\mu} = 66.04$ (`test.mu[which.max(lik)]`). But, the code above only worked because we held the SD to be known and the same as the sample SD. If we didn't,

we would have to search over a range of possible SD values, and find the combination of $\hat{\mu}$ and $\hat{\sigma}$ that maximized the likelihood. In that case, the profile of likelihoods would have been a surface, not a line.

```
# range of population means to test
test.mu <- seq(62, 70, length=100)

# range of population SDs to test
test.sigma <- seq(1, 5, length=100)

# matrix to hold likelihood values
Lmat <- matrix(NA, nrow=100, ncol=100)

# fill LMat in a loop
for(i in 1:100){
  mu.i <- test.mu[i]
  for(j in 1:100){
    sigma.j <- test.sigma[j]
    Lmat[i,j] <- prod(dnorm(a, mean=mu.i, sd=sigma.j))
  }#j
}#i
```

```
# make the heatmap

## sequence of breaks
brks <- seq(min(Lmat), max(Lmat), length=21)

## colors (length of brks - 1)
cols <- topo.colors(20)

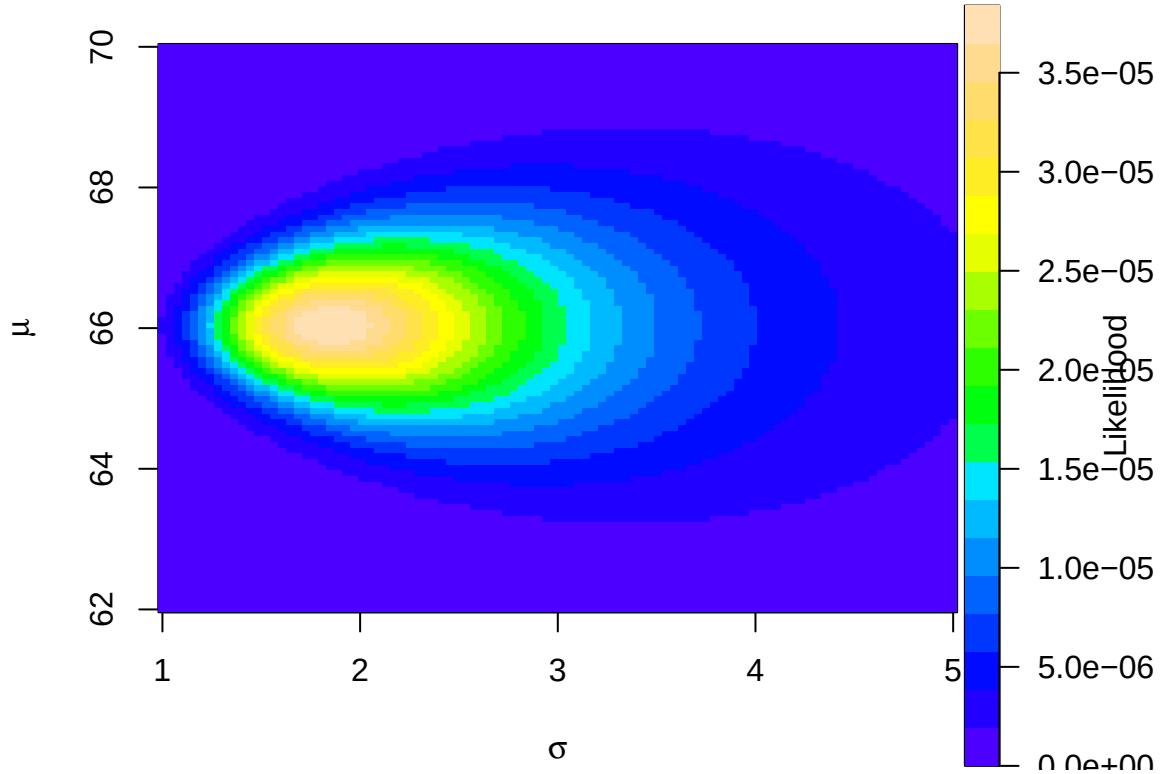
# wide panel for plot, narrow panel for legend
layout(matrix(c(1, 2), nrow=1), widths=c(5, 1))

# likelihood surface
par(mar=c(4.1, 4.1, 1.1, 0.1))
image(x=test.sigma, y=test.mu, z=t(Lmat),
      breaks=brks, col=cols,
      xlab=expression(sigma),
      ylab=expression(mu))

# color legend
par(mar=c(0.1, 0.1, 0.1, 4))
```

```
image(
  x=1,
  y=brks,
  z=matrix(brks[-1], nrow=1),
  breaks=brks,
  col=cols,
  xaxt="n", yaxt="n",
  xlab="",
  ylab="Likelihood"
)
axis(4, las=1)
mtext("Likelihood", side=4, line=2.5)
```

Figure 1.8: The likelihood surface of a model with 2 parameters, mu and $sigma$ resulting from a wide parameter search space, shown as a heatmap with `image()`.



The surface above shows that the joint likelihood peaks at around $\hat{\mu} = 66$ and $\hat{\sigma} = 2$. We can zoom in a bit and narrow our search:

```
# range of population means to test
test.mu2 <- seq(65.8, 66.2, length=100)

# range of population SDs to test
```

```r
test.sigma2 <- seq(1.8, 2.2, length=100)

# matrix to hold likelihood values
Lmat2 <- matrix(NA, nrow=100, ncol=100)

# fill LMat in a loop
for(i in 1:100){
  mu.i <- test.mu2[i]
  for(j in 1:100){
    sigma.j <- test.sigma2[j]
    Lmat2[i,j] <- prod(dnorm(a, mean=mu.i, sd=sigma.j))
  }#j
}#i

# make the heatmap

## sequence of breaks
brks2 <- seq(min(Lmat2), max(Lmat2), length=21)

## colors (length of brks - 1)
cols <- topo.colors(20)

# wide panel for plot, narrow panel for legend
layout(matrix(c(1, 2), nrow=1), widths=c(5, 1))

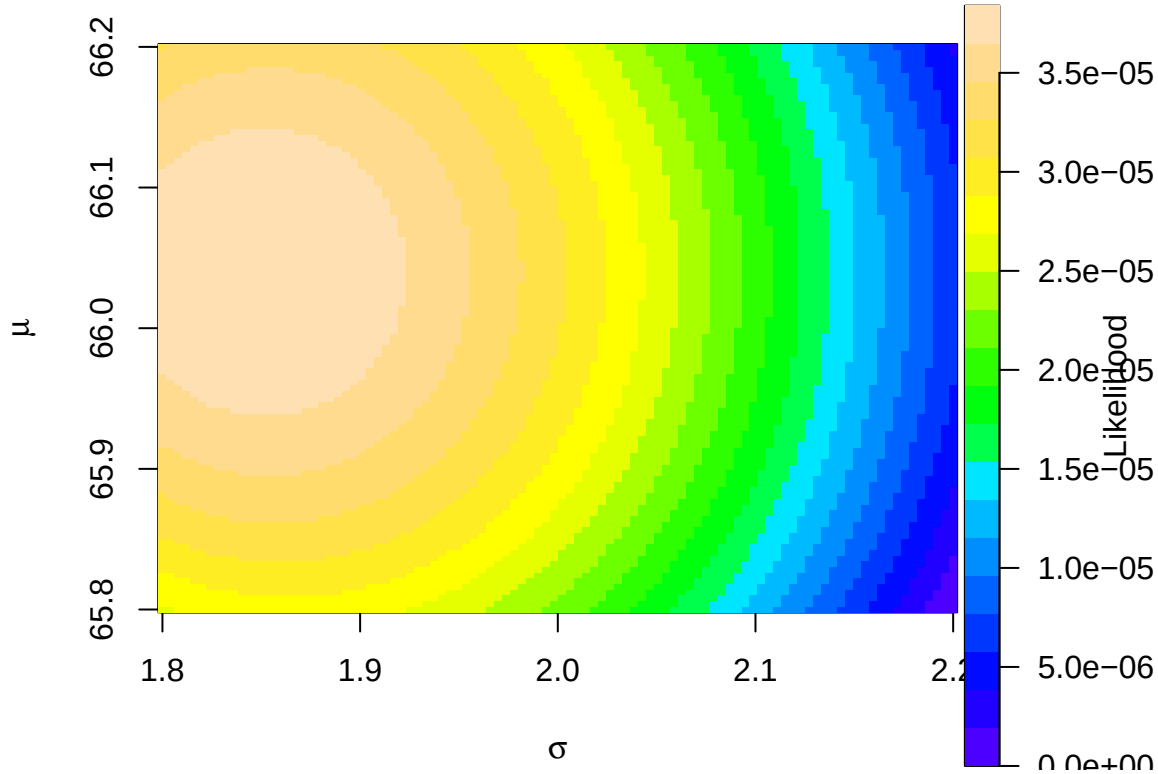
# likelihood surface
par(mar=c(4.1, 4.1, 1.1, 0.1))
image(x=test.sigma2, y=test.mu2, z=t(Lmat2),
      breaks=brks2, col=cols,
      xlab=expression(sigma),
      ylab=expression(mu))

# color legend
par(mar=c(0.1, 0.1, 0.1, 4))

image(
  x=1,
  y=brks,
  z=matrix(brks2[-1], nrow=1),
  breaks=brks2,
  col=cols,
```

```
  xaxt="n", yaxt="n",
  xlab="",
  ylab="Likelihood"
)
axis(4, las=1)
mtext("Likelihood", side=4, line=2.5)
```

Figure 1.9: The likelihood surface of a model with 2 parameters, mu and $sigma$, with a narrowed parameter search space, shhown as a heatmap with `image()`.



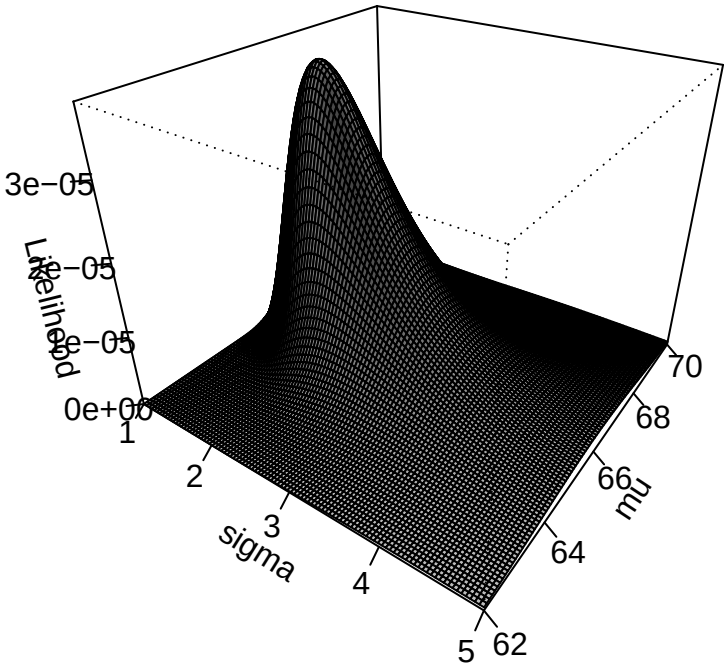
And we could keep going until we get a region we are happy with. Just as with the 1-d example, we would want the point where the likelihood surface *peaks*, and we could express the 95% confidence interval for the parameters as the ranges of $\hat{\mu}$ and $\hat{\sigma}$ where 95% of the volume under the surface is found.

Here is the likelihood *surface*:

```
par(mar=c(2,2,1,1))
persp(
  test.sigma, test.mu, t(Lmat),
  xlab=expression(sigma),
  ylab=expression(mu),
  zlab="Likelihood",
  theta=35,
  phi=30,
```

```
  expand=0.8,
  ticktype="detailed",
  shade=0.4
)
```

Figure 1.10: The likelihood surface for a model with 2 parameters, mu and $sigma$.



The maximum likelihood estimates for something like a linear regression model, or a Poisson GLM, are found by similar procedure, but with more than 2 parameters the resulting surface is 3-dimensional, or 4-dimensional, making the profiling much more complicated.

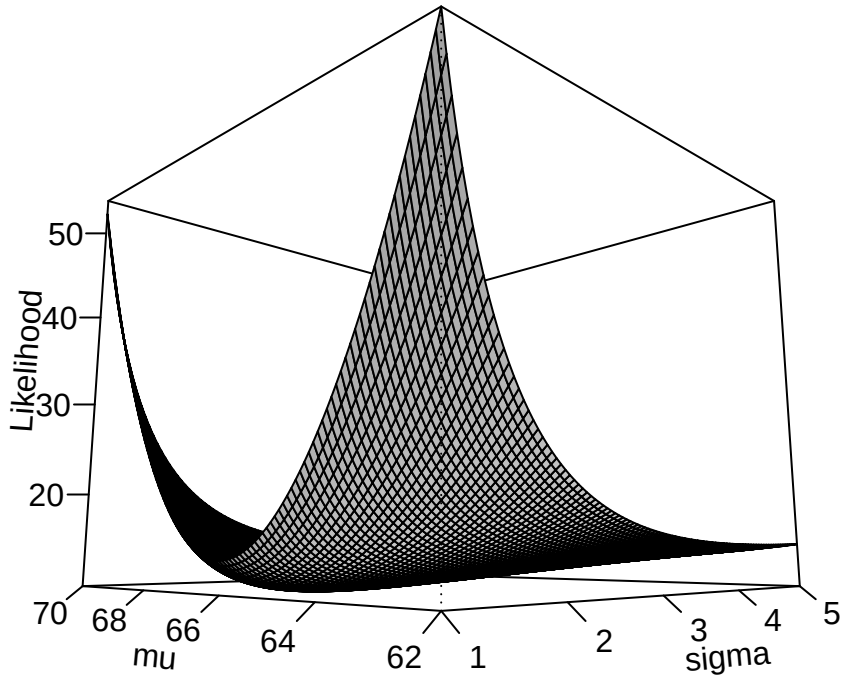
1.3.3 Negative log-likelihood

In practice, modern MLE seeks not to maximize the likelihood but to minimize the negative log-likelihood. That turns the surface in the previous figure into this:

```
NLL <- -1 * log(t(Lmat))
par(mar=c(2,2,1,1))
persp(
  test.sigma, test.mu, NLL,
  xlab=expression(sigma),
  ylab=expression(mu),
  zlab="Likelihood",
```

```
  theta=-45,
  phi=-10,
  expand=0.8,
  ticktype="detailed",
  shade=0.4
)
```

Figure 1.11: The log-likelihood surface for a model with 2 parameters, mu and $sigma$.



Why the negative log-likelihood? First, taking the logarithm makes likelihood calculations computationally easier: products become sums, and very small likelihoods are less likely to cause numerical problems (i.e., very small values won't unexpectedly round to 0). Because the logarithm is a strictly increasing function, the parameter values that maximize the likelihood also maximize the log-likelihood. Multiplying by -1 then turns this into an equivalent **minimization** problem.

For reasons outside the scope of this course, minimization problems are more convenient to solve numerically than maximization problems. At a smooth minimum, the slope of the negative log-likelihood function is 0, so finding the minimum can be approached as finding a root of its derivative (i.e., a place in the parameter space where the derivative of the negative log-likelihood function is 0). One classic approach is the Newton-Raphson algorithm, which iteratively uses derivatives to move toward such a point. Modern software has many optimization algorithms, but the basic goal is the same: find the parameter values that produce the smallest negative log-likelihood, and therefore the largest likelihood.

1.3.4 Conclusions

So at the end of the MLE analysis, we have still not established that a hypothesis is true. Instead, we have found how compatible our observed data (evidence) are with different possible values of the model parameters. We choose the parameter values under which the observed data have the greatest likelihood. In the language we will introduce shortly for Bayes' theorem, we have made $p\left(E|H\right)$ (or more precisely, the corresponding probability density) as large as possible. We still have not calculated or even mentioned $p\left(H|E\right)$: the probability that our hypothesis is correct given the available evidence. In other words:

$$\underbrace{p\left(E|H\right)}_{\text{likelihood}} \quad \neq \quad \underbrace{p\left(H|E\right)}_{\text{what we actually want to know}}$$

1.4 Bayes' Theorem

1.4.1 Introduction

So far, we have established that both classical and modern statistical practice are very good at two things: *p*-values, which are basically but not quite $p\left(E|\neg H\right)$; and likelihoods, which are $p\left(H|E\right)$. But,

$$\underbrace{p\left(E|\neg H\right)}_{\text{p-value}} \quad \neq \quad \underbrace{p\left(H|E\right)}_{\text{what we actually want to know}}$$

and

$$\underbrace{p\left(E|H\right)}_{\text{likelihood}} \quad \neq \quad \underbrace{p\left(H|E\right)}_{\text{what we actually want to know}}$$

Getting at what we really want to know, the probability of a hypothesis being true given the available evidence, is not really possible under classical or **frequentist** inference. It is, however, possible under **Bayesian inference**. The key difference between Bayesian and classical inference is that classical inference treats data as conditional on hypotheses, whereas Bayesian inference treats hypotheses conditional on evidence. The way that Bayesian inference closes the loop, philosophically, is to consider not just the probability of the evidence conditional on the hypothesis, but also the *unconditional* probability of observing the evidence. That unconditional probability, called the **prior**, is an important part of **Bayes' theorem**:

$$p\left(A|B\right) = \frac{p\left(B|A\right)p\left(A\right)}{p\left(B\right)}$$

This theorem relates several important probabilities:

- $p\left(A|B\right)$ is the **posterior probability**: the probability of event A occurring when B is true.
- $p\left(B|A\right)$ is the **likelihood**: the probability of event B occurring that A is true.

- $p(A)$ is the **prior probability**: the unconditional probability of event A. "Unconditional" here means "without considering B".
- $p(B)$ is the **marginal probability**: the unconditional probability of event B.

1.4.2 The language of evidence

In a scientific investigation, we are interested in the probability of a hypothesis being true, or $p(H)$. Now we have a way to relate $p(A)$ to the evidence E, which has its own set of probabilities. Just rewrite Bayes' theorem in the language of evidence and hypotheses:

$$p(H|E) = \frac{p(E|H)\,p(H)}{p(E)}$$

Now we have a relatively simple expression for what we really want: the probability that our hypothesis is correct given the evidence we have observed. We just need to know:

- $p(E|H)$ is the **likelihood**: the probability of the evidence occurring if the hypothesis is true. You might recall that this is what we calculated in the previous section.
- $p(H)$ is the **prior probability**: the unconditional probability of the hypothesis being true. This is the probability we place on a hypothesis before seeing any evidence and can be very large or very small.
- $p(E)$ is the **marginal probability**: the probability of observing the evidence whether or not the hypothesis is true.

The likelihood can be calculated from the data. The prior probability can be assigned arbitrarily or based on previous literature or evidence. However, the denominator is a bit trickier. We can make it easier by splitting $p(E)$ into two cases: $p(E|H)$ (the likelihood) and $p(E|\neg H)$. That give us:

$$p(H|E) = \frac{p(E|H)\,p(H)}{p(E|H)\,p(H) + p(E|\neg H)\,p(\neg H)}$$

Because observing $E|H$ is requires H and observing $E|\neg H$ requires $\neg H$ (that is why those probabilities occur as coefficients in the denominator). And now we can see that part of the denominator is the *p*-value, another piece we already know (Section 1.2.3), and also realize that $p(\neg H) = 1 - p(H)$.

Putting it all together:

$$\underbrace{p(H|E)}_{\text{what we want to know}} = \frac{\overbrace{p(E|H)}^{\text{likelihood}}\,\overbrace{p(H)}^{\text{prior}}}{\underbrace{p(E|H)}_{\text{likelihood}}\,\underbrace{p(H)}_{\text{prior}} + \underbrace{p(E|\neg H)}_{\text{p-value}}\,\underbrace{(1-p(H))}_{\text{1 - prior}}}$$

In practice, and for reasons beyond the scope of this course, calculating the denominator for a statistical model can require evaluating integrals in many dimensions (those reasons are related to finding the "area" or rather hypervolume beneath a likelihood surface in many dimensions…which makes "beneath" seem imprecise). We'll discuss the consequences of that in the next module.

1.4.3 Example

Let's consider a quick example of using Baye's theorem to evaluate the probability that a hypothesis is true given some evidence.

Here is a quick example of Bayesian reasoning. Imagine you go to the doctor and get tested for a rare condition, Niemann-Pick disease. Your doctor tells you that this disease affects about 1 in 250000 people. Unfortunately, you test positive. But, the doctor tells you not to worry because the test is only 99% reliable. Why is the doctor so sanguine, and what is the probability that you have Niemann-Pick disease?

The probability of having this disease can be estimated using Bayes' theorem. First, use what you know to assemble the terms:

- The overall incidence is 1 in 250000 people, or 0.0004%. So, the prior probability $P\left(H\right)$ is 0.000004.
- The probability of a positive test given that you have the disease, $P\left(E|H\right)$, is 0.99 because the test is 99% reliable. This is also called the **sensitivity** of the test.
- The probability of a positive test if someone does *not* have the disease, $p\left(E|\neg H\right)$, 1% because if 99% of positives are true positives, the remainder must be false positives.
- The denominator, $P\left(E\right)$, is the unconditional probability of a positive test.

The denominator includes two possibilities: either a positive test when someone has the disease, or a positive test when someone doesn't have the disease. That is, the probability of a true positive *or* a false positive. This can be calculated as:

$$\left(E|H\right)p\left(H\right)+p\left(E|\neg H\right)p\left(\neg H\right)$$

The first term is the **sensitivity**, or probability of a true positive ($p\left(E|H\right)$) scaled by the probability of having the disease (the prior $p\left(H\right)$). The second addend is the probability of a false positive ($p\left(E|\neg H\right)$) scaled by the probability of not having the disease ($p\left(\neg H\right)$). We can simplify a bit by reasoning that $p\left(\neg H\right)$ is really $1-p\left(H\right)$.

Plug every thing into Bayes' theorem:

$$\underbrace{p\left(H|E\right)}_{posterior}=\frac{\overbrace{\left(0.99\right)}^{likelihood}\overbrace{\left(0.000004\right)}^{prior}}{\underbrace{\left(0.99\right)}_{likelihood}\underbrace{\left(0.000004\right)}_{prior}+\underbrace{\left(0.01\right)}_{p(E|\neg H)}\underbrace{\left(0.999996\right)}_{1-prior}}$$

and simplify:

$$p\left(H|E\right)=\frac{0.00000396}{0.01000392}=0.000396$$

A 0.04% probability that you have the disease is not very worrying! In this example, the evidence from a 99% reliable test can't overcome the fact that the disease is so rare to begin with. In other words, even among people who get a positive test, it's far more likely that you don't have the disease but got a false

positive than that you have the disease and got a true positive. This is an example of a very strong prior being more influential than the evidence.

This might be easier to see if we imagine a representative population of 1 million people:

- The prior probability is still 0.000004, so $0.000004 \times 1000000 = 4$ people have the disease.
- The false positive rate is 1%, so we expect that $0.01 \times 999996 \approx 10000$ people without the disease will test positive.
- The true positive rate is 99%, so let's assume that all 4 people with the disease test positive.

So, the doctor should explain that, all things being equal, we should assume that it's more likely you are in the false positive group (10000 out of 1000000 or 1%) than in the positive group (4 out of 1000000 or 0.0004%). In fact, despite the positive test you are still 2500 times more likely to not have the disease than to have it. The good evidence (99% reliable) cannot overcome the extremely small prevalence of the disease.

But, there's an issue with this calculation. The prior probability of 0.000004 was based on the assumption that the test was conducted on a random person from the population. Do physicians conduct expensive tests for vanishingly rare diseases on random people? Of course not. Maybe the physician only orders this test if they think that there is a 10% chance that you really have the disease. This might be because, in their experience, 10% of people with your symptoms turn out to have the disease. In that case, the calculation becomes:

$$(H|E) = \frac{(0.99)\,(0.1)}{(0.99)\,(0.1) + (0.01)\,(0.9)}$$

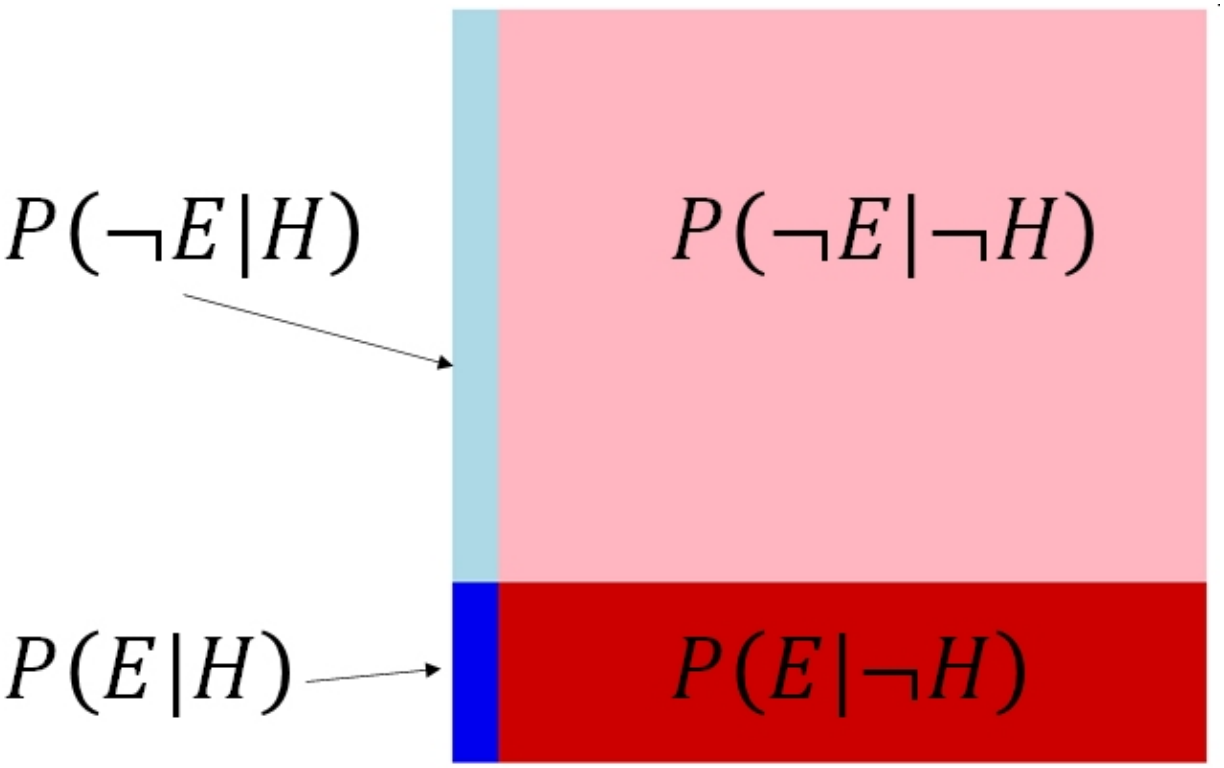
which simplifies to:

$$p\,(H|E) = \frac{0.099}{0.108} = 0.916$$

91.6% is a lot more worrying than 0.0004%. This example shows that the prior probability can have a huge effect on the posterior probability.

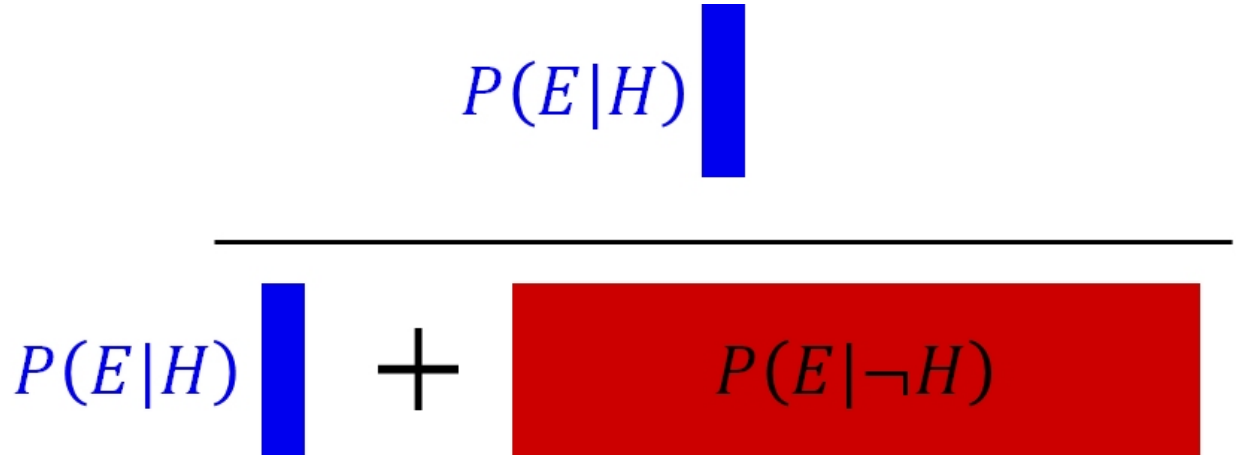
One way to visualize Bayes' rule is to think of the entire probability space as occupying a square with area 1. The square has 4 sections:

Figure 1.12: Bayes theorem helps us reason about the relative probabilities of observing a piece of evidence E or not when a hypothesis H is true or not true. The area of all four sections must add up to 1. According to Bayes' theorem, the probability of a hypothesis H being true, given evidence E, is the ratio of $P\left(E|H\right)$ to $P\left(E|H\right)+P\left(E|lnotH\right)$.



Notice that the prior probability $p\left(\ldots|H\right)$ (the two left, blue sections) is much smaller than the probability of $\neg H$ (the two right, red and pink sections with $p\left(\ldots|\neg H\right)$). Bayes' rule restricts our focus to the cases where we observe the evidence–the bottom two sections with $p\left(E|\ldots\right)$. Given that the evidence was observed, the sections with $\neg E$ are irrelevant. The calculation then becomes:

Figure 1.13: In Bayesian inference, the probability of the hypothesis being true given the observed evidence is the ratio of cases where the evidence is observed *and* the hypothesis is true, to all cases where the evidence is observed.



1.4.4 Estimating models under Bayesian inference

At this point you might be thinking, "Wait a minute! NHST and maximum likelihood were estimating parameters of models, but all you did with Bayesian inference was calculate probabilities!" And that's true. We still need to build the bridge from Bayes' theorem to model fitting.

Another way to look at the posteriors in Bayes theorem, $p\left(H|E\right)$, is $p\left(\theta|y\right)$, where θ is the set of parameters in some model. The parameters in θ are here serving as a statement of the hypothesis. This might–and should–remind you of the likelihood function $\mathcal{L}\left(\theta|x\right)$ that we saw earlier. Bayes theorem makes this:

$$p\left(\theta|y\right) = \frac{p\left(y|\theta\right)p\left(\theta\right)}{p\left(y\right)}$$

Because we're trying to make inferences about θ, we can just write

$$p\left(\theta|y\right) \propto p\left(y|\theta\right)p\left(\theta\right)$$

because $p\left(y\right)$ doesn't depend on θ. And here is the connection to model fitting. For every proposed value of the parameters in θ, we can use the prior to calculate:

$$\underbrace{p\left(y|\theta\right)}_{likelihood} \times \underbrace{p\left(\theta\right)}_{prior} = \underbrace{\text{something proportional to } p\left(\theta|y\right)}_{posterior}$$

The denominator from earlier, $p\left(y\right)$, is difficult to calculate. But, we don't have to. Because the same $p\left(y\right)$ occurs everywhere, we can ignore it and instead compare the **relative posterior density of different parameter values**. In other words, we use Bayes' theorem to help us estimate a **posterior distribution** for each model parameter.

In a problem like the MLE example above, with a small number of parameters, this is feasible through a brute force grid search. Instead of just calculating the likelihood function of the data in `a` as

```
Lmat[i,j] <- prod(dnorm(a, mean=mu.i, sd=sd.sigma.j))
```

we could have also specified priors and calculated a posterior probability density for each combination of μ and σ up to a proportionality constant as

```
posterior.matrix[i,j] <- prod(
  dnorm(a, mean=mu.i, sd=sd.sigma.j)) *
  dnorm(mu.i, ...) *    # prior on mu
  dgamma(sigma.j, ...) # prior on sigma
```

If the search grid was sufficiently precise, we could have also normalized these values so that they integrated to 1, which by definition would make them a probability distribution–the posterior distribution.

1.4.5 MCMC and Metropolis-Hastings

In practice, calculating posterior distributions of many parameters simultaneously is computationally difficult. The surface of possible values and their joint likelihoods (Figure 1.10), defined by Bayes' theorem, often can't be defined explicitly. Instead, we can use different techniques to explore the surface. That is, to **sample from the posterior distributions**.

One strategy for sampling from the posterior distribution is really a family of techniques known as **Markov chain Monte Carlo (MCMC)**. A **Markov chain** is a sequence of values, where each value depends only on the value before it. The **Monte Carlo** part refers to a European city famous for its casinos. So, MCMC involves a series or *chain* of values that involves at least some random number generation. One of the most common MCMC algorithms is the **Metropolis-Hastings** algorithm.

Suppose we are using a Markov chain to explore the posterior distributions of the parameters in a model. At the current step, the chain is at state θ. Using some process (it doesn't matter what), a proposed new set of parameter values θ'. Whether θ' is an improvement depends on the ratio

$$\frac{p\left(\theta'|y\right)}{p\left(\theta|y\right)}$$

Substitute the expressions for $p\left(\theta'|y\right)$ and $p\left(\theta|y\right)$ that we get from Bayes' theorem:

$$\frac{p\left(y|\theta'\right)p\left(\theta'\right)/p\left(y\right)}{p\left(y|\theta\right)p\left(\theta\right)/p\left(y\right)}$$

The $1/p\left(y\right)$ terms cancel, leaving us with:

$$\frac{p\left(y|\theta'\right)p\left(\theta'\right)}{p\left(y|\theta\right)p\left(\theta\right)}$$

If the proposed parameters in θ' have higher posterior density than those in θ, the algorithm can accept them. If they have lower posterior density, the algorithm *sometimes* accepts them, with a probability equal to that posterior density ratio.

After enough iterations, the chain will spend more time in some regions of the parameter space than in other parts. The amount of time it spends in each region is proportional to the posterior probability densities for the parameters.

So that's the connection:

> Bayes' theorem **constructs** the posterior surface. MCMC is how we **explore** that surface.

MCMC with Metropolis-Hastings is just one algorithm for sampling from posterior distributions. A variant called **Gibbs sampling** omits the accept/reject step. More modern algorithms like HMC/NUTS are more sophisticated still, but they solve the same basic problem: efficiently sampling a posterior distribution defined by a prior distribution and a likelihood. And unlike MLE estimates, which we use to make inferences about parameters, posterior samples literally are the inferential result. Summaries of the posterior draws, like their mean, median, SD, and so on, tell us about the parameters and their uncertainties.

1.4.6 Credible intervals

The distinction between MLE estimates and posterior draws becomes especially important when we construct intervals around parameter estimates to express our uncertainty about them. In frequentist inference, the parameter is treated as fixed but unknown. Because the parameter value is fixed, we cannot use probability to describe uncertainty about the parameter (because definitionally, there is none). Instead, we can construct a **95% confidence interval (CI)** to describe uncertainty that arises from how we arrived at the estimate. If we repeatedly collected data and constructed intervals in the same way, 95% of those intervals would contain the true parameter value. We *cannot* say that the 95% CI is 95% certain to contain the true parameter value.

In Bayesian inference, the posterior draws are our sample of the posterior distribution. Given enough draws, we can calculate a **95% credible interval (CRI)** that contains 95% of the posterior probability for that parameter. Given the model, prior, and observed data, we *can* say that there is a 95% probability that the parameter lies within the 95% CRI. For example, given a set of 10,000 draws for an unknown population mean μ:

$$\mu_{(1)}, \mu_{(2)}, \mu_{(3)}, \cdots, \mu_{(10000)}$$

we can calculate the 2.5 and 97.5 percentiles:

```
quantile(mu.draws, c(0.025, 0.975))
```

If that command returns $[64.2, 67.8]$, then the 95% of the posterior distribution lies between 64.2 and 67.8 and we could legitimately say "There is 95% probability that μ lies between 64.2 and 67.8".

This point is worth reiterating:

- A frequentist 95% confidence interval is a statement about the long run performance of a series of repeated experiments. If we repeatedly collected new samples and calculated intervals the same way, about 95% of those intervals would contain the true parameter.
- A Bayesian 95% credible interval is a statement about the parameter itself. Given the data, model, and prior, there is a 95% probability that the parameter lies within the interval. That is, a credible interval does what most people mistakenly think the confidence interval does!

1.4.7 Conclusions

Bayesian inference is a framework for evaluating evidence and *updating beliefs based on evidence*. When conducting Bayesian inference, researchers start with some initial idea about their model system: a **prior**. They then collect evidence (data), and update their idea based on the evidence. The updated idea is called the **posterior**.

- When the prior is strong, or the evidence weak, their ideas will not be updated much and the posterior will be strongly influenced by the prior.
- When the evidence is strong, or the prior weak, the posterior will be updated more and thus less influenced by the prior.

Some researchers prefer to use naïve or uninformative priors, so that their conclusions are influenced mostly by the evidence. Bayesian statistical methods will return essentially the same parameter estimates (e.g., differences in means or slopes) as other methods when uninformative priors are used. This means that almost any traditional statistical analysis can be replaced with an equivalent Bayesian one. The statistical models that are fit, such as linear regression or ANOVA, are the same. All that really changes is how the researcher interprets the relationship between the model and the data.

In frequentist inference, researchers consider parameters like slopes, intercepts, effect sizes, etc., as fixed, unknown parameters and consider the data to occur stochastically following probability distributions. Bayesian inference flips that script, treating the data as fixed, and assuming that the model parameters are unknown quantities following probability distributions.

Bayesian inference is attractive to many biologists because it mimics the way that scientists actually think: starting with one idea (a prior), collecting evidence, then updating that idea (posterior). Part of the Bayesian calculation is $p\left(H|E\right)$, or the probability of the hypothesis being true given the evidence that is observed. Contrast this with the frequentist *p*-value, which is $p\left(E|\neg H\right)$. Most people instinctively want the first, but have to settle with the almost backwards logic of the *p*-value.

1.5 Literature cited

Gould SJ. 1983. Hen's teeth and horse's toes. W.W. Norton, New York.

Chapter 2

Probability and Bayes' theorem

2.1 Bayesian thinking

Most of the time when we fit a model in R, we need to supply the software with enough information to specify which model to fit, what variables to include, and where to find those variables. All the rest–i.e., the machinery of sums of squares or maximum likelihood–takes place under the hood. Once you specify, say, a logistic regression model, there really isn't much else to do. For example:

```
# fit a logistic regression model
mod1 <- glm(y~x1, data=my.data, family=binomial(link="logit"))
```

fits a logistic regression model, and the command is very concise:

- `mod1` is (or will be?) the fitted model object.
- `glm()` is the command that fits GLMs.
- `y~x1` is the model formula, such that `y` is the response variable and `x1` is the predictor variable.
- `my.data` is the data frame that contains all of the variables in the model.
- `binomal()` is the response distribution (i.e., the `family` of the GLM).
- `"logit"` is the link function.

You don't need to specify a distribution for the intercept or slope: those are baked into the GLM. Likewise, you don't need to make a decision about when the gradient descent methods have reached a valid solution. That too is baked into `glm()`.

The same is not true when using Bayesian inference. Note that I said, "Bayesian inference" and not "Bayesian models". There isn't really such a thing as a "Bayesian model", only a Bayesian approach to testing a model.

When working in Bayesian, we need to specify more pieces of the model upfront, and we need to make more decisions on the back end.

2.1.1 Priors

2.1.1.1 Role in Bayesian inference

As we saw in Section 1.4, the prior $p(H)$ is an important part of Bayes' theorem:

$$p(H|E) = \frac{p(E|H)\,p(H)}{p(E)}$$

The prior is used to scale the ratio of the likelihood $p(E|H)$ to the probability. This means that priors can exert varying degrees of influence: when the prior is strong, the prior dominates the calculation. When the prior is weak or uninformative, the data dominate.

2.1.1.2 Informative priors

An **informative** prior is a prior that contains a lot of information, or a very strong guess, about a parameter or a model. For example, if you were building a model to predict the body mass of an animal a weak ahead, your prior distribution might be centered on the animal's current weight with a variance equal to the historical average of week-to-week body mass fluctuations. Or, you could use a distribution of body masses for animals of that precise age.

Using informative priors is using *pre-existing knowledge* to help fit your model. This does not necessarily mean that your prior distribution will determine the fitted model. It means that more evidence will be needed to find an outcome that is very unlikely under the prior distribution. Put another way, informative priors can help you reach a robust conclusion with fewer additional data. The latter reason is part of why Bayesian methods are becoming more and more common in clinical trials.

For example, if a drug developer is testing a new, higher dose of a medicine (say, 10 mg instead of 5 mg), there is no reason to start from the assumption that the 10 mg version will have 0 effect if the 5 mg version is already known to be clinically efficacious. They could set their prior for the effect of the 10 mg version to slightly larger than the known effect of the 5 mg version, and obtain tiral results with fewer data.

A less dramatic example is to use informative priors when the sign of a coefficient is known. In a recent study published here at KSU, a student used Bayesian path analysis models to explore how environmental conditions affected fire behavior. After consultation with his PI (and me), he defined the prior distributions of some parameters to be either positive or negative based on strong evidence from the literature. This gave his analysis a boost in power that allowed him to discover some neat effects–without having to waste sample units eliminating possibilities that were already known to be false.

2.1.1.3 Uninformative (vague) priors

2.1.1.4 Conjugate priors

2.1.2 Evidence (Data)

2.1.3 Convergence

2.1.4 Posteriors

2.2 Updating beliefs and Bayes' factors

Chapter 3

Introduction to JAGS and MCMC

3.1 Overview and objectives

This week we will explore the basics of JAGS, a common software package for fitting models under Bayesian inference. Despite its conceptual elegance, Bayesian inference in practice is far more computationally intensive than frequentist inference. While computational methods for frequentist inference were optimized decades ago, methods for Bayesian inference are growing and improving rapidly. For these reasons most entry-level statistical packages such as R or SAS rely on extensions or separate programs to conduct the computations for Bayesian methods. In this module we will go over the steps of running a simple model in JAGS from R. This workflow will serve as a template for more complicated analyses throughout the semester.

By the end of this module you should be able to:

1. Install R, JAGS, and R packages.
2. Call and run JAGS from R.
3. Specify simple models in the BUGS/JAGS language.
4. Interpret outputs from JAGS to make inferences about biological systems.

For this module you will need:

```
library(R2jags)
library(rjags)
```

3.2 Preliminary steps

3.2.1 Installing R

R is a software package used primarily for statistical analysis, as well as a programming language used to interact with the program. It is widely used in academia, government, and private industry for data analysis. Many on-campus computers at KSU already have R installed on them.

First, install the latest version of R from here (accessed 2026-07-24). Select a mirror (download server) from the list. You will be taken to a page where you will select the version of R that corresponds to your operating system. Click on the link for your OS and follow the instructions from there. Like many programs from the internet, R comes with an installer that will largely take care of everything automatically. Just let the installer run and accept the default options.

Optionally, you can install RStudio. It can be downloaded from here (accessed 2026-07-24). As with R, select the version that corresponds to your operating system. As you did when you installed R above, just let the installer run and accept the default options. In this class we will use the base R graphical user interface (GUI) for simplicity, but some people prefer RStudio.

This course assumes some basic familiarity with R and R programming; alternatively, some level of comfort with any command-line or programming interface. If you can write an Excel formula, you will probably be fine. If you need an introduction to or refresher on R, there are many free tutorials for R available on the web.

3.2.2 Install JAGS

JAGS, short for "Just Another Gibbs Sampler", is a program for fitting models using Markov chain Monte Carlo (Plummer 2003). It is one of the most popular programs for Bayesian inference.

JAGS is hosted on SourceForge here (accessed 2023-01-05). Click the big green download button. After the program downloads, follow the on-screen instructions for the installation process. JAGS comes with its own installer much like any other program you download from the internet.

3.2.3 Installing R packages

JAGS is mainly designed to be called from other programs, so it doesn't have a GUI. In order to call JAGS from R, you will need two packages: rjags (Plummer 2022) and R2jags (Su and Yajima 2021). Install these in R using the following commands. Click Yes or OK on any pop-up messages from R.

```
install.packages("rjags")
install.packages("R2jags")
```

These packages will be installed to your personal R library. This means that they are only available under your operating system profile (log-in). If someone else logs into the same machine, they may have R installed but won't have any add-on packages installed. This is because packages are stored in a folder tied to your operating system profile.

These packages are also specific to the version of R used to install them. Each version of R installed on your machine has its own library, or folder containing add-on packages. If you update R, you will need to reinstall any add-on packages.

3.3 What you need to know about MCMC

We introduced the basics of MCMC in Section 1.4.5. The more practical things you need to know are about MCMC are:

Chains can take many steps to find the posterior distribution

MCMC chains often start at initial parameter values that are essentially random and not representative of the posterior distribution. So, we discard the first part of a chain as burn-in. Many people use 10% or 20% as burn-in, but there is no universally agreed-upon rule. MCMC chains often need to be very long, many thousands of steps, to find robust approximations of the posterior distributions of all the parameters. In my own work I usually use $\geq$ 100,000 steps, or $\geq$ 250,000 for complicated models.

Chains are autocorrelated

The values in an MCMC chain can be correlated with values in nearby steps; in other words, chains are autocorrelated. To reduce the influence of this autocorrelation, some people **thin** the chains by only keeping every so many observations. Ratios of 1 in 20 or 1 in 100 are common. I usually thin my chains so that they end up with 1000 to 2000 posterior values. Note that in some software (e.g., stan), thinning is not recommended. However, thinning is still standard practice for using JAGS.

Chains might not find the posterior distribution

There is no guarantee that and MCMC chain will adequately explore the correct posterior distribution. Even if the initial prior distribution is appropriate and the data robust, there are still some chain pathologies to watch out for. Chains can fail to converge; become trapped near a local mode (but not the global mode); or mix poorly (the chains do not agree).

For this reason, most MCMC analyses use multiple chains. If the chains converge to the same stationary distribution, then we say the chains have converged. Most analyses use 3 chains, but some researchers use other numbers.

MCMC model fitting requires tuning

The parameters of the MCMC process often need to be optimized for an analysis. This process is referred to as **tuning**. Some people will try fitting models with different numbers of chains, different chain lengths, different thinning intervals, and different burn-in periods until they find a consensus. In my experience, this is usually overkill. My approach (which is very common) is to simply use a conservative set of tuning parameters—that is, a set of tuning parameters that is probably overkill and is thus very likely to find a good solution. As they say, brute force always works if you use enough.

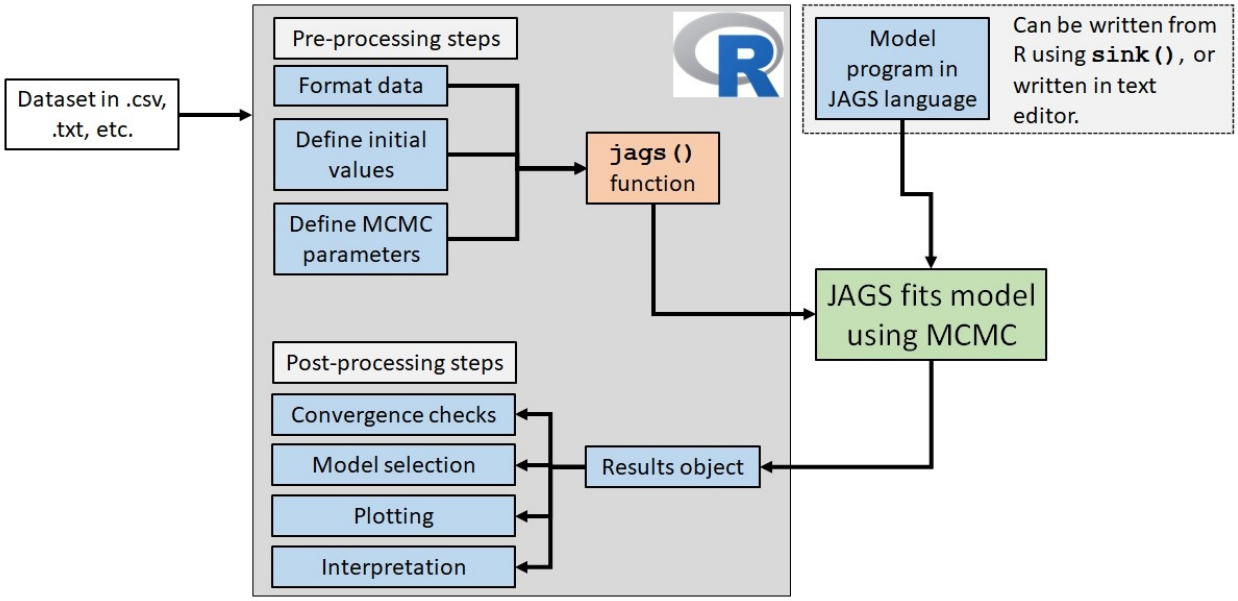
One chain or many chains?

Some people argue that you should use 1 very long chain instead of 3 shorter chains. For example, 1 chain of 300,000 iterations instead of 3 chains of 100,000 iterations. Both approaches would get you the same number of posterior samples. In my own work I prefer the multiple chain approach, mostly because this helps avoid issues related to the choice of initial value.

3.4 JAGS workflow

The workflow for fitting Bayesian models in JAGS, called from R, is summarized in the flowchart below. Most of the steps take place within R, using R code (central grey box).

Figure 3.1: Workflow for fitting Bayesian models in JAGS using R. Data are first imported into R and pre-processed by formatting the data, specifying initial parameter values, and choosing MCMC settings. These inputs are passed to the `jags()` function, which calls a model written in the JAGS language. JAGS fits the model using MCMC and returns a results object to R, where convergence is assessed and the posterior distributions are used for model selection, plotting, and interpretation.



3.4.1 Step 1: Data import

The dataset should be set up for import to R just like it would for a frequentist analysis. Ideally this is in a plain-text format like comma-separated values (.csv file) or tab-delimited values (.txt). The data should follow "tidy" data principles (Broman and Woo 2018; Wickham 2014), where each observation is a row, and each variable (column) contains only one kind of value. Note that some common R conventions, like using character strings to identify levels of a factor, are not compatible with the JAGS language. These issues will be addressed in the next step.

3.4.2 Step 2: Pre-processing steps

Before fitting a model in JAGS, we need to set up some objects that will tell JAGS what to do.

3.4.2.1 Format data

Data in R are usually held in a data frame. In order to go to JAGS, the data need to be repackaged into a list. Some kinds of data need to be reformatted:

- Character strings need to be replaced with integers. For example, a factor with levels `level1`, `level2`, and `level3` needs to be recoded to `1`, `2`, and `3`. It is very important that you record how the values were recoded.
- Values intended to be integers (e.g., counts) should be rounded (e.g., `round(x, 0)`). This is not always necessary, but sometimes non-integer values can sneak in if the data file was not saved correctly. Doing this is an extra safety check.
- Variable names should be made simple, and not include any spaces.

3.4.2.2 Define initial values

Any stochastic value in the model—a model **parameter**—must be supplied with an initial value for each chain. These values should be drawn randomly from the prior distribution. The most common way I've seen is to write a function in R that randomly draws initial values and stores them in an object with the following structure:

3.4.2.3 Set MCMC parameters

Markov-chain Monte Carlo (MCMC) is inherently a stochastic process, so fitting models with MCMC involves some tuning. This means adjusting the parameters of the run so that the results make sense and are consistent from run to run. The following parameters need to be set for JAGS run:

Table 3.1: Typical MCMC tuning parameters used when fitting Bayesian models in JAGS.

Parameter	Typical settings	Meaning
Number of chains	≥ 3	Number of independent MCMC chains. Multiple chains are run to reduce the influence of initial values and reduce the chances of getting stuck in local optima.
Number of iterations	≥ 50000	Number of steps in each MCMC chain.
Burn-in	20% of iterations	The initial iterations of each chain are discarded as *burn-in* to reduce the influence of initial values and increase the likelihood of chain convergence.

Parameter	Typical settings	Meaning
Thinning interval	≥ 20 or ≥ 100	MCMC chains are *thinned* by keeping only iterations separated by the thinning interval. This reduces the influence of autocorrelation within a chain.

For example, a researcher might run a model with 3 chains of 100,000 iterations, a burn-in period of 20,000, and a thinning interval of 100. They would end up with $3 \times (100000 - 20000)/100 = 2400$ samples from the posterior distribution of each parameter. These 2,400 samples would be the basis for inference about the parameters. I try to make sure that I have at least 1000 to 2000 posterior samples for each parameter, but there is no universally agreed-upon rule.

Finally, the parameters to be "monitored" by JAGS need to be identified as a character vector. These are the parameters whose posterior distributions JAGS will keep track of. For example, if you fit a regression model, you need to tell JAGS to monitor the intercept, slope, and residual variance term. Otherwise, JAGS will fit the model but not return any posterior chains.

3.4.3 Step 3: Writing the model file

Models in JAGS are coded using a dialect of the BUGS modeling language (Lunn et al. 2000). The model specification must define the prior distribution of each parameter, the relationships between the parameters and the data, and any derived quantities of interest (more on this later). JAGS reads a model specification from a text file (.txt) or BUGS file (.bug). I prefer to use text files because they can be written easily from within R, and read on any machine. The model text file, as its file path, is provided to the `jags()` function that calls JAGS from R so JAGS knows where to look.

As we'll see later, the BUGS/JAGS language is declarative, not imperative. This means that a model is specified by defining how the variables relate to each other, not by issuing a series of instructions to be executed in order.The order of most model statements does not determine the order of execution, because JAGS uses them to construct a directed acyclic graph of dependencies.

This can be a bit confusing to new users. Most of us are used to writing code like this:

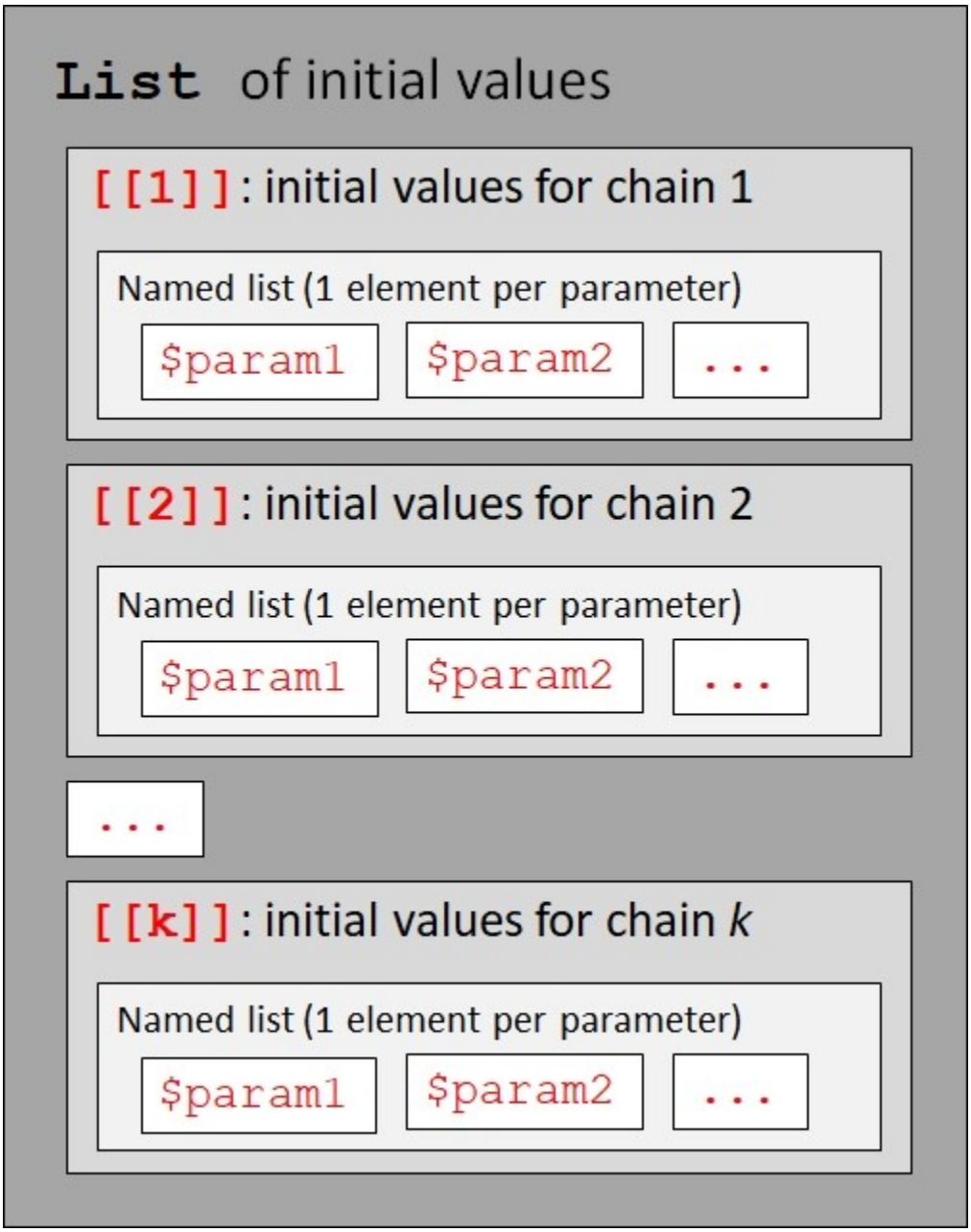
```
Do thing 1

Do thing 2

Do thing 3
```

But JAGS code works like this:

Figure 3.2: Structure of the initial values object supplied to JAGS. The object is a list with one element for each MCMC chain. Each element is itself a named list containing one initial value for every model parameter. This allows each chain to begin from a different set of starting parameter values.



```
Object A depends on Object B

Values in Object B are drawn from a normal distribution with mean = 2 and sd = 1

Variable C is calculated from Object A
```

3.4.4 Step 4: Running JAGS from R

All of the objects and files produced in the pre-processing steps are included as inputs to the function `jags()`. This function will package everything up and send it to JAGS. JAGS will then run the model, occasionally printing status updates to the R console. Depending on the size of the dataset and the complexity of the model, this step can eat up a lot of memory and processor capacity. So, save everything before you start JAGS!

When JAGS is done it will return the results to R so they can be saved in an object. This is why you should always assign the output of `jags()` to an object:

```
resultobject <- jags(...)
```

There is nothing so frustrating as waiting 3 days for a model to run only to find that the output wasn't saved anywhere!

3.4.5 Step 5: Post-processing steps

After you run the model in JAGS and get an output, the real work begins. Unlike a frequentist model that has essentially everything calculated for you automatically, a Bayesian model output needs to be carefully inspected. This is, in no small part, because there are no universally accepted criteria for model acceptance or rejection the way there are for frequentist models.

Every project's workflow is a little different, but most involve at least these steps:

- **Convergence:** checking to see whether the MCMC chains converged on the same posterior distribution. Basically, this means that the chains agree with each other . If chains do not agree, this could indicate any number of problems: (1) the model needs to be run for longer; (2) the tuning parameters need to be adjusted; (3) the model is poorly defined or does not fit; (4) the initial values have a lot of influence; or some combination of these (or another problem all together…here be dragons).
- **Model selection:** Just as in frequentist inference, we often fit several models representing alternative hypotheses, and try to identify which model best fits the data. JAGS calculates a metric called the deviance information criterion (DIC) that can be used in a manner analogous to the Akaike information criterion (AIC) .
- **Plotting:** Plotting the model and its predictions (and uncertainty) is one of the best ways to present the results of any analysis. Predicted values and the uncertainty surrounding those predictions are trivially easy to calculate with posterior distributions—because we have thousands of samples from the posterior distribution of each parameter, we can automatically get a distribution of the response

variable conditional on those distributions. This neatly solves the error propagation problem inherent in frequentist models .

- **Interpretation:** The values from each chain are samples from the posterior distributions of each parameter. These samples, and summaries thereof, can be used to make inferences about the parameters.

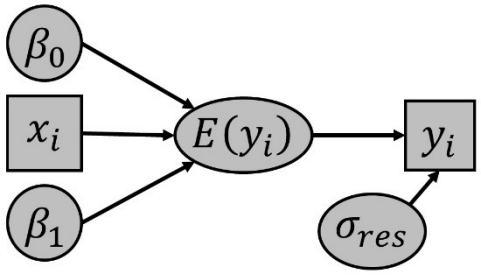
3.5 Specifying models in JAGS

JAGS has its own language for programming models. This language is almost identical to the language developed for an older family of programs, WinBUGS and OpenBUGS ("BUGS" is short for "Bayesian inference Using Gibbs Sampling"). So, almost any code example written for WinBUGS or OpenBUGS should also work for JAGS . Much of the syntax of JAGS is like the syntax of R, so R users often have an easy time learning BUGS and JAGS.

The biggest difference between the JAGS and R languages are in how commands are interpreted. R is usually imperative: commands are executed in the order in which they are written, one after another. JAGS, however, is declarative: commands define the relationships between objects and parameters. This difference can be very confusing to new users, especially users without much programming experience.

One way to think of a JAGS program is as the definition of a directed acyclic graph (DAG). This is a way of representing the connections between components of a system (like a statistical model). For example, the directed acyclic graph for an ordinary linear regression model is shown below:

Figure 3.3: Directed acyclic graph (DAG) for a Bayesian linear regression model. The intercept (β_0), slope (β_0), and predictor (x_i) determine the expected value of the response, $E\left(y_i\right)$. The observed response y_i) is generated from this expectation together with the residual standard deviation σ_{res}.



The graph shows that each observed y value y_i is determined by its expected value $E\left(y_i\right)$ and the residual standard deviation σ_{res}. The expected value is determined by the observed x value x_i and the model parameters β_0 and β_0.

The key components of JAGS code are deterministic nodes and stochastic nodes.

Deterministic nodes have no random component. They are defined with a $<-$, just like assignment in R.

```
eta[i] <- beta0 + beta1 * x[i]
```

Stochastic nodes have a random component. They are defined with a $\sim$. Think of this symbol as "distributed as".

```
y[i] ~ dnorm(eta[i], tau[i])
```

One important thing to remember is that each node can only be defined **once**. This is because defining a node more than once is tantamount to having a cycle in the graph, which is not allowed in JAGS.

Another key difference between R and JAGS is that R is vectorized, while JAGS is not. For example, the following R code is valid, no matter how many elements are in x:

```
eta <- 0.5 + 2 * x
```

The same expression or relationship in JAGS must be coded in a `for()` loop:

```
for(i in 1:n){eta[i] <- 0.5 + 2 * x[i]}
```

When expressions like this are used, the size of the deterministic node (`eta`) is defined implicitly by the loop length `n`. This can be a bit confusing if you are used to declaring object sizes explicitly.

3.6 Example analysis: two-sample *t*-test

Let's put all of this together and run a simple analysis—a two-sample *t*-test—in JAGS. We will simulate our own data so that we can check the results.

Some of this tutorial is inspired by chapter 7 of Kéry (2010), but implemented differently.

First, simulate two sets of normally distributed values with known means (`mu1` and `mu2`) and SD (`sigma1` and `sigma2`):

```
set.seed(123)
# true parameters
n <- 20
mu1 <- 10
mu2 <- 20
sigma1 <- 3
sigma2 <- 6

# draw values
y1 <- rnorm(n, mu1, sigma1)
y2 <- rnorm(n, mu2, sigma2)

# assemble data frame
# note grouping variable x is
# coded as a number because
# that's what works in JAGS
y <- c(y1, y2)
x <- rep(1:2, each=n)
```

Next, we need to define the model file. There are several ways to code the two-sample *t*-test—we'll use the means parameterization. **Means parameterization** refers to the fact that the model is specified in terms of the group-level means, rather than in terms of the between-group effect. This is a little different than the way that most frequentist methods describe models. We'll try the effects parameterization later.

```
# means parameterization of two-sample t-test
# with equal variances
mod.name <- "jags_ttest_01.txt"
sink(mod.name)
cat("
    model{
        # priors

        for(i in 1:2){
            ## group means
            mu[i] ~ dnorm(0, 0.001)
            ## residual and precision
            sigma[i] ~ dunif(0.0001, 10)
            tau[i] <- 1 / (sigma[i]^2)
        }#i

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(mu[x[i]], tau[x[i]])
        }#i

        # derived parameters
        effect <- mu[2] - mu[1]
    }#model

", fill=TRUE)
sink()
```

The pair of `sink()` commands will write everything between them to the specified file…in this case, a text file in the R home directory.

Next, make a vector of names of parameters to monitor. These parameters will be reported in the results table at the end.

```
# parameters to monitor
params <- c("mu", "effect", "sigma")
```

Set MCMC parameters: number of iterations, length of burn-in, thinning interval, and number of chains.

```
# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3
```

Every stochastic node in the model must be supplied an initial value. JAGS can guess at appropriate initial values, but these guesses tend to cause trouble. The best way, in my experience, is to write a function that returns a list with an element for each chain. Each element is itself a list, with a named element for each stochastic node that needs an initial value.

```
# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(mu=rnorm(2, 0, 100),
                         sigma=runif(2, 0.001, 10))
    }
    return(res)
}
inits <- init.fun(nchains)
```

Package up the data for JAGS in a list. Each element of the list must be named, and these names must match the variables in the model file exactly. JAGS, like R, is case-sensitive.

```
# package data for JAGS
in.data <- list(y=y, x=x, n=length(x))
```

Finally, run the model with the `jags()` command. All of the objects we've created so far, including the text file with the model code, are inputs to this function. If there is something wrong with your model or the inputs, an error message will appear at this stage.

```
mod01 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 40
   Unobserved stochastic nodes: 4
   Total graph size: 96
```

```
Initializing model
```

The model should take ≤ 1 minute to run. Once it's done, print the model object to the R console to see the output.

```
mod01
```

```
Inference for Bugs model at "jags_ttest_01.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 2.27 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
effect     9.276   1.427   6.379   8.360   9.286  10.225  11.950 1.007   280
mu[1]     10.421   0.691   9.047   9.983  10.439  10.882  11.733 1.006   350
mu[2]     19.697   1.243  17.117  18.937  19.694  20.506  21.999 1.004   510
sigma[1]   3.122   0.555   2.264   2.758   3.037   3.418   4.431 1.002   840
sigma[2]   5.346   0.948   3.818   4.682   5.219   5.887   7.525 1.002   840
deviance 222.972   3.172 218.981 220.670 222.141 224.621 230.495 1.003   670

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 5.0 and DIC = 228.0
DIC is an estimate of expected predictive error (lower deviance is better).
```

The first column to check is `Rhat` ($\hat{R}$, "R hat"), the Gelman-Rubin statistic. This statistic is a heuristic for assessing whether the chains have converged on the same posterior distribution for a parameter. There is no universally-accepted rule, but most people require $\hat{R} < 1.01$ or < 1.001 before they consider a parameter converged. Models with unconverged parameters should be treated as highly suspect (or better yet, rerun with longer chains). Convergence does not indicate anything about "statistical significance", only how consistent the model's estimate of the posterior distribution is.

Second, we check that the number of effective samples (`n.eff`) is close to `n.sims` (i.e., number of iterations saved). If not, then there may be autocorrelation in the chains. The chains should be inspected, and the model may need to be rerun with a longer thinning interval. In our case, most of the parameters have `n.eff` = 1200, the number of iterations saved, so we're probably fine.

Finally, check out the summaries of the posterior distributions for each parameter. The estimated means are about 10.42 ± 0.607 for group 1 (i.e., `y1`) and 19.870 ± 0.643 for group 2. Is this correct? R also prints various quantiles to help us get a sense of the posterior distributions.

The estimated residual SD was 2.8 ± 0.33. Is this correct? Finally, the estimated effect (μ1 – μ2) was 9.448 ± 0.883, with a 95% credible interval (CRI, not CI) of [7.72, 11.21]. Compare all of this to the output of a conventional one-sample *t*-test:

```
t.test(y~x)
```

```
    Welch Two Sample t-test

data:  y by x
t = -7.181, df = 30.672, p-value = 4.784e-08
alternative hypothesis: true difference in means between group 1 and group 2 i
95 percent confidence interval:
 -11.900862  -6.634309
sample estimates:
mean in group 1 mean in group 2
       10.42487        19.69246
```

The Bayesian estimate of the difference in means is essentially the same as the estimate from the frequentist version of the test. This illustrates an important point: given non-informative (vague) priors, a Bayesian analysis will return essentially the same parameter point estimates as a frequentist analysis.

3.7 Basic model diagnostics

3.7.1 Convergence: Rhat ($\hat{R}$)

The Gelman-Rubin statistic, usually abbreviated `Rhat` ($\hat{R}$), is a heuristic for assessing whether the chains have converged on the same posterior distribution for a parameter. There is no universally-accepted rule, but most people require $\hat{R}$ <1.01 or <1.001 before they consider a parameter converged. Models with unconverged parameters should be treated as highly suspect (or better yet, rerun with longer chains).

Many people mistake $\hat{R}$ for a kind of *p*-value that indicates whether parameters are "statistically significant". However, this is a mistake. First and foremost, the concept of "statistically significant" is inherently a frequentist notion. It is not well- or easily-defined in Bayesian inference. Even so, think of what it means in a frequentist context for a parameter to be statistically significant. A "significant" effect is inferred to be different from 0. This doesn't say anything about whether the parameter has converged!

The point here is that parameter convergence has nothing to do with parameter significance. Convergence means only that the model reached a stable, consistent solution (i.e., posterior distribution). That solution may or may not include 0 in its credible interval. I.e., may or not be "significant" from a frequentist perspective (more on this below). Lack of convergence says more about whether a parameter is well-defined, or is even identifiable, than what its value is.

If a model has non-converged parameters, you have three options. You should try these in order. First, try tuning the model by running longer chains, varying the thinning interval, and varying the burn-in period (or some combination of these). Some models just take more sampling to converge, and that's okay.

Second, try reformulating the model without the non-converging parameter. If a parameter cannot be reliably identified (i.e., a unique solution found), then it probably shouldn't be in the model anyway. This often

happens when a parameter only occurs in conjunction with another parameter. E.g., in the classic Cormack-Jolly-Seber (CJS) mark-recapture model, the survival probability and capture probability for the last time interval (ϕ_t and p_t) are not identifiable because they occur in the model likelihood only as a product (i.e., $\phi_t p_t$). The problem of identifiability can also arise in designed experiments when certain treatments or combinations of treatments are not sufficiently replicated.

Third, you can use your biological judgment. Some non-converged parameters are not problematic because of what they represent. For example, non-converged group-level random effects may not matter because these effects are not of biological interest—that is what it means to model something as a random effect!

3.7.2 Convergence: MCMC chains

While the `Rhat` offers a quick, numerical summary of convergence, you can also inspect the MCMC chains themselves for a visual check on convergence. Well-mixed chains that fluctuate around a common stationary distribution provide visual evidence that the sampler has converged.

This needs to be done one parameter at a time. Usually I will produce one plot for each parameter. The examples below uses the effect parameter from model 1, but the procedure is the same for any parameter.

One of the primary visual tools for convergence assessment is the traceplot. You can get one with function `traceplot(` in package `R2jags`.

```
traceplot(mod01, varname="effect")
```

This traceplot shows that the distributions within the chains overlap (are “well-mixed”), and the chains are essentially stationary (do not move up or down).

If you want a more customized traceplot, you can extract the values from the model output. The MCMC chains are stored in the model output in several formats. For plotting the chains, we want the values in the `$BUGSoutput$sims.array`. These are the raw MCMC chains in a 3-way array that have not been permuted (i.e., the values are in the same order as they were sampled in the MCMC process).

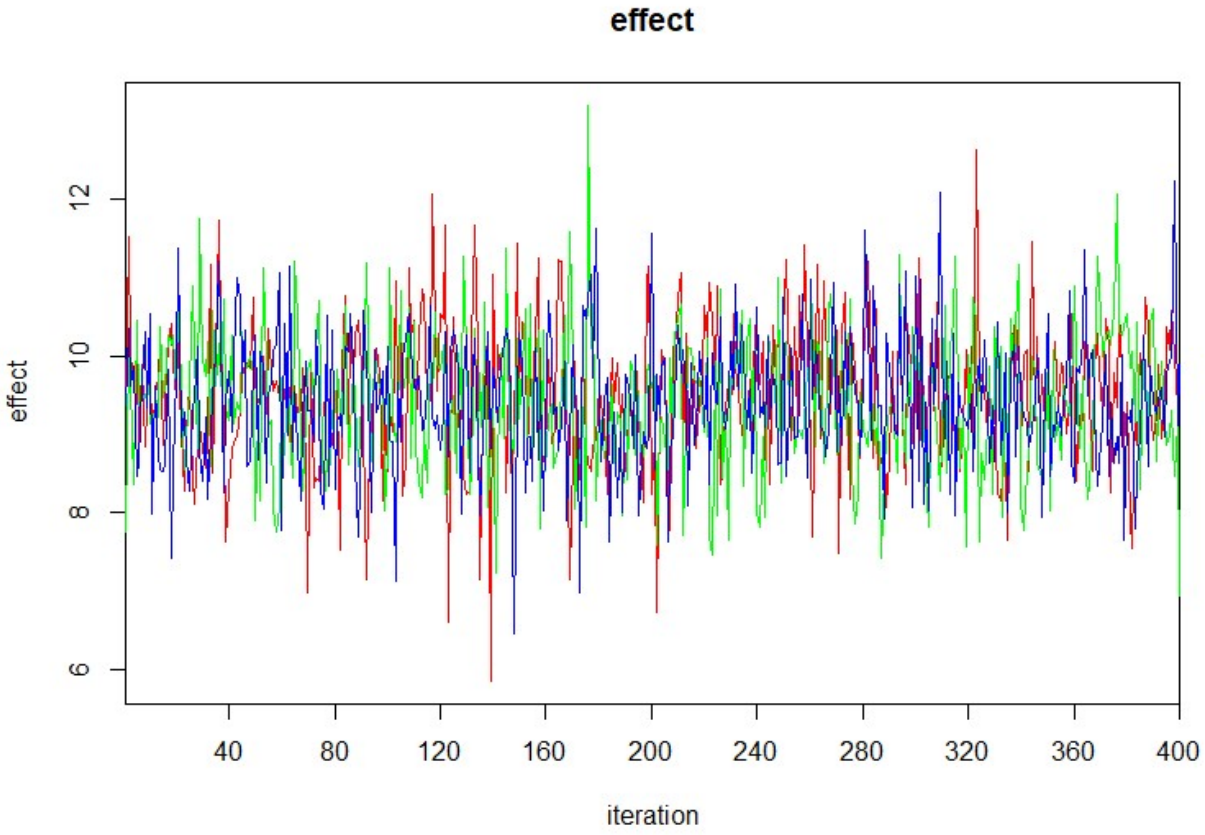
3.7.3 Posterior distributions

Often the model outputs we are most interested in are the posterior distributions. These represent the estimated distributions of each parameter as supported by the data. We can learn a lot from the model outputs printed to the R console.

For each parameter, JAGS returns and R prints the posterior mean and SD (mu.vect and sd.vect) as well as various quantiles of the posterior distribution. These quantiles are sample quantiles of the values from the 3 MCMC chains.

The nature of the posterior distribution depends in large part on the prior distribution. Most Bayesian models are formulated using **conjugate priors**: prior distributions that, under certain conditions, are guaranteed to result in certain posterior distributions. In model 1, the prior distributions for parameters other than sigma were normal. The normal is conjugate prior to itself: a normal prior will update to a normal posterior. For sigma, we used a uniform (which does not have a conjugate posterior) because we wanted to guarantee the

Figure 3.4: Trace plot for three Markov chain Monte Carlo (MCMC) chains sampling the posterior distribution of a model parameter. The three chains fluctuate around the same mean value, overlap extensively, and show no obvious trends over time, indicating good mixing and suggesting that the chains have converged to the same posterior distribution.



domain of the variable (≥ 0.0001). It may have been more appropriate to use an overdispersed gamma distribution or a lognormal distribution (some textbooks recommend this for variance or SD terms). However, in practical terms, a uniform distribution will be very similar to a normal with very large variance within the range of the uniform.

Practically speaking, if an appropriate conjugate prior is used then the posterior distribution will almost always be normal. Thus, the posterior distribution can be summarized by the posterior mean and SD (μ_{post} and σ_{post}). If the normal distribution is not appropriate for a variable (e.g., for term which must be non-negative), then plotting its posterior distribution may be helpful. It's worth noting that non-negative distributions can be modeled well by the lognormal distribution, which is its own conjugate prior.

For plotting posterior distributions, we can use any of the sets of values stored in the model output.

```
mx <- mod01$BUGSoutput$sims.array[,,"sigma"]
# convert matrix to vector (just in case)
mx <- as.numeric(mx)
hist(mx, main="residual SD (sigma)")
```

Sometimes it can be useful to plot two parameters on the same plot. This time we'll use a kernel density plot instead of a histogram.

```
mx1 <- mod01$BUGSoutput$sims.array[,,"mu[1]"]
mx2 <- mod01$BUGSoutput$sims.array[,,"mu[2]"]

# convert matrix to vector (just in case)
plot(density(mx1), lwd=3, xlim=c(0, 30),
    main="")
title(main=expression(Posterior~densities~of~mu[1]~and~mu[2]))
lines(density(mx2), lwd=3, col="red")
```

3.7.4 Predicted values

Finally, we can produce plots of predicted values and their credible intervals. This isn't really useful for a t-test (where a garden variety R boxplot works just fine), but can be extremely useful for more complicated models (especially GLMs and mixed models). We'll explore how to do that in the next module when we explore other linear models.

Figure 3.5: Posterior distribution of the residual standard deviation (( )) estimated by an MCMC analysis. The histogram summarizes the posterior samples, showing the range of plausible values for the residual standard deviation after combining the prior information with the observed data.

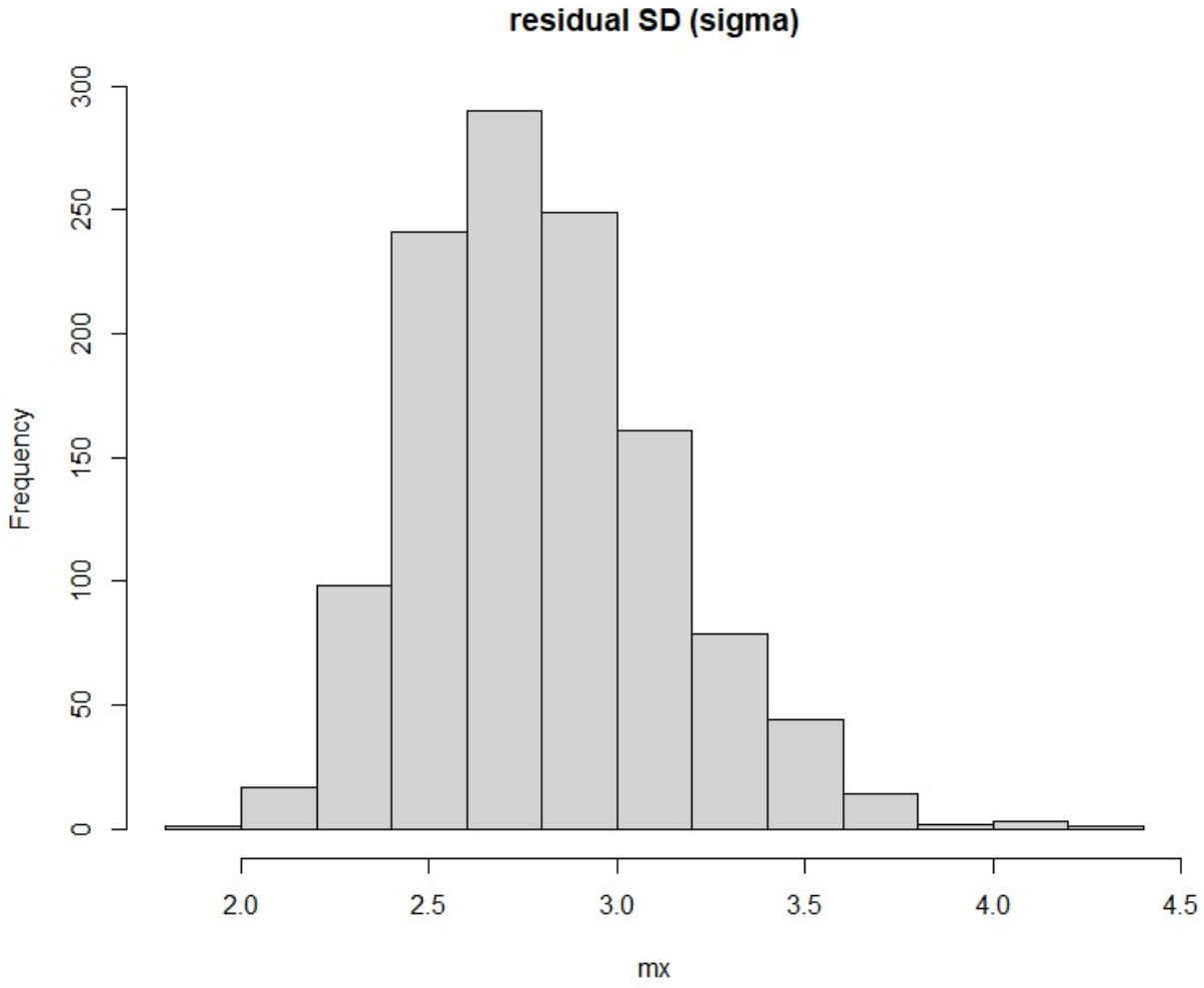
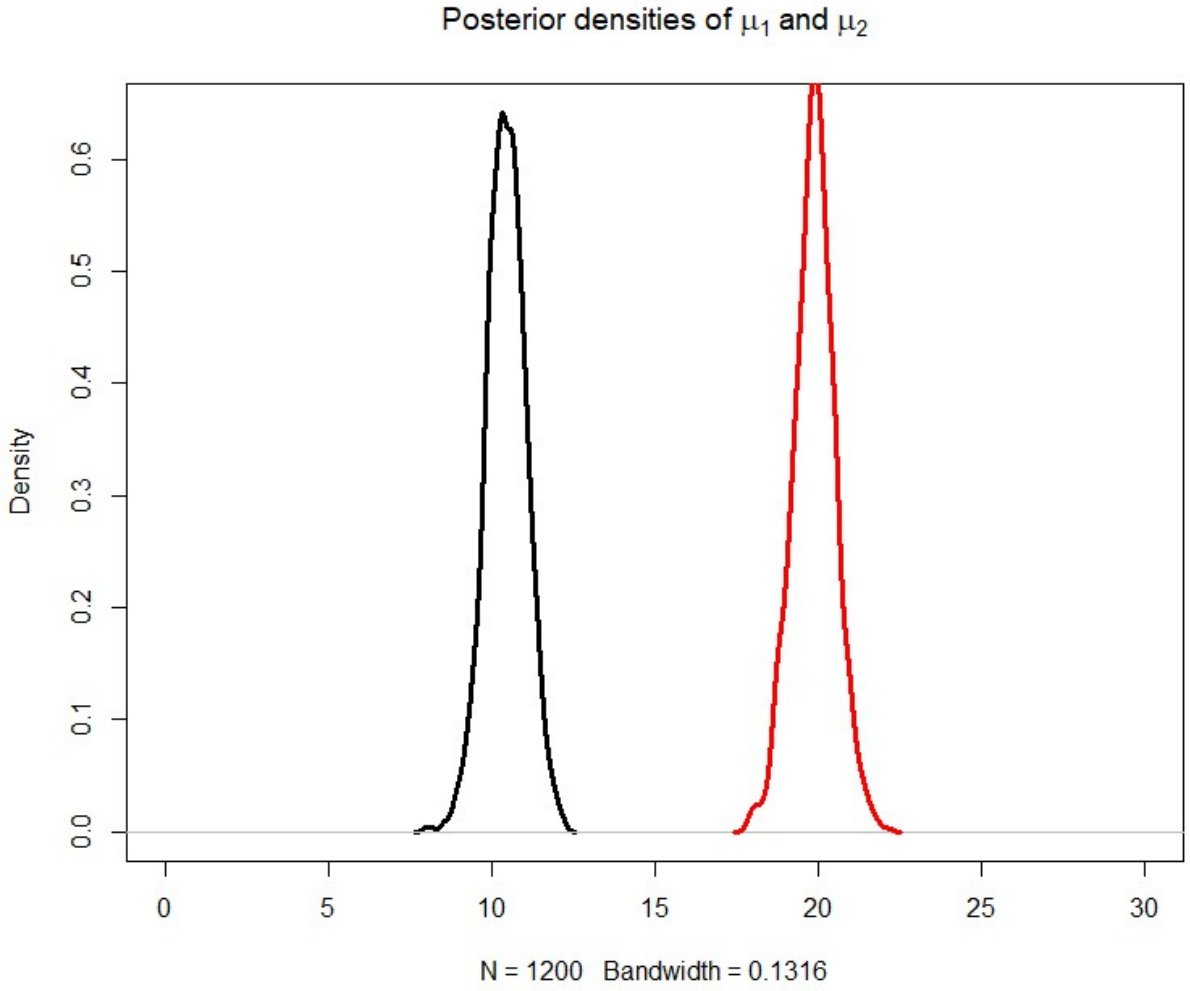


Figure 3.6: Posterior density estimates for two model parameters, μ_1 and μ_2. Each curve represents the posterior probability density of a parameter estimated by MCMC. The peaks indicate the most probable values, while the widths of the curves reflect the uncertainty in each parameter estimate.



Part II

Example analyses with JAGS and R

LM, GLM, NLS, and more!

Chapter 4

Linear models, Bayesian style

This week we will continue learning the JAGS language and interface by exploring an important class of statistical methods: the linear model. This module includes the following common linear models:

- Student's *t*-test
- ANOVA
- Linear regression (including multiple linear regression)
- Analysis of covariance (ANCOVA)
- …and probably many more.

In one sense, these are all different "tests" because they test different null hypotheses, use different test statistics, and arrive at their *p*-values differently. However, as we shall see, the distinctions between many of the classic linear models are largely an artifact of the different ways that *p*-values are calculated. In Bayesian inference, where P-values are not used, there is no need to calculate a test statistic. Thus, the Bayesian modeler can look behind the curtain and see that these are really all the same model. So, while "what's the difference between a *t*-test and ANOVA?" is a good exam question for an introductory (and thus frequentist) statistics course, the question doesn't even make sense to a Bayesian. In Bayesian inference, this question is like asking about the difference between 3 and $2 + 1$.

By the end of this module you should be able to:

1. Specify common linear models in the BUGS/JAGS language.
2. Interpret outputs from JAGS to make inferences about biological systems.

For this module you will need a recent versions of R, JAGS, and the R packages `rjags` and `R2jags` (Plummer 2003, Plummer 2022, R Core Team 2022, Su and Yajima 2021). This module and all code examples were developed and tested using R version 4.2.2, JAGS 4.3.1, `rjags` version 4-13, and `R2jags` version 0.7-1.

For this module you will need:

```
library(rjags)
library(R2jags)
```

4.1 Linear models refresher

4.1.1 Enter the matrix

Linear models are called linear models because they assume **linearity**. This means that a change in some response variable Y is described by an additive change in some predictor X multiplied by a constant called the **slope**. The slope encodes the change in in Y resulting from adding 1 to X.

$$Y = f(X)$$

$$f(X+1) = f(X) + \beta_1$$

In practice, we also include a *y*-intercept to define the value of the function at $X = 0$.

$$f(0) = \beta_0$$

All linear models can be written very compactly in matrix notation:

$$\mathbf{Y} = \mathbf{XB} + \mathbf{U}$$

The boldface variables indicate that each term is really a matrix, not a variable in the usual sense. In most biological analyses, **Y**, **B**, and **U** are column vectors, or matrices with 1 column. The matrix **X** is the **design matrix**, which contains the explanatory variables. This notation isn't very helpful for us biologists, but it describes what the computer is doing when it fits a model. The matrix notation above can be expanded as:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

where n is the number of observations; y_1, y_2, and so on up to y_n is a column vector of observed values of the dependent variable (aka: response variable); p is the number of model coefficients including the intercept; x_{ij} is the i-th value of predictor j; β_0, β_1, up to β_p is the vector of regression coefficients; and ε_1, ε_2, up to ε_n is a vector of *i.i.d.* normally distributed residuals. This matrix notation illustrates how linear regression can be extended to models with multiple predictor variables or even multiple response variables.

4.1.2 Straight lines (somehow)

Another meaning of the name "linear" refers to the fact that the change in the response variable Y per unit change in some predictor variable X is a scalar (aka: constant). This means that a given change in X will

always produce the same change in Y. Changing X from 1 to 2 will have the same effect on Y as changing X from 101 to 102, or 1,000,001 to 1,000,002, which is decidedly not the case with nonlinear functions. In other words, the plot of Y vs. X will be a straight line with a constant slope. If it sounds like I'm belaboring this point, it's because I am…it's really important!

Many models that initially appear nonlinear can nevertheless be analyzed using linear-model methods. The important distinction is not whether the predictors are transformed, but whether the model is linear in its parameters.

For example, a polynomial model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3$$

is still a linear model. Although the relationship between Y and X is curved, the quantities X^2 and X^3 are simply additional predictor variables once the data are observed. The unknown parameters β_0, β_1, β_2, and β_3 enter the model only as coefficients.

Similarly, some nonlinear models can be transformed into linear ones. For example, the power law

$$Y = aX^b$$

becomes

$$log\,(Y) = log\,(a) + blog\,(X)$$

after taking logarithms of both sides, producing a linear relationship between $log\,(Y)$ and $log\,(X)$.

I think of these models as "linear with extra steps." They may require transformed variables or transformed equations, but once those transformations are complete, they can be fit using the same machinery as ordinary linear models.

This notion of linearity is what makes many analyses with factors (categories or grouping variables) as predictors "linear". For example, the *t*-test is a linear model, but it doesn't really describe a line on a plot of y vs. x the way that linear regression does. However, if you encode the factor (grouping variable) as 0s and 1s and think about the underlying linear algebra, it becomes clear what the difference in group means really means:

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

For any observation Y_i, the expected value (without any error term) is:

$$E\left(y_i\right) = \left(1 \times \beta_0\right) + \left(x_i \times \beta_1\right)$$

For group 1, $x_i = 0$, so the expected value is β_0. For group 2, $x_i = 0$, so the β_1 term comes into play as the difference in means between groups 1 and 2. This is what R, SAS, and other programs are doing internally when you analyze data with a factor variable.

4.1.3 Other details and assumptions

Linear models have some other features and assumptions that you need to be aware of. If your data or variables violate the assumptions below, you may need to use something other than linear models to analyze your data.

4.1.3.1 Normal residuals

A model's **residuals** are the random error, or differences between observed and predicted values. Linear models assume that residuals follow a normal distribution. Models for data that do not have normal residuals exist, but they are not linear models. We'll explore those in a later module.

The assumption of normal residuals explains the last term in the linear model equation, ε_i:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

In this equation, y_i is observation i of the response variable y; β_0 is the intercept; β_1 is the slope; x_i is the predictor variable x for observation i, and ε_i is the residual for observation i (ε is the Greek letter "epsilon" and usually stands for error or residual terms). The residual term can be expanded like this:

$$y_i = \beta_0 + \beta_1 x_i + Normal\left(0, \sigma_{res}\right)$$

This expansion makes it clear that each residual ε_i comes independently from a normal distribution with mean 0 and standard deviation (SD) σ_{res}. For convenience, we sometimes separate the deterministic part and the stochastic part of the model and write the state space form of the linear regression model:

$$y_i \sim Normal\left(\eta_i,\ \sigma_{res}\right)$$

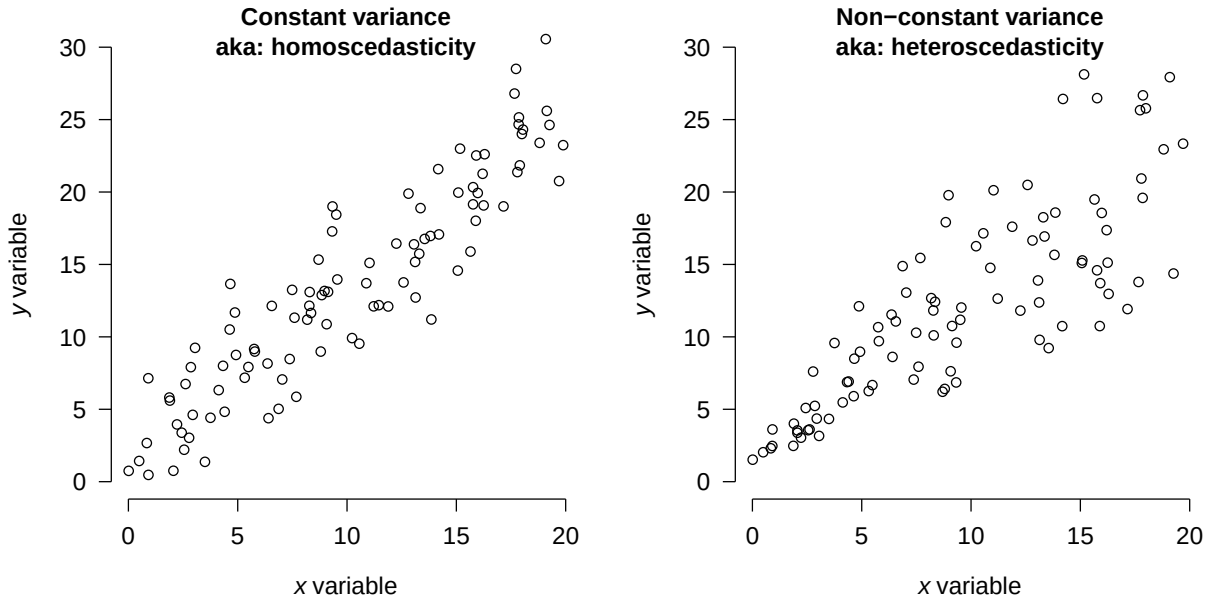
$$\eta_i = \beta_0 + \beta_1 x_i$$

In this version, the symbol $□$ means "distributed as" and denotes a stochastic relationship—a relationship that contains some element of randomness. This is in contrast to a deterministic relationship, which does not (equality, aka: identity, = is the most famous kind of deterministic relationship).

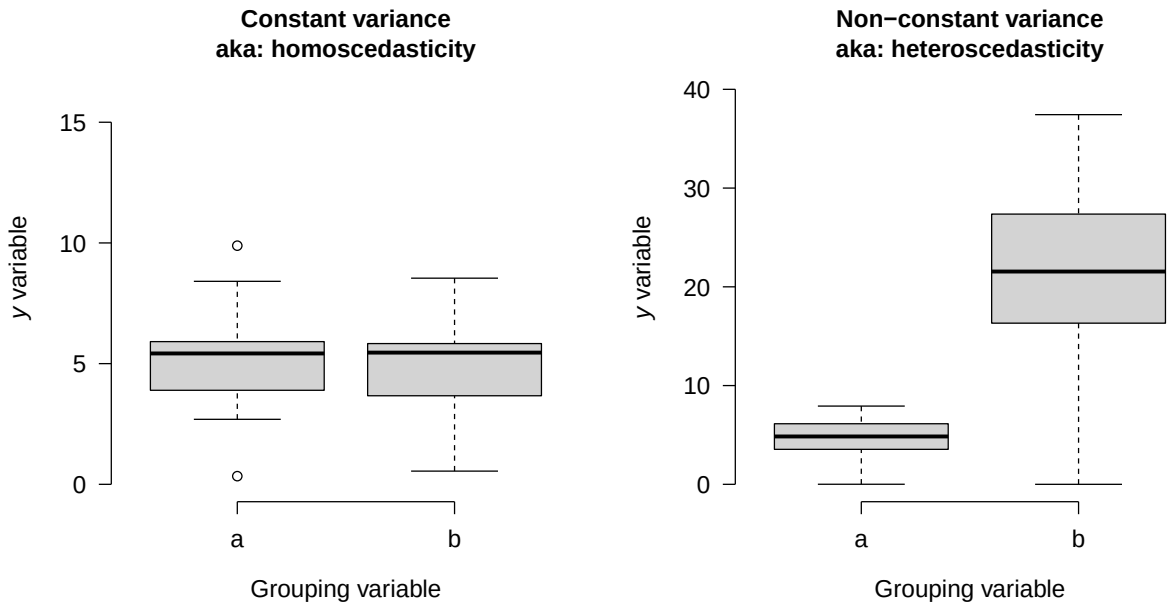
4.1.3.2 Constant variance (aka: heteroscedasticity)

Linear models assume **homoscedasticity**, or constant variance. This means that the residual variance does not depend on either x or y. This is part of the definition of a variable being independent and identically distributed (i.i.d.). If the variance in the response variable depends on some predictor variable, then this should be incorporated into the model (resulting in something other than a linear model). If the variance appears to depend on the response variable, then there is probably an issue with the assumed response distribution (more on this below).

The figure below shows two datasets with a linear relationship between x and y. In the left panel, the variance is the same everywhere (homoscedastic). The right panel shows a heteroscedastic relationship where the variance increases at larger x.



The figure below shows two datasets with a categorical predictor, suitable for a *t*-test. The left panel shows a relationship with equal variances in each group, while the right panel shows a situation with unequal variances.



4.1.3.3 Fixed and independent predictors

Linear models assume that predictor values are precisely known ("fixed") and independent of each other (i.e., not autocorrelated). If there is uncertainty in the predictor variables, this adds uncertainty to the response values that linear models cannot account for. Simply put, linear models have a term for uncertainty in *y*, σ_{res}, but no such term for uncertainty in *x*.

4.1.3.4 Independent and identically distributed errors (i.i.d.)

I've mentioned this assumption before, but it bears repeating because it is so often violated in biology. The assumption of independently and identically distributed (i.i.d.) errors is ***very*** important. It means that the residual, or predictive error, for each observation depends only on the parameters of the residual distribution and not on predictor variables or other observations. When the assumption of independence is violated, the degrees of freedom in the analysis is artificially inflated because the number of unique pieces of information is smaller than the nominal number of samples. In frequentist analyses, this deflates the P-value and increases the chance of a type I error (false positive). In Bayesian analyses, this tends to reduce the width of the credible intervals and bias parameter estimates. There are methods to deal with errors that are not independent, but linear models are not among them.

4.2 Comparing groups

One common class of analysis is to compare the means, or central tendencies, of groups to each other. In frequentist statistics, this is the realm of the *t*-test and ANOVA. In Bayesian statistics, they are the same thing.

Here we have options:

- Compare the mean of a single set of numbers to a target value: **one-sample *t*-test**.
- Compare the means of two sets of numbers: **two-sample *t*-test**.
- Test whether the differences in paired data are different from 0 (or a target value): **paired *t*-test**.
- Compare the means of two or more sets of numbers: **analysis of variance (ANOVA)**.

4.2.1 Student's *t*-test

We'll start with one of the most famous linear models, the Student's *t*-test. Most people have some vague idea that the *t*-test compares the means of two groups. However, the "*t*-test" is actually a family of tests that evaluate null hypotheses regarding sample means. Some of the most common *t*-tests are shown below:

- One-sample *t*-test, one-tailed: Tests whether population mean of a single set of values is less than, or greater than, a specified values μ_0. Only appropriate in when differences in one direction only are of interest.
- One-sample *t*-test, two-tailed: Tests whether population mean is equal to some specified value μ_0 (null hypothesis) or not (alternative hypothesis).
- Two-sample *t*-test, one-tailed: Tests whether population mean of a one group (μ_1) is less than, or greater than, the mean of another group (μ_2). Only appropriate in when differences in one direction only are of interest.
- Two-sample *t*-test, two-tailed: Tests whether the means in two groups (μ_1 and μ_2) are equal. Can also be thought of testing whether difference in means is equal to 0. Both equal variance (original) and unequal variance (Welch-Satterthwaite) versions exist…biologists should use the unequal variance version.
- Paired *t*-test, one-tailed: Tests whether the mean of differences between paired samples ($\bar{x}_D = \frac{1}{n}\sum_{i=1}^{n}\left(x_{1,i} - x_{2,i}\right)$) is less than, or greater than, some specified mean difference D_0. Only appropriate in when differences in one direction only are of interest.
- Paired *t*-test, one-tailed: Tests whether the mean of differences between paired samples ($\bar{x}_D = \frac{1}{n}\sum_{i=1}^{n}\left(x_{1,i} - x_{2,i}\right)$) is equal to 0.

These tests share a name because they revolve around the same test-statistic, *t*. The modern version of *t* is the Welch's *t*, which is an improvement over the original Student's *t* that can account for unequal group sizes and group variances:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Then the Welch-Satterthwaite equation is used to calculate the appropriate number of degrees of freedom (ν, "nu"):

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

and then *t* and ν are used to calculate the *p*-value. In both equations, $\bar{x}_i$ is the sample mean of group i, S_i^2 is the sample variance of group i, and n_i is the sample size of group i. In a Bayesian analysis, we don't do any of that.

In R, the different *t*-tests are obtained with different arguments to the function `t.test()`. Notice that the outputs have most of the same components; the only real difference is in how the output is interpreted. In particular, notice that the test estimates group means and a confidence interval (CI) for either the group mean or the difference in group means. Our Bayesian models below will also estimate those means and differences, but they will take a more central role.

4.2.1.1 One-sample *t*-test

The one sample test compares whether the mean of a set of numbers is significantly different from some target value. By default, that target is 0, but you can specify whatever target you like. Here are some quick examples in R.

```
# generate some data for demonstration
set.seed(123)
a <- rnorm(20, 5, 1)

# one-sample test, two tails
## Is mean of a == 0?
t.test(a)
```

```
    One Sample t-test

data:  a
t = 23.64, df = 19, p-value = 1.494e-15
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 4.686402 5.596845
sample estimates:
mean of x
 5.141624
```

```
# one-sample test, one tail
## Is mean of a > 0?
t.test(a, alternative="greater")
```

```
    One Sample t-test

data:  a
t = 23.64, df = 19, p-value = 7.468e-16
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
```

```
 4.765547      Inf
sample estimates:
mean of x
 5.141624
```

```
# one-sample test, one tail
## Is mean of a > 2?
t.test(a, alternative="greater", mu=2)
```

```
    One Sample t-test

data:  a
t = 14.445, df = 19, p-value = 5.334e-12
alternative hypothesis: true mean is greater than 2
95 percent confidence interval:
 4.765547      Inf
sample estimates:
mean of x
 5.141624
```

To run a Bayesian one-sample *t*-test, we need to set up our JAGS run as outlined in Chapter 3. Then, write this model to text file and run JAGS. All one-sample *t*-tests are the same model in JAGS. The only difference is in how the credible interval of the mean is interpreted.

```
# generate data
set.seed(123)
mu <- 10
sigma <- 3
n <- 20
y <- rnorm(n, mu, sigma)

# means parameterization of one-sample t-test
mod.name <- "jags_ttest_onesample.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## population mean
        mu ~ dnorm(0, 0.001)

        ## precision and residual SD
        sigma ~ dunif(0.0001, 10)
        tau <- 1/(sigma^2)
```

```
        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(mu, tau)
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("mu", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(mu=rnorm(1, 0, 100),
                         sigma=runif(1, 0.001, 10))
    }
    return(res)
}
inits <- init.fun(nchains)

# package data for JAGS
in.data <- list(y=y, n=length(y))

mod01 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
```

```
Graph information:
   Observed stochastic nodes: 20
   Unobserved stochastic nodes: 2
   Total graph size: 31

Initializing model
```

```
mod01
```

```
Inference for Bugs model at "jags_ttest_onesample.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 1.22 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
mu         10.449   0.712  9.006 10.002 10.457 10.904 11.828 1.002  1200
sigma       3.118   0.564  2.232  2.741  3.037  3.429  4.470 1.001  1200
deviance  100.812   2.259 98.628 99.186 100.115 101.767 107.008 1.009   540

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 2.5 and DIC = 103.4
DIC is an estimate of expected predictive error (lower deviance is better).
```

The most important output is the mean and 95% CRI for `mu`. If this interval includes the target mean `μ0`, then this is analogous to a non-significant frequentist test and we cannot conclude that $\mu \neq \mu 0$. The 95% CRI for `mu` is about `[9.010, 11.846]`. This means, in frequentist terms, that we can conclude that `mu` is not <9.010, and not >11.846. This covers any number of one-tailed tests. Conversely, we cannot conclude that `mu` is different from any `μ0` found within the 95% CRI.

4.2.1.2 Two-sample *t*-test

The two-sample *t*-test is just an extension of the one-sample *t*-test. We just need to include terms for the means, sample sizes, and variances a second group. The data that are generated below have unequal group variances, but the model works just as well with data that have equal variances. The unequal variances model is highly likely to be appropriate in just about any real dataset…so much so that it is the default in R.

```
# generate data
set.seed(123)
mu1 <- 10
mu2 <- 15
sigma1 <- 2
sigma2 <- 1
n <- 20
```

```
y1 <- rnorm(n, mu1, sigma1)
y2 <- rnorm(n, mu2, sigma2)
y <- c(y1, y2)
x <- rep(1:2, each=n)

# means parameterization of two-sample t-test
mod.name <- "jags_ttest_twosample.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## population means
        for(i in 1:ngroups){
            mu[i] ~ dnorm(0, 0.001)
            sigma[i] ~ dlnorm(1, 0.25)
            tau[i] <- 1/(sigma[i]^2)
        }#i

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(mu[x[i]], tau[x[i]])
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("mu", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
```

```
        res[[i]] <- list(mu=rnorm(2, 0, 100),
                         sigma=runif(2, 0.001, 10))
    }
    return(res)
}
inits <- init.fun(nchains)

# package data for JAGS
in.data <- list(y=y, x=x, n=length(y), ngroups=2)

mod02 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 40
   Unobserved stochastic nodes: 4
   Total graph size: 95

Initializing model
```

```
# inspect results
mod02
```

```
Inference for Bugs model at "jags_ttest_twosample.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 2.5 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
mu[1]      10.287   0.462  9.360  9.987 10.307 10.594 11.209 1.001  1200
mu[2]      14.936   0.189 14.580 14.814 14.930 15.054 15.313 1.001  1200
sigma[1]    2.021   0.337  1.487  1.767  1.979  2.230  2.790 1.001  1200
sigma[2]    0.872   0.155  0.628  0.768  0.847  0.956  1.260 1.000  1200
deviance  134.778   2.844 131.146 132.662 134.139 136.302 142.119 1.003   960

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
```

```
pV = 4.0 and DIC = 138.8
DIC is an estimate of expected predictive error (lower deviance is better).
```

For a standard two-sample, two-tailed *t*-test, we can compare the 95% CRI of mu[1] and mu[2]: [9.485, 11.223] and [14.571, 15.333]. These CRI do not overlap, so we can conclude that $\mu_1 \neq \mu_2$.

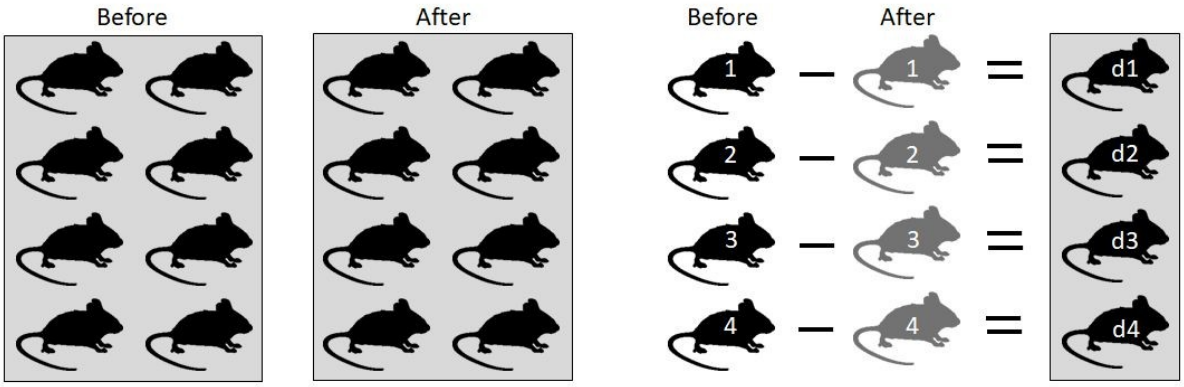
For a two-sample, one-tailed test, we also compare the 95% CRI of `mu[1]` and `mu[2]`. The upper limit of `mu[1]`'s CRI is less than the lower limit of the CRI for `mu[2]`. So, for any number of one-tailed tests we could conclude that $\mu_1 < \mu_2$.

If we are unsure whether or not a standard or Welch-Satterthwaite (unequal variances) *t*-test was warranted, we can compare the 95% CRI of `sigma[1]` and `sigma[2]`. These are [1.512, 2.796] and [0.646, 1.222]. Because these CRI do not overlap, we can conclude that groups 1 and 2 had different variances (strictly speaking, they had different SD; you can get the variances and their CRI by squaring the posteriors if that's what you want).

4.2.1.3 Paired *t*-tests

A paired *t*-test is essentially a one-sample *t*-test on differences between paired samples. For example, mean body weights of individual lab rats before and after some feeding trial.

Figure 4.1: Comparison of two-sample and paired *t*-tests. A two-sample *t*-test compares the means of two independent groups, whereas a paired *t*-test tests whether the mean within-individual change differs from zero.



Two-sample *t*-test: compare mean of mouse masses before to mean of mouse masses after.

Paired *t*-test: test whether mean of individual-level differences (before vs. after) is 0.

4.2.1.3.1 Method 1: One sample test on differences

One way to fit a paired *t*-test is to calculate the differences between the paired samples outside of JAGS, then conduct a one-sample *t*-test on the differences. Basically, calculate the differences and feed that difference to the `jags_ttest_onesample` model from Section 4.2.1.1 as `y`.

4.2.1.3.2 Method 2: Paired test (direct)

Alternatively, we could code the paired *t*-test as its own model. Instead of fitting group means, we can fit the mean difference directly as `dif` and inspect its posteriors.

```
# generate data
set.seed(123)
n <- 20
dif <- 4
y1 <- rnorm(n, 5, 1)
y2 <- y1 + rnorm(n, dif, 0.5)

# means parameterization of paired t-test
mod.name <- "jags_ttest_paired_v1.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## difference in means
        dif ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma.res ~ dlnorm(1, 0.25)
        tau <- 1/(sigma.res^2)

        # likelihood
        for(i in 1:n){
            y2[i] ~ dnorm(eta[i], tau)
            eta[i] <- y1[i] + dif
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("dif", "sigma.res")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3
```

```
# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(dif=rnorm(1, 0, 100),
                         sigma.res=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# package data for JAGS
in.data <- list(y1=y1, y2=y2, n=n)

mod03 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 20
   Unobserved stochastic nodes: 2
   Total graph size: 70

Initializing model
```

```
mod03
```

```
Inference for Bugs model at "jags_ttest_paired_v1.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 1.24 secs
          mu.vect sd.vect   2.5%    25%   50%    75%  97.5%  Rhat n.eff
dif        3.972   0.096  3.776  3.908  3.97  4.032  4.160 1.000  1200
sigma.res  0.441   0.077  0.319  0.385  0.43  0.485  0.610 1.000  1200
deviance  22.653   2.079 20.614 21.162 22.03 23.364 28.083 1.001  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).
```

```
DIC info (using the rule: pV = var(deviance)/2)
pV = 2.2 and DIC = 24.8
DIC is an estimate of expected predictive error (lower deviance is better).
```

The result is spot-on what is estimated by the base R `t.test()` function:

```
t.test(y1, y2, paired=TRUE)
```

```
    Paired t-test

data:  y1 and y2
t = -42.832, df = 19, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -4.168583 -3.780160
sample estimates:
mean difference
      -3.974371
```

4.2.2 Analysis of variance (ANOVA)

The analysis of variance (ANOVA) is another case of the linear model. It is usually used to compare the means of 3 or more groups. In this way it can be thought of as an extension of the *t*-test from 2 groups to >2, but this is incorrect from a frequentist perspective because of how differently the *t*-test and ANOVA arrive at a *p*-value. It's better to think of the frequentist *t*-test and ANOVA as cousins. Frequentist ANOVA assesses statistical significance using the statistic *F*, which is related to the amount of variance associated with different sources of variance, relative to the number of degrees of freedom. This is different from t, which is a difference in means scaled by variances and sample sizes, *F*. Either way, *t* and *F* (and all test statistics) are just contrivances to obtain a *p*-value. In Bayesian statistics, we don't need that.

Consider this model:

$$y_i \sim Normal\left(\mu_{x_i}, \sigma_{res.}\right)$$

If μ_{x_i} is the mean for the group defined by xi, then what is this model saying? It's saying that the value of y for observation *i* is drawn from a normal distribution with a mean that depends on group of observation *i*, and some residual variation $\sigma_{res.}$. This single model covers a lot of situations:

- If the number of groups is 1, then this is a one-sample *t*-test
- If the number of groups is 2, then this is a two-sample *t*-test. Or, a one-way ANOVA with two factor levels.
- If the number of groups is 3, then this is a one-way ANOVA with three factor levels.
- If the number of groups is 1, but the values represented paired differences, then this is a paired *t*-test.

All four of these methods, which have different names and different fitting methods in frequentist statistics, are the same model for a Bayesian and can be fit with the same code in JAGS. This kind of generalization is one of the greatest things about Bayesian inference and MCMC model fitting. By getting away from a hyper-fixation on test statistics and *p*-values, we can take a step back and think about what our models really mean.

4.2.2.1 ANOVA basics

In a typical frequentist ANOVA, one first conducts the omnibus test, which assesses whether any of the group means differ from each other. Then, a post-hoc test is performed to check which groups differ from each other. The code below illustrates this process in R using the built-in `iris` dataset.

```
# fit the model for anova()
mod1 <- aov(Petal.Length~Species, data=iris)

# omnibus test to get ANOVA table:
summary(mod1)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
Species       2  437.1  218.55    1180 <2e-16 ***
Residuals   147   27.2    0.19
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This is the omnibus test result, or "ANOVA table".The ANOVA table shows us that the factor species uses 2 degrees of freedom (Df or d.f.) to account for 437.10 of the squared errors; the residuals use 147 d.f. to account for 27.22 of the squared errors. The ratio of the mean squared error per d.f. associated with each source, 218.551 / 0.185 = 1180.2, is the test statistic F. The is used to calculate the *p*-value. We can also see that about 94.1% of variation (437.10 / (437.10+27.22)) is associated with species.

However, all the *p*-value in the omnibus test tells us is that the means of at least one pair of species differ. To find out which pair or pairs, we need to use a post-hoc test. My favorite is the Tukey's honest significant difference (HSD) test because it automatically adjusts for multiple comparisons.

```
# post-hoc test:
TukeyHSD(mod1)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = Petal.Length ~ Species, data = iris)

$Species
                     diff     lwr     upr p adj
versicolor-setosa    2.798 2.59422 3.00178     0
```

```
virginica-setosa     4.090 3.88622 4.29378     0
virginica-versicolor 1.292 1.08822 1.49578     0
```

This result shows us that all of the species differ from each other. We can see this because all of the *p*-values (`p adj`) for the pairwise comparisons are <0.05 (they are printed by R as 0, but are really just approximated to very tiny value). We can also see this because the 95% CI for the pairwise differences do not include 0. For example, the difference in means for species `versicolor` and `setosa` is 2.798, with 95% CI = [2.59, 3.00]. The value `diff` is literally the mean of `versicolor` minus the mean of `setosa`, as printed on the left of the table.

This is a lot of work. Remember, most of the testing and interpretation in this ANOVA was really just steps taken to get a *p*-value. By using a means parameterization in JAGS, we can get everything in one step in JAGS.

4.2.2.2 One-way ANOVA in JAGS

The previous example with the iris data is an example of one-way ANOVA. The "-way" part just defines how many explanatory factors there are. In the next section we'll try "two-way" ANOVA, which has 2 factors.

A basic one-way ANOVA model can be fitted in JAGS using essentially the same code as for the two-sample *t*-test (see above). All that we have to adjust are some of the inputs, including `ngroups` (which was 2 for the *t*-test, but can be any positive integer in ANOVA). We'll use the `iris` dataset as in the previous example

```
mod.name <- "jags_anova_oneway.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## population means
        for(i in 1:ngroups){
            mu[i] ~ dnorm(0, 0.001)
            sigma[i] ~ dlnorm(1, 0.25)
            tau[i] <- 1/(sigma[i]^2)
        }#i

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(mu[x[i]], tau[x[i]])
        }#i

    }#model

", fill=TRUE)
```

```r
sink()

# parameters to monitor
params <- c("mu", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# function to define initial values for MCMC chains
init.fun <- function(nc, ngroups=3){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(mu=rnorm(ngroups, 0, 100),
                         sigma=rlnorm(ngroups, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains, 3)

# package data for JAGS
xfac <- match(iris$Species, sort(unique(iris$Species)))
ngroups <- max(xfac)
in.data <- list(y=iris$Petal.Length,
    x=xfac, n=nrow(iris), ngroups=ngroups)

mod05 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 150
   Unobserved stochastic nodes: 6
   Total graph size: 319
```

```
Initializing model
```

```
mod05
```

```
Inference for Bugs model at "jags_anova_oneway.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 8.14 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
mu[1]       1.462   0.025  1.412  1.445  1.462  1.479  1.509 1.002  1100
mu[2]       4.257   0.065  4.133  4.211  4.256  4.301  4.382 1.002  1200
mu[3]       5.552   0.081  5.391  5.498  5.554  5.605  5.719 1.012   160
sigma[1]    0.178   0.018  0.147  0.166  0.177  0.189  0.217 1.000  1200
sigma[2]    0.481   0.049  0.397  0.447  0.476  0.511  0.583 1.003   570
sigma[3]    0.560   0.055  0.465  0.522  0.556  0.595  0.678 1.000  1200
deviance  118.592   3.441 113.758 116.068 117.958 120.434 126.862 1.000  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 5.9 and DIC = 124.5
DIC is an estimate of expected predictive error (lower deviance is better).
```

Unlike the frequentist ANOVA we performed in R, there is no need for post-hoc test in the Bayesian version of the model. Instead of performing a separate test, we can compare the 95% CRI of the group means directly. For example, species 1 and 2 have different petal lengths because their 95% CRI don't overlap.

4.2.2.3 Two-way ANOVA in JAGS

In a two-way ANOVA, the variation is partitioned among two potentially explanatory factors—i.e., the variation is split in two ways. There is also three-way ANOVA, four-way ANOVA, and so on. We'll focus on two-way ANOVA because the code is relatively simple to generalize to other situations.

In base R, we can add more factors to the model with the plus sign +.

```
# make a spare copy of iris, and add a variable for color
# (which doesn't explain anything)
iris2 <- iris
iris2$color <- c("red", "white")

# fit the two-way anova
mod2 <- aov(Petal.Length~Species+color,
            data=iris2)
anova(mod2)
```

```
Analysis of Variance Table

Response: Petal.Length
           Df Sum Sq Mean Sq   F value Pr(>F)
Species     2 437.10 218.551 1174.2292 <2e-16 ***
color       1   0.05   0.049    0.2611 0.6101
Residuals 146  27.17   0.186
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(mod2)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = Petal.Length ~ Species + color, data = iris2)

$Species
                       diff      lwr      upr p adj
versicolor-setosa    2.798 2.593692 3.002308     0
virginica-setosa     4.090 3.885692 4.294308     0
virginica-versicolor 1.292 1.087692 1.496308     0

$color
            diff        lwr       upr     p adj
white-red -0.036 -0.1752347 0.1032347 0.6101255
```

In JAGS, we can get everything in one step. We'll fit the effects parameterization because it's easier to define independent effects of species and color this way. Notice that we have to play a little trick in the priors for beta1 and beta2. Technically, the "effect" for the baseline level of a factor should be 0. However, this is not allowed in JAGS because it would have required having some elements of the vectors beta1 and beta2 being deterministic (i.e., `beta1[1] <- 0`), while having other elements of those vectors be stochastic (e.g., `beta1[2] ~ dnorm(0, 0.001)`). This provokes the error message, "`Cannot set value of non-variable node`".

The solution is to define the prior for the baseline levels to have essentially 0 variance about the true value of 0. We used a normal with $\mu = 0$ and $\sigma^2 = 0.000000001$. Another way to look at this is that we place a very strong prior on this parameter, so strong that no amount of data would be sufficient to update it.

```
iris2 <- iris
iris2$color <- c("red", "white")

# effects parameterization of two-way ANOVA
# without interaction (main effects)
```

```
mod.name <- "jags_anova2way_main.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## intercept
        beta0 ~ dnorm(0, 0.001)

        ## effects of factor 1
        beta1[1] ~ dnorm(0, 1e9) # set first level to 0
        beta1[2] ~ dnorm(0, 0.001)
        beta1[3] ~ dnorm(0, 0.001)

        ## effects of factor 2
        beta2[1] ~ dnorm(0, 1e9) # set first level to 0
        beta2[2] ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(eta[i], tau)
            eta[i] <- beta0 + beta1[fac1[i]] + beta2[fac2[i]]
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("beta0", "beta1", "beta2", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3
```

```
# reformat data for JAGS
fac1 <- match(iris2$Species, sort(unique(iris2$Species)))
nfac1 <- max(fac1)
fac2 <- match(iris2$color, sort(unique(iris2$color)))
nfac2 <- max(fac2)

# package data for JAGS
in.data <- list(y=iris2$Petal.Length,
    n=nrow(iris2),
    fac1=fac1, fac2=fac2)

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(beta0=rnorm(1, 0, 100),
                         beta1=c(0, rnorm(nfac1-1, 0, 100)),
                         beta2=c(0, rnorm(nfac2-1, 0, 100)),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod06 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 150
   Unobserved stochastic nodes: 7
   Total graph size: 472

Initializing model
```

```
mod06
```

```
Inference for Bugs model at "jags_anova2way_main.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 10.05 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
beta0      1.481   0.072  1.339  1.433  1.479  1.527  1.632 1.002   750
beta1[1]   0.000   0.000  0.000  0.000  0.000  0.000  0.000 1.000     1
beta1[2]   2.799   0.088  2.633  2.740  2.797  2.857  2.968 1.004   510
beta1[3]   4.089   0.091  3.914  4.028  4.088  4.150  4.266 1.002  1100
beta2[1]   0.000   0.000  0.000  0.000  0.000  0.000  0.000 1.000     1
beta2[2]  -0.037   0.069 -0.173 -0.081 -0.040  0.008  0.100 1.001  1200
sigma      0.436   0.026  0.389  0.418  0.434  0.452  0.489 1.000  1200
deviance 174.594   3.300 170.318 172.208 173.877 176.355 182.549 1.001  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 5.4 and DIC = 180.0
DIC is an estimate of expected predictive error (lower deviance is better).
```

The output tells us that the effect of being in level 2 of species (`versicolor`) instead of level 1 (`setosa`) is to have a petals about 2.799 cm longer, with 95% CRI approximately [2.614, 2.967] (numbers may differ a bit because of the random sampling at render time). Likewise, the effect of being `white` instead of `red` (color 2 instead of color 1) is -0.039, with 95% CRI = [-0.182, 0.095]. As always, whether or not the 95% CRI includes 0 gives us an indication of whether or not that factor differs significantly from the baseline…analogous to the main effect P-value in a frequentist analysis.

4.2.2.4 Two-way ANOVA with interaction

Interestingly, fitting a two-way ANOVA with an interaction term is easier than fitting the model without interactions. Easier, that is, so long as you understand JAGS syntax. We're going to use the bracket notation for subsetting objects. This notation works the same way in JAGS as it works in R. In a nutshell:

- For some vector `x`, `x[a]` is the `a`-th element of `x`. So `x[1]` is first, `x[3]` is third, and so on.
- For some matrix `z`, `z[a,b]` is the element in row `a`, column `b`. So, `z[1,3]` is in row `1`, column `3`. A blank index selects all elements of that dimension: `z[,2]` selects all elements of column `2`, while `x[4,]` selects all elements from row `4`. Depending on the context, selecting an entire row or column like this may return a vector instead of a matrix.
- A matrix with > 2 dimensions is called an **array**, and the bracket notation extends to arrays of any dimension . So, `a[j,k,l]` is the value in row `j`, column `k`, and layer `l` of array `a`. A blank index selects all elements of a dimension.

It's all arrays

Technically, an array is any ordered arrangement of elements. Vectors are 1-dimensional arrays (length), matrices are 2-dimensional arrays (rows and columns), and anything with >2 layers is usually just called an array. There is no technical name for the third or higher dimensions of an array, but "layer" makes sense for the third dimension of a 3d array.

Clever use of high-dimensional arrays can simplify the programming of some very complex models…we'll see some examples later in the semester .

Let's set up a data frame with an interaction.

```
set.seed(123)
n <- 20
dx <- expand.grid(species=1:3,
    loc=1:2, ind=1:n)
dx$y <- NA

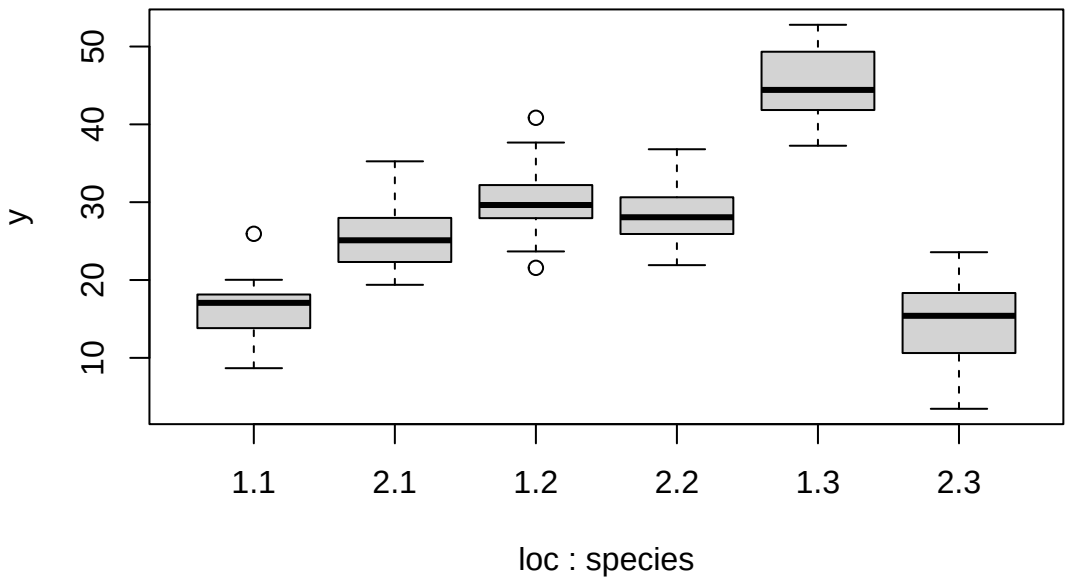
n <- nrow(dx)
nspp <- 3
nloc <- 2

# "true" means with interaction between spp and loc
mu <- matrix(c(15, 30, 45, 25, 30, 15),
    nrow=nspp, ncol=nloc,
    dimnames=list(
        paste0("spp", 1:nspp),
        paste("loc", 1:nloc)))

# residual SD
sigma.res <- 5

# draw y values given interaction and SD
for(i in 1:n){
    dx$y[i] <- rnorm(1, mu[dx$species[i], dx$loc[i]], sigma.res)
}

# quick check:
boxplot(y~loc+species, data=dx)
```

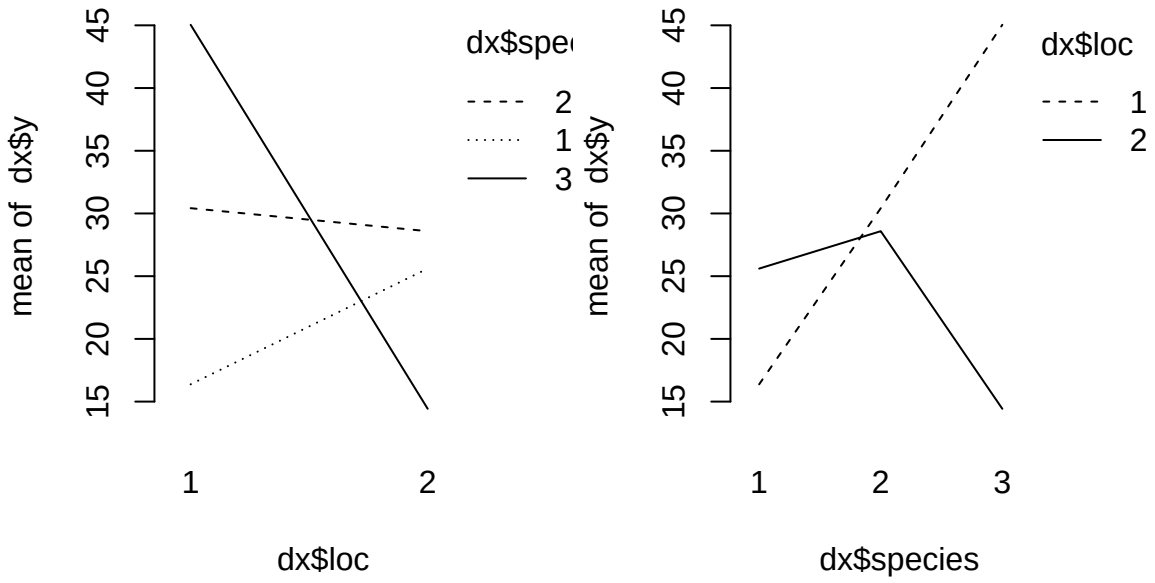


The boxplot shows us what the interaction does. The effect of one factor depends on the effect of another factor. Here, the effect of location is different for each species:

- Location 2 has greater values than location 1 for species 1, smaller values than location 1 for species 3, and the same values as location 1 for species 2.
- In location 1, species 1 has the smallest values and species 3 has the greatest values. In location 2, species 3 has the smallest values and species 2 has the greatest values.

Another way to visualize this is with an **interaction plot**. This shows the group means across every combination of factor levels. When lines are parallel (or do not cross), then there is probably no interaction. However, if lines do cross, then there is likely an interaction and you need to test for it.

```
par(mfrow=c(1,2), bty="n")
interaction.plot(dx$loc, dx$species, dx$y)
interaction.plot(dx$species, dx$loc, dx$y)
```



We can fit a Bayesian ANOVA with interaction by using the means parameterization of the model. This model can also be used to fit ANOVA without interaction as well. Just as with some of the *t*-tests above, different readings of the model outputs can lead to different versions of the test.

```
#| label: fit-twowayanovainter-jags
#| cache: true

set.seed(123)
n <- 20
dx <- expand.grid(species=1:3,
    loc=1:2, ind=1:n)
dx$y <- NA

n <- nrow(dx)
nspp <- 3
nloc <- 2

# "true" means with interaction between spp and loc
mu <- matrix(c(15, 30, 45, 25, 30, 15),
    nrow=nspp, ncol=nloc,
    dimnames=list(
        paste0("spp", 1:nspp),
        paste("loc", 1:nloc)))
```

```
# residual SD
sigma.res <- 5

# draw y values given interaction and SD
for(i in 1:n){
    dx$y[i] <- rnorm(1, mu[dx$species[i], dx$loc[i]], sigma.res)
}

# means parameterization of two-way ANOVA
# with interaction (works for without also)
mod.name <- "jags_anova2way_means.txt"
sink(mod.name)
cat("
    model{
        # priors
        for(i in 1:nfac1){
            for(j in 1:nfac2){
                mu[i,j] ~ dnorm(0, 0.001)
            }#j
        }#i

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(mu[fac1[i], fac2[i]], tau)
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("mu", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
```

```
n.thin <- 100
nchains <- 3

# reformat data for JAGS
fac1 <- dx$species
nfac1 <- max(fac1)
fac2 <- dx$loc
nfac2 <- max(fac2)

# package data for JAGS
in.data <- list(y=dx$y,
    n=n,
    fac1=fac1, fac2=fac2,
    nfac1=nfac1, nfac2=nfac2)

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(mu=matrix(rnorm(nfac1*nfac2, 0, 100),
                                   nrow=nfac1, ncol=nfac2),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod07 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 120
   Unobserved stochastic nodes: 7
   Total graph size: 377
```

```
Initializing model
```

```
mod07
```

```
Inference for Bugs model at "jags_anova2way_means.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 6.77 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
mu[1,1]   16.416   1.005  14.441 15.734 16.450 17.110 18.233 1.004   550
mu[2,1]   30.378   0.994  28.510 29.673 30.373 31.028 32.372 1.004   680
mu[3,1]   45.018   1.020  43.084 44.307 45.013 45.691 47.008 1.002   920
mu[1,2]   25.626   0.977  23.778 24.948 25.610 26.285 27.510 1.001  1200
mu[2,2]   28.568   1.034  26.602 27.855 28.540 29.252 30.620 1.001  1200
mu[3,2]   14.395   1.026  12.443 13.681 14.385 15.084 16.431 1.001  1200
sigma      4.507   0.311   3.949  4.291  4.478  4.698  5.172 1.002   830
deviance 701.406   3.929 695.840 698.631 700.632 703.412 710.592 1.000  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

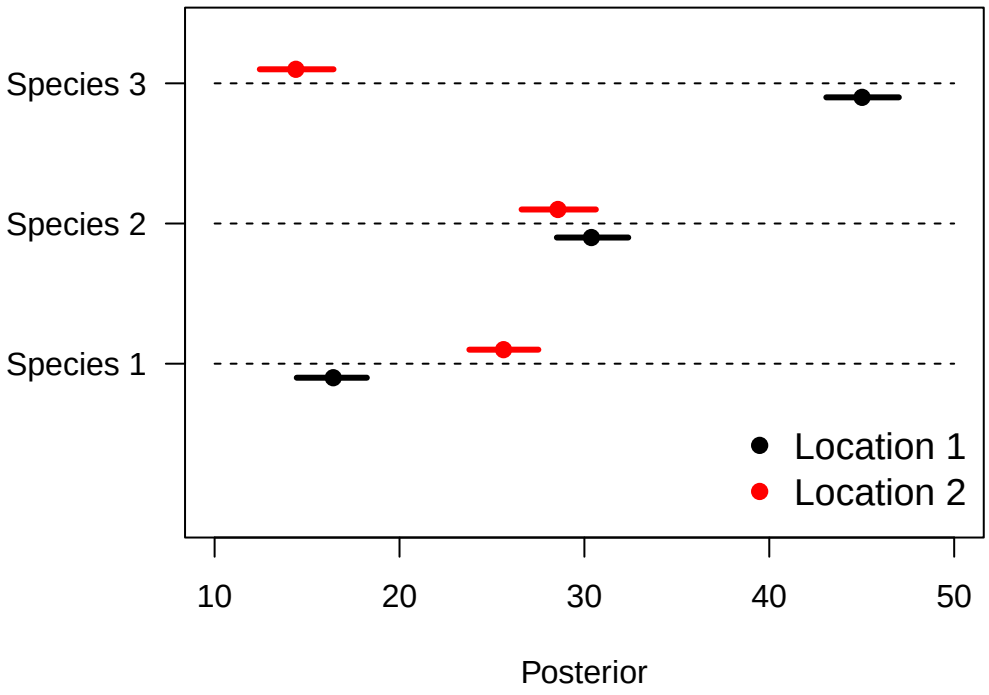
DIC info (using the rule: pV = var(deviance)/2)
pV = 7.7 and DIC = 709.1
DIC is an estimate of expected predictive error (lower deviance is better).
```

We can inspect the posteriors to see how the means differed among the 6 combinations of factor levels: species 1, location 1 (mu[1,1]); species 1, location 2 (mu[1,2]); and so on. Doing so would lead us to discover the same interaction we suspected based on the interaction plot above. A better way might be to visualize the group means using the posteriors. Below is one way to do it:

```
# get means and 95% CRI for plot
res <- mod07$BUGSoutput$summary
res1 <- res[2:4, c(1, 3, 7)]
res2 <- res[5:7, c(1, 3, 7)]

par(mfrow=c(1,1), mar=c(5.1, 8.1, 1.1, 1.1))
plot(1:2, type="n", xlim=c(10, 50),
    ylim=c(0, 3.5), xlab="Posterior",
    yaxt="n", ylab="")
segments(res1[,2], 1:3, res1[,3], 1:3, lwd=3)
segments(res2[,2], 1:3+0.2, res2[,3], 1:3+0.2, lwd=3, col="red")
points(res1[,1], 1:3, pch=16, cex=1.2)
points(res2[,1], 1:3+0.2, pch=16, cex=1.2, col="red")
legend("bottomright", legend=paste("Location", 1:2),
    pch=16, col=c("black", "red"), cex=1.2, bty="n")
```

```
axis(side=2, at=1:3+0.1, labels=paste("Species", 1:3), las=1)
segments(10, 1:3+0.1, 50, 1:3+0.1, lty=2)
```



4.3 Continuous patterns

4.3.1 Linear regression

One of the most well-known of all statistical analyses, the linear regression model describes a pattern where some response variable y varies as a linear function of a predictor variable x. The equation is:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

where: - y is the response or dependent variable - x is the explanatory, predictor, or independent variable - β_0 is the *y*-intercept (i.e., the value of y when $x = 0$) ("beta zero" or "beta naught") - β_1 is the slope or regression coefficient (i.e., the change in y per unit change in x) ("beta one"). If x increases by 1, then y increases by β_1. - ε is a random error term that describes residual variation not explained by the model ("epsilon").

When we specify the model in JAGS, we do so using an observation-wise, state-space notation.

$$y_i \; Normal\left(\eta_i,\; \sigma_{res}\right)$$

$$\eta_i = \beta_0 + \beta_1 x_i$$

This format makes it clear that each y value has its own expected value η_i ("eta sub i"), dependent on that observation's x value x_i. This relates to a key assumption of the linear model, that residuals are independently and identically distributed (i.i.d.). It also echoes the way we program the model likelihood in JAGS, a for loop.

The "state-space" part of the formulation above refers to the fact that the model equation explicitly spearates the true, unobserved, "state" of the system from the observed values which occur in a "space" defined by the state and the uncertainty about the observation process.

4.3.1.1 Linear regression in JAGS

The procedure for linear regression in JAGS is very similar to the procedure for the effects parameterization of the two-way ANOVA that we saw in section X. As we'll soon see, this is also very similar to how we'll fit a wide variety of models for the rest of the course—linear or otherwise.

```
library(rjags)
library(R2jags)
# simulate some data
set.seed(42)
n <- 50
dx <- data.frame(x=runif(n, 0, 20))

beta0 <- 4
beta1 <- 1.8
sigma.res <- 4
dx$y <- beta0 + beta1*dx$x + rnorm(n, 0, sigma.res)

# linear regression with 1 predictor
mod.name <- "jags_linreg_1x.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## intercept
        beta0 ~ dnorm(0, 0.001)

        ## slope
        beta1 ~ dnorm(0, 0.001)

        ## residual SD and precision
```

```
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        # (for loop)
        for(i in 1:n){
            y[i] ~ dnorm(eta[i], tau)
            eta[i] <- beta0 + beta1 * x[i]
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("beta0", "beta1", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# package data for JAGS
in.data <- list(y=dx$y,
    n=n, x=dx$x)

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(beta0=rnorm(1, 0, 100),
                         beta1=rnorm(1, 0, 100),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
```

```
mod08 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 50
   Unobserved stochastic nodes: 3
   Total graph size: 211

Initializing model
```

```
mod08
```

```
Inference for Bugs model at "jags_linreg_1x.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 4 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
beta0       3.854   1.313  1.312  3.002  3.863  4.724  6.458 1.001  1200
beta1       1.791   0.096  1.602  1.725  1.790  1.855  1.984 1.001  1200
sigma       3.995   0.439  3.227  3.681  3.970  4.260  4.965 1.000  1200
deviance  279.763   2.600 276.746 277.844 279.052 281.041 286.526 1.003  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 3.4 and DIC = 283.1
DIC is an estimate of expected predictive error (lower deviance is better).
```

The outputs show that the posterior means for beta0 and beta1 are reasonably close to the truth (3.843 vs. 4, and 1.791 vs. 1.8) and the estimates from the frequentist model fit in R. The true values are well within the 95% CRI.

The next step is to produce a plot of the model's predicted values and 95% CRI. As we'll see, this is a little more involved than the process in R. But, we'll practice the process so much that it will seem easy by the end of the course!

4.3.1.2 Plotting model predictions in R

The basic procedure for plotting a linear model and its predictions in R is as follows: 1. Fit the model and save it to a model object. 1. Generate predicted values and SE of predicted values over a range of x values that cover the range of observed x. 1. Use the predicted values and SE to calculate 95% confidence intervals (CI) of the predictions. 1. Plot the predictions and 95% CI.

Here is what that looks like:

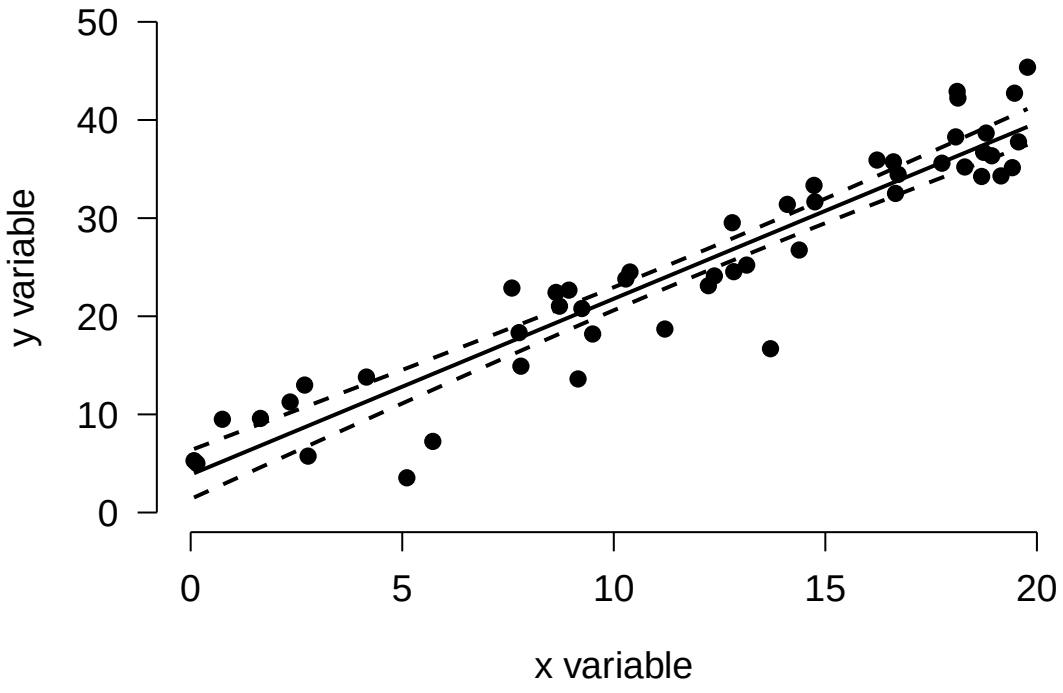
```
# fit the model
mod08fr <- lm(y~x, data=dx)

# new data for prediction
px <- seq(min(dx$x), max(dx$x), length=100)

# prediction with expected mean and SE of Y
pred <- predict(mod08fr, newdata=data.frame(x=px),
                se.fit=TRUE, interval="confidence")

# predicted mean and 95% CI
y.lo <- pred$fit[,2]
y.mn <- pred$fit[,1]
y.up <- pred$fit[,3]

# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
# make the plot and set basic options
plot(dx$x, dx$y, type="n",
    ylim=c(0, 50), xlim=c(0, 20),
    xlab="x variable", ylab="y variable")
# add lines for lower, upper conf. limits
# and predicted mean
points(px, y.lo, type="l", lwd=2, lty=2)
points(px, y.up, type="l", lwd=2, lty=2)
points(px, y.mn, type="l", lwd=2)
# put original data on top, so not covered by
# lines and etc.
points(dx$x, dx$y, pch=16, cex=1.2)
```



We will have to perform the same basic steps to produce a similar plot from JAGS, but the implementation is a bit different because there isn't a good `predict()` method in R for these kinds of models .

We are going to use the posterior draws for each parameter. To avoid issues with autocorrelation in the chains, we will pull these from `sims.matrix` instead of `sims.array`. These values have been permuted from the raw MCMC outputs so that predicted values do not suffer from any potential issues of autocorrelation in the chains or cross-correlations between the posteriors of different parameters.

```
# matrix of coefficients
prx <- mod08$BUGSoutput$sims.matrix
## don't need deviance or sigma
prx <- prx[,c("beta0", "beta1")]
post.n <- nrow(prx)
```

Next, we need a vector of x values at which to predict using the model. As before, we'll set up a series of values at small intervals over the full range of observed x.

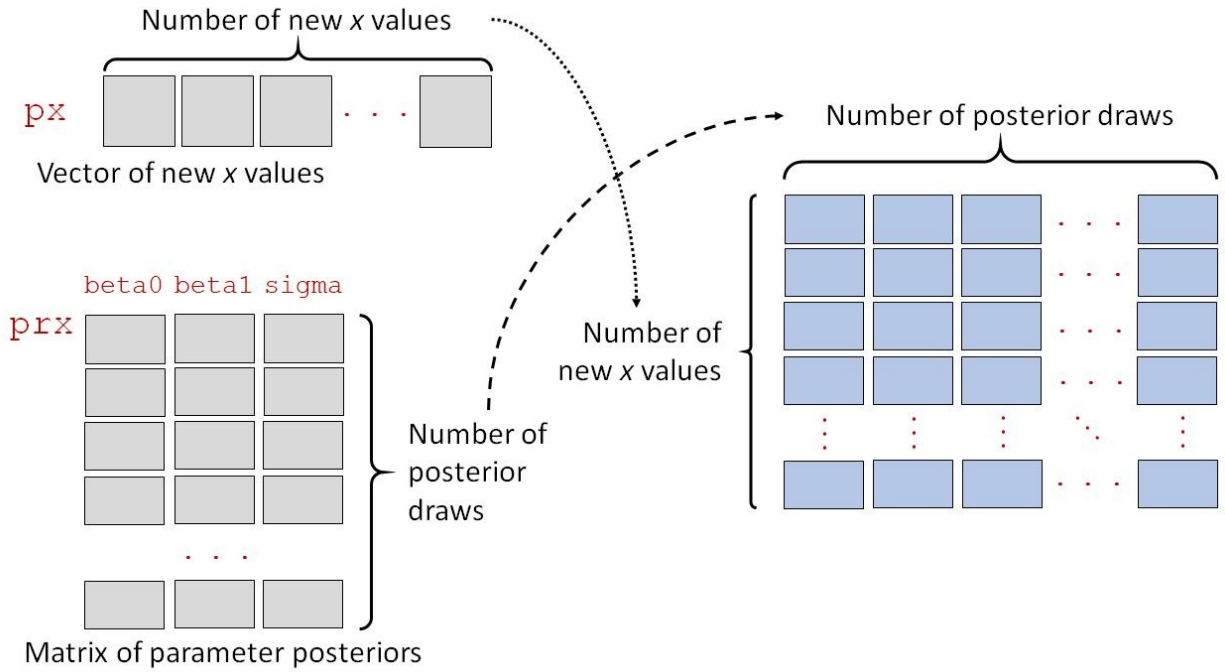
```
# x values at which to predict
pred.n <- 100
px <- seq(min(dx$x), max(dx$x), length=pred.n)
```

If you are familiar with matrix multiplication, the matrix of predictions can be obtained very quickly by constructing the design matrix and multiplying it by the transpose of the coefficients matrix:

```
design.mat <- matrix(c(rep(1, pred.n), px), ncol=2)
pred.mat <- design.mat %*% t(prx)
```

In my experience, the easier and more flexible approach is to calculate the predicted values in a loop (the above only works with ordinary linear models). Set up a matrix to hold the predicted values for each posterior draw of the predictors, and for each value of the predictor variable at which we want a prediction. This part is a bit tricky, so consider the figure below. The plan here is to end up with the predictor values along one margin (rows) and the replicates along the other (columns) . Once the matrix is filled, we can use `apply()` to get quantiles of the predicted values for each x.

Figure 4.2: Relationship between objects used to calculate model predictions and their 95% confidence interval (CI).




```
# set up matrix
pred.mat <- matrix(NA, nrow=pred.n, ncol=post.n)

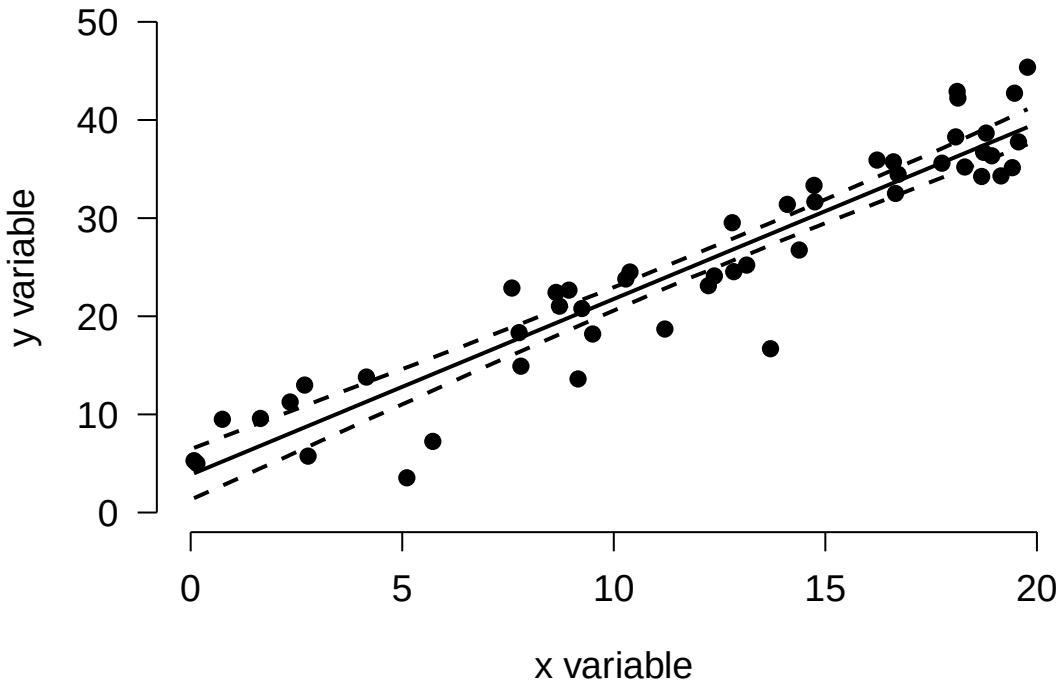
# fill in a loop
for(i in 1:post.n){
    pred.mat[,i] <- prx[i,"beta0"] + prx[i,"beta1"] * px
}#i
```

No matter how the predicted values for every new x were calculated, we can get the mean and 95% CRI using the mean and quantile functions. Sometimes I'll use the posterior median instead of the mean. For linear models it doesn't matter, but for models with other response distributions the median can be more representative of the central tendency.

```
# set up data frame with mean and CRI
plotdf <- data.frame(x=px,
    mn=apply(pred.mat, 1, mean),
    lo=apply(pred.mat, 1, quantile, 0.025),
    up=apply(pred.mat, 1, quantile, 0.975))
```

We can use our new object `plotdf` to make a nice plot of the model predictions. The code is almost identical to the code for the frequentist model outputs shown above because we set up the objects containing the plot values in the same way.

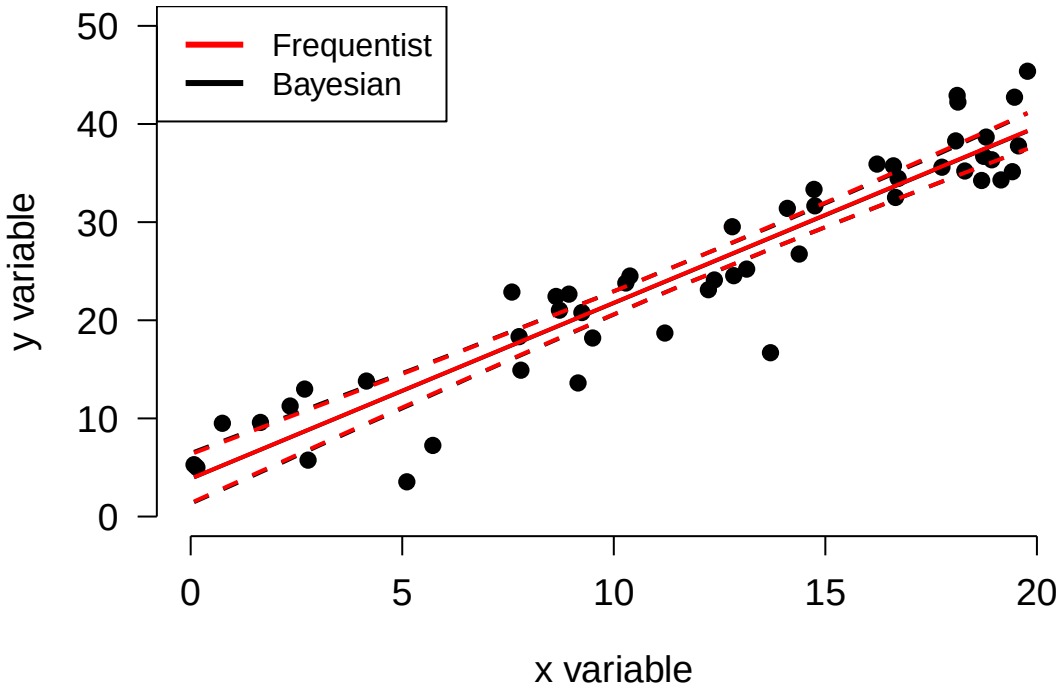
```
# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
# make the plot and set basic options
plot(dx$x, dx$y, type="n",
    ylim=c(0, 50), xlim=c(0, 20),
    xlab="x variable", ylab="y variable")
# add lines for lower, upper conf. limits
# and predicted mean
points(plotdf$x, plotdf$lo, type="l", lwd=2, lty=2)
points(plotdf$x, plotdf$up, type="l", lwd=2, lty=2)
points(plotdf$x, plotdf$mn, type="l", lwd=2)
# put original data on top, so not covered by
# lines and etc.
points(dx$x, dx$y, pch=16, cex=1.2)
```



If you add the frequentist predictions and CI with the following commands, you'll see that they are nearly the same as the Bayesian prediction and CRI.

```
# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
# make the plot and set basic options
plot(dx$x, dx$y, type="n",
    ylim=c(0, 50), xlim=c(0, 20),
    xlab="x variable", ylab="y variable")
# add lines for lower, upper conf. limits
# and predicted mean
points(plotdf$x, plotdf$lo, type="l", lwd=2, lty=2)
points(plotdf$x, plotdf$up, type="l", lwd=2, lty=2)
points(plotdf$x, plotdf$mn, type="l", lwd=2)
# put original data on top, so not covered by
# lines and etc.
points(dx$x, dx$y, pch=16, cex=1.2)
# add frequentist interval for comparison
points(px, y.lo, type="l", lwd=2, lty=2, col="red")
points(px, y.up, type="l", lwd=2, lty=2, col="red")
points(px, y.mn, type="l", lwd=2, col="red")
```

```
legend("topleft", legend=c("Frequentist", "Bayesian"),
    lwd=3, col=c("red", "black"))
```



4.3.2 Multiple linear regression

In multiple linear regression, or multiple regression, Y is predicted by more than one X variable. The X variables are assuemed to be orthogonal, or uncorrelated, with each other. If predictors are correlated, it may be difficult to accurately partition sums of squares between them and get a precise estimate of each one's effect. This problem is also called model **identifiability**.

Adding additional variables to a regression in JAGS is straightforward; it just requires a little more bookkeeping than in R.

```
# simulate some data
set.seed(42)
n <- 100
dx <- data.frame(x1=runif(n, 0, 20),
    x2=runif(n, 0, 20))

beta0 <- 4
beta1 <- 1.8
beta2 <- -2.1
sigma.res <- 4
```

```
dx$y <- beta0 + beta1*dx$x1 + beta2*dx$x2 +
    rnorm(n, 0, sigma.res)

# linear regression with 2 predictors
mod.name <- "jags_linreg_2x.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## intercept
        beta0 ~ dnorm(0, 0.001)

        ## slopes
        beta1 ~ dnorm(0, 0.001)
        beta2 ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(eta[i], tau)
            eta[i] <- beta0 + beta1 * x1[i] + beta2 * x2[i]
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("beta0", "beta1", "beta2", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# package data for JAGS
```

```
in.data <- list(y=dx$y,
    n=n, x1=dx$x1, x2=dx$x2)

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(beta0=rnorm(1, 0, 100),
                         beta1=rnorm(1, 0, 100),
                         beta2=rnorm(1, 0, 100),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod09 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 100
   Unobserved stochastic nodes: 4
   Total graph size: 612

Initializing model
```

```
mod09
```

```
Inference for Bugs model at "jags_linreg_2x.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 7.22 secs
      mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
beta0    3.622   1.006  1.547  2.969  3.620  4.324  5.548 1.003   660
beta1    1.786   0.060  1.675  1.745  1.785  1.828  1.907 1.002   950
beta2   -2.083   0.066 -2.207 -2.127 -2.083 -2.041 -1.947 1.001  1200
sigma    3.689   0.256  3.235  3.511  3.664  3.857  4.233 1.000  1200
```

```
deviance 543.678   2.870 540.217 541.630 542.983 545.093 551.413 1.003   700

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 4.1 and DIC = 547.8
DIC is an estimate of expected predictive error (lower deviance is better).
```

####Plotting predicted values from multiple regression

Getting predicted values is also straightforward. The biggest change from the model with one predictor is that we need to generate two sets of predictions:

- Predict over range of `x1` while holding `x2` at its median
- Predict over range of `x2` while holding `x1` at its median

Because the effects of `x1` and `x2` are independent of each other (at least, they are assumed to be), these two sets of predictions will allow us to see the effects of each variable while holding the other constant at a "typical" value. In your write-up, you should state explicitly that this is how you generated predictions.

```
# matrix of coefficients
prx <- mod09$BUGSoutput$sims.matrix
## don't need deviance or sigma
prx <- prx[,c("beta0", "beta1", "beta2")]

# x values at which to predict
pred.n <- 100
px1 <- seq(min(dx$x1), max(dx$x1), length=pred.n)
px2 <- seq(min(dx$x2), max(dx$x2), length=pred.n)

# set up matrices for predictions
pred2 <- pred1 <- matrix(NA, nrow=pred.n, ncol=post.n)

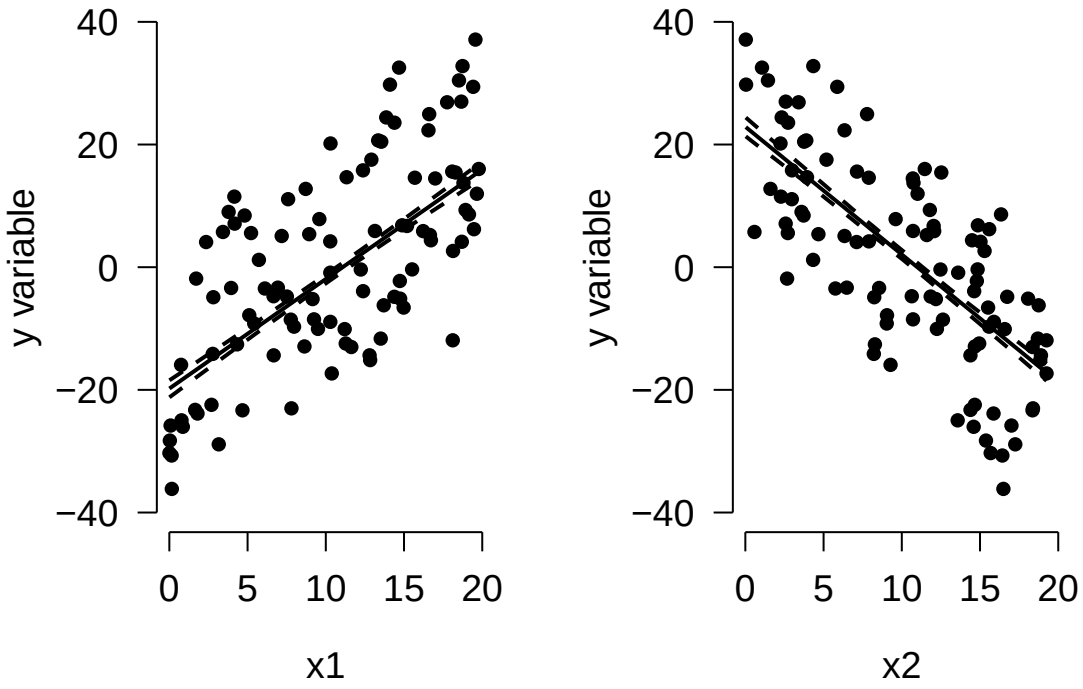
# fill in a loop
for(i in 1:post.n){
    pred1[,i] <- prx[i,"beta0"] +
        prx[i,"beta1"] * px1 +
        prx[i,"beta2"] * median(dx$x2)
    pred2[,i] <- prx[i,"beta0"] +
        prx[i,"beta1"] * median(dx$x1) +
        prx[i,"beta2"] * px2
}#i
```

```
pr1 <- data.frame(x1=px1,
    mn=apply(pred1, 1, mean),
    lo=apply(pred1, 1, quantile, 0.025),
    up=apply(pred1, 1, quantile, 0.975))

pr2 <- data.frame(x2=px2,
    mn=apply(pred2, 1, mean),
    lo=apply(pred2, 1, quantile, 0.025),
    up=apply(pred2, 1, quantile, 0.975))

# fancy plot options with par()
par(mfrow=c(1,2), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
# x1 plot
plot(dx$x1, dx$y, type="n",
    ylim=c(-40, 40), xlim=c(0, 20),
    xlab="x1", ylab="y variable")
points(pr1$x1, pr1$lo, type="l", lwd=2, lty=2)
points(pr1$x1, pr1$up, type="l", lwd=2, lty=2)
points(pr1$x1, pr1$mn, type="l", lwd=2)
# put original data on top, so not covered by
# lines and etc.
points(dx$x1, dx$y, pch=16)

# x2 plot
plot(dx$x2, dx$y, type="n",
    ylim=c(-40, 40), xlim=c(0, 20),
    xlab="x2", ylab="y variable")
points(pr2$x2, pr2$lo, type="l", lwd=2, lty=2)
points(pr2$x2, pr2$up, type="l", lwd=2, lty=2)
points(pr2$x2, pr2$mn, type="l", lwd=2)
# put original data on top, so not covered by
# lines and etc.
points(dx$x2, dx$y, pch=16)
```



4.4 Analysis of covariance (ANCOVA)

In **analysis of covariance (ANCOVA)**, a continuous response variable is modeled using a continuous explanatory variable and a factor (grouping variable). You've probably heard of ANCOVA before without realizing it. When scientific studies are described on the news, the reporter might say that the researchers "controlled for" the effects of additional variables. That often means that the researchers used a type of ANCOVA in their analysis.

The figure below shows two examples of what ANCOVA looks like in practice. Both types of ANCOVA are extremely common in biology, so we'll fit both (although, as we'll see, they can be the same model in JAGS).

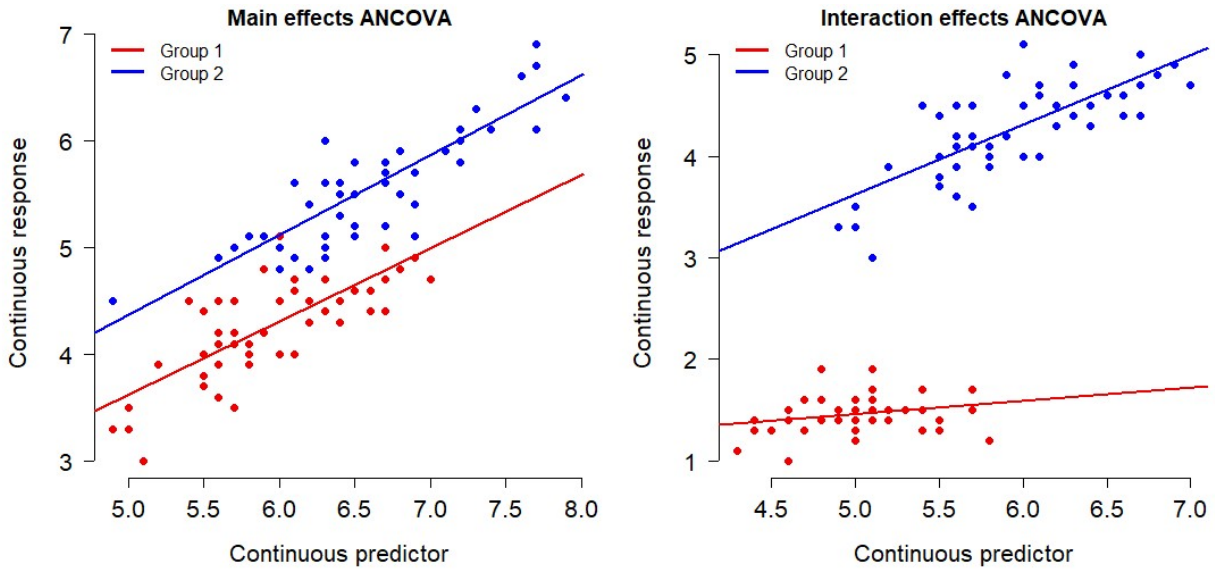
- Left: y increases as a linear function of x, and values in group 2 tend to be larger than values in group 1. Groups 1 and 2 share the same slope, so x has the same effect on y in both groups.
- Right: y varies as a function of x, and the effect of x on y is different in each group (i.e., the slopes are different). This is "interaction effects" ANCOVA, or "ANCOVA with interaction". An "interaction" means that the effect of one variable affects the effect of another variable.

Put figure here

4.4.1 ANCOVA with main effects only

As always, we'll simulate a dataset and then analyze it. Before that we need to think about how we want to write the ANCOVA model. The traditional way is the effects parameterization:

Figure 4.3: Illustration of ANCOVA without an interaction (different intercepts, same slope) and with an interaction (different intercepts and slopes.)



$$y_i = Normal\left(\eta_i, \sigma_{res}\right)$$

$$\eta_i = \beta_0 + \beta_1 x1_i + \beta_2 x2_i$$

where x1 is the continuous predictor variable, and x2 is a variable (or matrix of dummy variables) that describes group membership (i.e., levels of the factor). This model has an intercept (β_0) and a term that describes group effects (β_0). The downside of writing the model this way is in how effects are represented: as differences from the baseline level. This can be confusing to add up.

Grouping the factor effects and the intercept gives us something more like a means parameterization of ANCOVA:

$$y_i = Normal\left(\eta_i, \sigma_{res}\right)$$

$$\eta_i = \beta_{0,fac_i} + \beta_1 x1_i$$

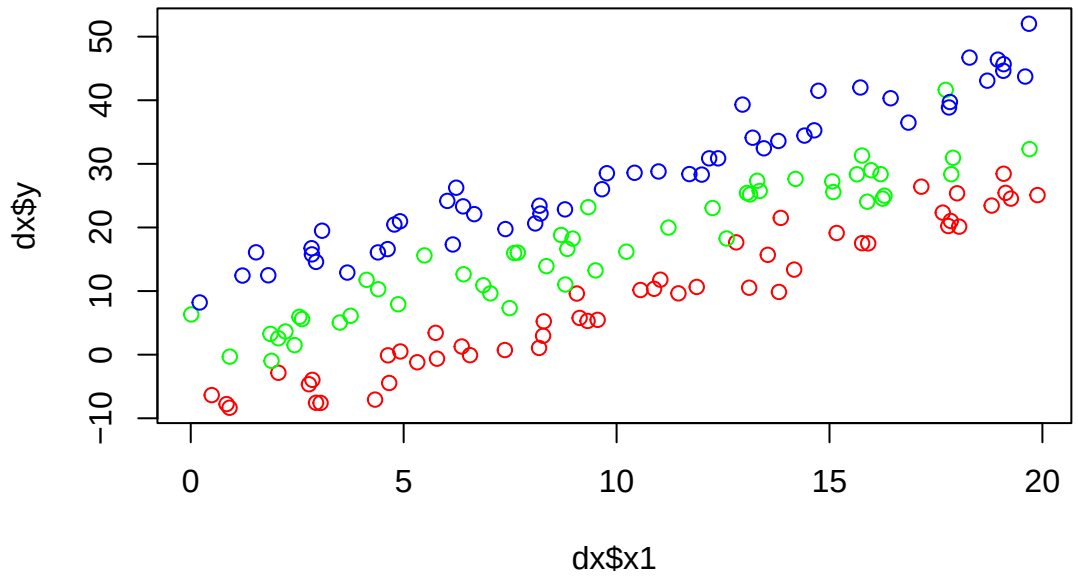
Here the β_2 term is combined with the intercept, such that the intercept varies between groups: β_{0,fac_i}. This version of the model avoids the awkwardness of having to add parameters together to see what the values in each group should be.

The means parameterization of ANCOVA is easier to deal with in JAGS than the effects parameterization, and is easy to extend to other situations, including mixed models.

```r
# simulate some data
set.seed(123)
n <- 50
nfac <- 3
dx <- data.frame(x1=runif(n*nfac, 0, 20),
    fac1=rep(1:nfac, each=n))

beta0 <- c(-10, 0, 10)
beta1 <- 1.8
sigma.res <- 3
dx$y <- beta0[dx$fac1] + beta1*dx$x1 +
    rnorm(nrow(dx), 0, sigma.res)

dx$col <- rainbow(3)[dx$fac1]
plot(dx$x1, dx$y, col=dx$col)
```




```r
summary(lm(y~x1+factor(fac1), data=dx))
```

```
Call:
lm(formula = y ~ x1 + factor(fac1), data = dx)

Residuals:
    Min      1Q  Median      3Q     Max
```

```
-6.1114 -1.8415 -0.3887  1.7631  9.6479

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -10.11686    0.59274  -17.07   <2e-16 ***
x1              1.81207    0.04129   43.88   <2e-16 ***
factor(fac1)2   9.97812    0.57875   17.24   <2e-16 ***
factor(fac1)3  19.94729    0.57767   34.53   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.888 on 146 degrees of freedom
Multiple R-squared:  0.9551,    Adjusted R-squared:  0.9542
F-statistic:  1035 on 3 and 146 DF,  p-value: < 2.2e-16
```

```
# linear regression with 2 predictors
mod.name <- "jags_ancova_main_1x1f.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## intercept
        for(i in 1:nfac1){
            beta0[i] ~ dnorm(0, 0.001)
        }#i

        ## slope
        beta1 ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(eta[i], tau)
            eta[i] <- beta0[fac1[i]] + beta1 * x1[i]
        }#i

    }#model

", fill=TRUE)
```

```r
sink()

# parameters to monitor
params <- c("beta0", "beta1", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# package data for JAGS
in.data <- list(y=dx$y, n=nrow(dx),
    x1=dx$x1,
    fac1=dx$fac1, nfac1=nfac)


# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(beta0=rnorm(nfac, 0, 100),
                         beta1=rnorm(1, 0, 100),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod10 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 150
   Unobserved stochastic nodes: 5
```

```
   Total graph size: 764

Initializing model
```

```
mod10
```

```
Inference for Bugs model at "jags_ancova_main_1x1f.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 10.67 secs
          mu.vect sd.vect    2.5%     25%     50%     75%   97.5%  Rhat n.eff
beta0[1]  -10.126   0.604 -11.292 -10.553 -10.107  -9.728  -8.898 1.002   780
beta0[2]   -0.147   0.585  -1.268  -0.556  -0.164   0.258   1.029 1.003   680
beta0[3]    9.817   0.571   8.727   9.419   9.834  10.200  10.890 1.003   700
beta1       1.813   0.041   1.725   1.787   1.813   1.839   1.895 1.002   950
sigma       2.901   0.169   2.601   2.784   2.888   3.006   3.237 1.000  1200
deviance  744.936   3.126 740.660 742.604 744.365 746.445 752.664 1.000  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 4.9 and DIC = 749.8
DIC is an estimate of expected predictive error (lower deviance is better).
```

The estimated slope and group-level means are basically correct, up to sampling error.

4.4.1.1 Plotting ANCOVA with main effects only

Making a plot with predictions and their 95% CRI is similar to what we did for multiple linear regression. We need to generate predictions within each group of the factor `fac1` using the group-specific parameters, and present the groups with different colors, symbols, line styles, etc.

```
# matrix of coefficients
prx <- mod10$BUGSoutput$sims.matrix
## don't need deviance or sigma
prx <- prx[,grep("beta", colnames(prx))]

# x values at which to predict
pred.n <- 100
px <- seq(min(dx$x1), max(dx$x1), length=pred.n)

# set up matrices for predictions
# one for each group (level of fac1)
preddf <- expand.grid(x1=px, fac1=1:nfac)
```

```
npred <- nrow(preddf)

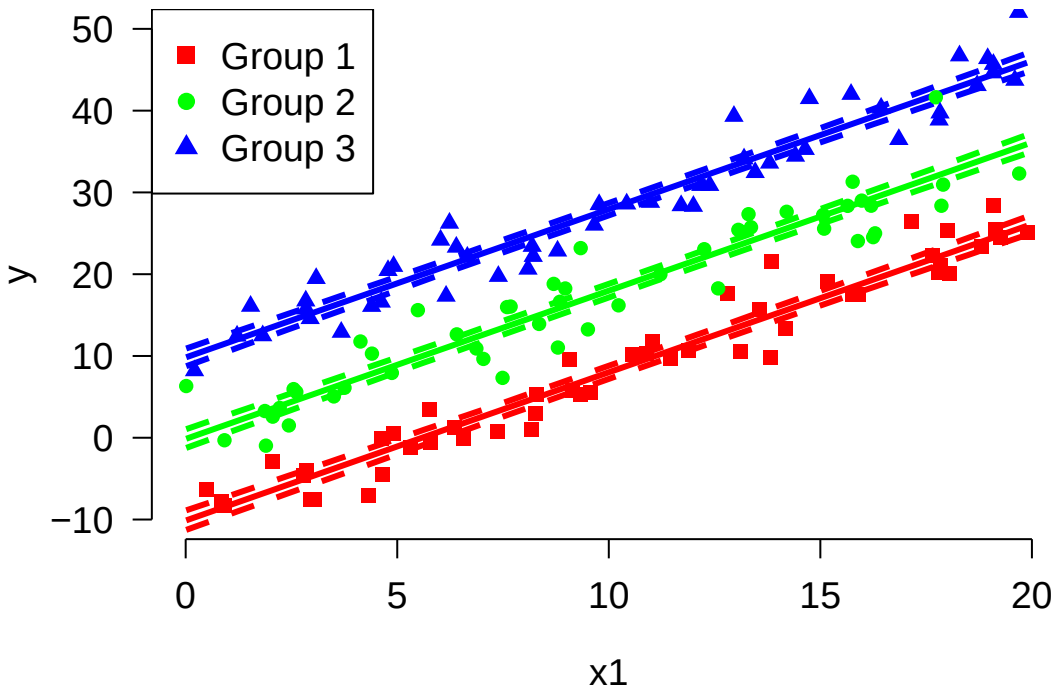
pred <- matrix(NA, nrow=nrow(preddf), ncol=post.n)

# fill in a loop
for(i in 1:post.n){
    pred[,i] <- prx[i,preddf$fac1] +
        prx[i,"beta1"] * preddf$x1
}#i

# apply to get mean and CRI
preddf$mn <- apply(pred, 1, mean)
preddf$lo <- apply(pred, 1, quantile, 0.025)
preddf$up <- apply(pred, 1, quantile, 0.975)

# define some colors and shapes:
cols <- rainbow(nfac)
pchs <- 15:17 # square, circle, triangle

# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
# x1 plot
plot(dx$x1, dx$y, type="n",
    ylim=c(-10, 50), xlim=c(0, 20),
    xlab="x1", ylab="y")
for(i in 1:nfac){
    flag <- which(preddf$fac1==i)
    points(px, preddf$mn[flag], type="l",
        lwd=3, col=cols[i])
    points(px, preddf$lo[flag], type="l",
        lwd=3, lty=2, col=cols[i])
    points(px, preddf$up[flag], type="l",
        lwd=3, lty=2, col=cols[i])
}#i
# put original data on top, so not covered by
# lines and etc.
points(dx$x1, dx$y, pch=pchs[dx$fac1], col=cols[dx$fac1])
legend("topleft", legend=paste("Group", 1:nfac),
       pch=pchs, col=cols, cex=1.2)
```



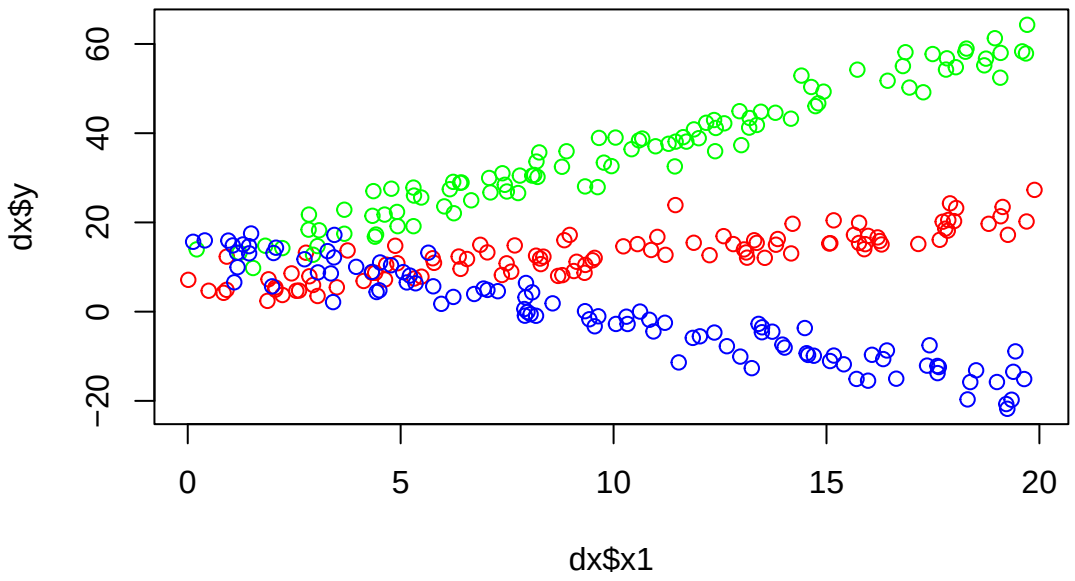
4.4.2 ANCOVA with interaction

The ANCOVA with interaction effect model will be constructed by combining bits of the main effects ANCOVA model with the "parameters as arrays" trick we learned in the ANOVA with interaction model. The key is that each group has its own slope and intercept.

```
# simulate some data
set.seed(123)
n <- 100
nfac <- 3
dx <- data.frame(x1=runif(n*nfac, 0, 20),
    fac1=rep(1:nfac, each=n))

beta0 <- c(5, 10, 15)
beta1 <- c(0.8, 2.5, -1.6)
sigma.res <- 3
dx$y <- beta0[dx$fac1] + beta1[dx$fac1]*dx$x1 +
    rnorm(nrow(dx), 0, sigma.res)

dx$col <- rainbow(3)[dx$fac1]
plot(dx$x1, dx$y, col=dx$col)
```




```
summary(lm(y~x1*factor(fac1), data=dx))
```

```
Call:
lm(formula = y ~ x1 * factor(fac1), data = dx)

Residuals:
    Min      1Q  Median      3Q     Max
-7.6183 -1.9110 -0.0861  1.7883  9.7024

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)        5.21178    0.60045   8.680 2.77e-16 ***
x1                 0.78332    0.05234  14.965  < 2e-16 ***
factor(fac1)2      4.82355    0.88711   5.437 1.14e-07 ***
factor(fac1)3     10.24185    0.83151  12.317  < 2e-16 ***
x1:factor(fac1)2   1.74993    0.07706  22.708  < 2e-16 ***
x1:factor(fac1)3  -2.45160    0.07286 -33.648  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.969 on 294 degrees of freedom
Multiple R-squared:  0.9744,    Adjusted R-squared:  0.9739
F-statistic:  2236 on 5 and 294 DF,  p-value: < 2.2e-16
```

```
# linear regression with 2 predictors
mod.name <- "jags_ancova_inter_1x1f.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## intercept and slope
        for(i in 1:nfac1){
            beta0[i] ~ dnorm(0, 0.001)
            beta1[i] ~ dnorm(0, 0.001)
        }#i

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            y[i] ~ dnorm(eta[i], tau)
            eta[i] <- beta0[fac1[i]] + beta1[fac1[i]] * x1[i]
        }#i

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("beta0", "beta1", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# package data for JAGS
in.data <- list(y=dx$y, n=nrow(dx),
    x1=dx$x1,
    fac1=dx$fac1, nfac1=nfac)
```

```
# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(beta0=rnorm(nfac, 0, 100),
                         beta1=rnorm(nfac, 0, 100),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod11 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 300
   Unobserved stochastic nodes: 7
   Total graph size: 1516

Initializing model
```

```
mod11
```

```
Inference for Bugs model at "jags_ancova_inter_1x1f.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 22.18 secs
          mu.vect sd.vect    2.5%     25%     50%     75%   97.5%  Rhat
beta0[1]   5.203   0.639   3.968   4.747   5.203   5.634   6.451 1.001
beta0[2]  10.045   0.670   8.731   9.600  10.033  10.485  11.448 1.002
beta0[3]  15.428   0.581  14.323  15.050  15.425  15.844  16.502 1.003
beta1[1]   0.783   0.055   0.677   0.745   0.784   0.823   0.887 1.000
beta1[2]   2.532   0.057   2.424   2.493   2.532   2.567   2.647 1.003
beta1[3]  -1.667   0.051  -1.764  -1.702  -1.668  -1.632  -1.567 1.006
sigma      2.972   0.124   2.735   2.886   2.969   3.052   3.227 1.001
```

```
deviance 1505.402  3.887 1500.017 1502.500 1504.658 1507.450 1514.401 1.000
         n.eff
beta0[1]  1200
beta0[2]  1200
beta0[3]   830
beta1[1]  1200
beta1[2]   990
beta1[3]   300
sigma     1200
deviance  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 7.6 and DIC = 1513.0
DIC is an estimate of expected predictive error (lower deviance is better).
```

4.4.2.1 Plotting ANCOVA with interactions

And the plot, much as for the ANCOVA without interactions:

```
# matrix of coefficients
prx <- mod11$BUGSoutput$sims.matrix
## don't need deviance or sigma
prx <- prx[,grep("beta", colnames(prx))]

# x values at which to predict
pred.n <- 100
px <- seq(min(dx$x1), max(dx$x1), length=pred.n)

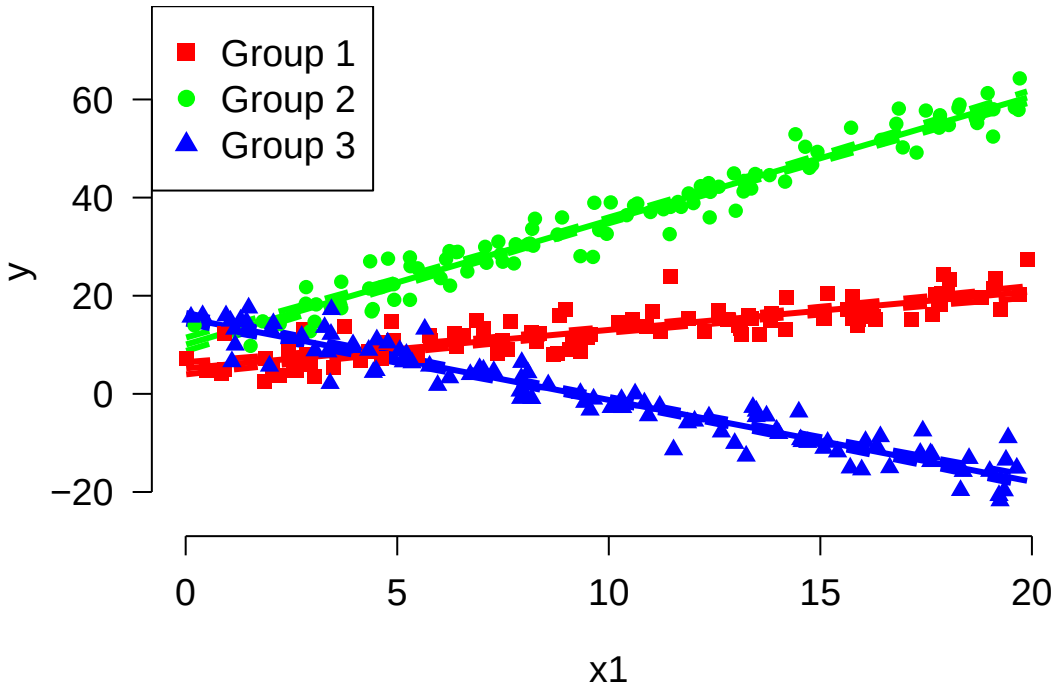
# set up matrices for predictions
# one for each group (level of fac1)
preddf <- expand.grid(x1=px, fac1=1:nfac)
npred <- nrow(preddf)

pred <- matrix(NA, nrow=nrow(preddf), ncol=post.n)

# fill in a loop
for(i in 1:post.n){
   pred[,i] <- prx[i,grep("beta0", colnames(prx))[preddf$fac1]] +
      prx[i,grep("beta1", colnames(prx))[preddf$fac1]] * preddf$x1
}#i
```

```
# apply to get mean and CRI
preddf$mn <- apply(pred, 1, mean)
preddf$lo <- apply(pred, 1, quantile, 0.025)
preddf$up <- apply(pred, 1, quantile, 0.975)
# define some colors and shapes:
cols <- rainbow(nfac)
pchs <- 15:17 # square, circle, triangle

# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
plot(dx$x1, dx$y, type="n",
    ylim=c(-25, 75), xlim=c(0, 20),
    xlab="x1", ylab="y")
for(i in 1:nfac){
    flag <- which(preddf$fac1==i)
    points(px, preddf$mn[flag], type="l",
        lwd=3, col=cols[i])
    points(px, preddf$lo[flag], type="l",
        lwd=3, lty=2, col=cols[i])
    points(px, preddf$up[flag], type="l",
        lwd=3, lty=2, col=cols[i])
}#i
# put original data on top, so not covered by
# lines and etc.
points(dx$x1, dx$y, pch=pchs[dx$fac1], col=cols[dx$fac1])
legend("topleft", legend=paste("Group", 1:nfac),
    pch=pchs, col=cols, cex=1.2)
```



4.5 Linear with extra steps

Earlier we discussed how some nonlinear curves can be fit statistically as linear models, by transforming or otherwise rewriting them. I like to think of these as "linear with extra steps".

4.5.1 Log-linear models

Log-linear models are essentially linear models where the logarithm of the response variable is a linear function of the predictors. In other words, it's a linear model where the response is $log\,(y)$. For this reason, the response variable in a log-linear model is strictly positive ($y > 0$) because e^a is positive for any real number a.

In state-space notation:

$$y_i \sim Lognormal\,(\eta_i,\, \sigma_{res})$$

$$\eta_i = \beta_0 + \beta_1 x_i$$

Or sometimes:

$$\log\,(y_i) \sim Normal\,(\eta_i,\, \sigma_{res})$$

$$\eta_i = \beta_0 + \beta_1 x_i$$

Or even:

$$y_i \sim e^{Normal(\eta_i,\, \sigma_{res})}$$

$$\eta_i = \beta_0 + \beta_1 x_i$$

No matter how you write it, the log-linear model has the deterministic part on the log scale, and the response variable on the non-logarithmic data scale. This means that either y follows a lognormal distribution, or the logarithm of y follows a normal distribution. These statements are equivalent. You can code a log-linear model either way you want in JAGS.

4.5.1.1 Method 1: *y* follows lognormal distribution

Let's simulate some data. Notice that x and y are both on the linear scale (not logarithmic). The x domain and model parameters were chosen with the fact that they would be exponentiated in mind.

```
# simulate some data
set.seed(74656)
n <- 40
dx <- data.frame(x=runif(n, 0, 5))

beta0 <- 0.5
beta1 <- 1.02
sigma.res <- 0.3
dx$y <- exp(beta0 + beta1*dx$x + rnorm(n, 0, sigma.res))

# log-linear model
mod.name <- "jags_loglinear.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## intercept
        beta0 ~ dnorm(0, 0.001)
        beta1 ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)
```

```
        # likelihood
        for(i in 1:n){
            # expected value on LOG scale
            eta[i] <- beta0 + beta1 * x[i]
            # NOTE: uses lognormal, not normal
            y[i] ~ dlnorm(eta[i], tau)
        }#i
    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("beta0", "beta1", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# package data for JAGS
in.data <- list(y=dx$y,
    n=n, x=dx$x)

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(beta0=rnorm(1, 0, 100),
                         beta1=rnorm(1, 0, 100),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod12 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
```

```
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 40
   Unobserved stochastic nodes: 3
   Total graph size: 171

Initializing model
```

```
mod12
```

```
Inference for Bugs model at "jags_loglinear.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 10.72 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
beta0       0.397   0.112  0.178  0.325  0.398  0.469  0.624 1.005   390
beta1       1.060   0.042  0.980  1.034  1.060  1.089  1.141 1.003   610
sigma       0.364   0.045  0.293  0.333  0.359  0.389  0.464 1.005   960
deviance  256.942   2.794 253.868 254.962 256.214 258.111 263.865 1.007  390

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 3.9 and DIC = 260.8
DIC is an estimate of expected predictive error (lower deviance is better).
```

4.5.1.2 Method 2: log(y) follows normal distribution

The other method is to log-transform y in a data block, and use a normal distribution to describe the response variable. The model file below uses the log-transformed `z` in place of the original y in the model likelihood, and the normal instead of the lognormal distribution. Only the model file has changed; everything else (i.e., the steps in R) is the same.

```
# log-linear model version 2, with transform
mod.name <- "jags_loglinear_v2.txt"
sink(mod.name)
cat("
    data{
        for(i in 1:n){
```

```
            z[i] <- log(y[i])
        }#i
    }#data

    model{
        # priors
        ## intercept
        beta0 ~ dnorm(0, 0.001)
        beta1 ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            # expected value on LOG scale
            eta[i] <- beta0 + beta1 * x[i]
            # NOTE: uses normal, not lognormal
            # and z, which is on log scale
            z[i] ~ dnorm(eta[i], tau)
        }#i
    }#model
", fill=TRUE)
sink()
```

The alternative version got essentially the same results, but the intercept beta0 hasn't quite converged. The researchers would need to run longer chains or otherwise tune the model to get convergent outputs.

4.5.1.3 Plotting a log-linear model

Preparing a figure to show the data, model predictions, and 95% CRI is very similar to preparing the figure for a linear regression. The only difference is that we need to remember that the model predictions will be on the logarithmic scale, so we have to exponentiate the predicted mean and 95% CRI limits with R function `exp()`.

```
# matrix of coefficients
prx <- mod12$BUGSoutput$sims.matrix
## don't need deviance or sigma
prx <- prx[,c("beta0", "beta1")]
post.n <- nrow(prx)

# x values at which to predict
```

```
pred.n <- 100
px <- seq(min(dx$x), max(dx$x), length=pred.n)

# set up matrix
pred.mat <- matrix(NA, nrow=pred.n, ncol=post.n)

# fill in a loop
for(i in 1:post.n){
    pred.mat[,i] <- prx[i,"beta0"] + prx[i,"beta1"] * px
}#i

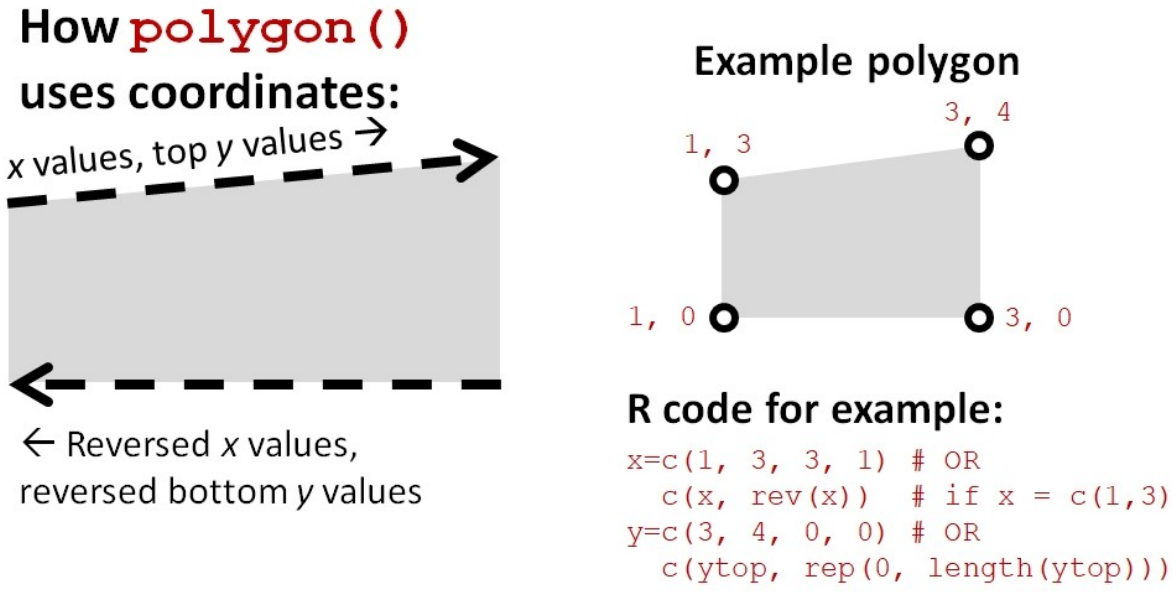
# set up data frame with mean and CRI
plotdf <- data.frame(x=px,
    mn=apply(pred.mat, 1, mean),
    lo=apply(pred.mat, 1, quantile, 0.025),
    up=apply(pred.mat, 1, quantile, 0.975))

# put predictions back on original scale:
# up to now, the code is EXACTLY the same as
# for linear regression
plotdf$mn <- exp(plotdf$mn)
plotdf$lo <- exp(plotdf$lo)
plotdf$up <- exp(plotdf$up)
```

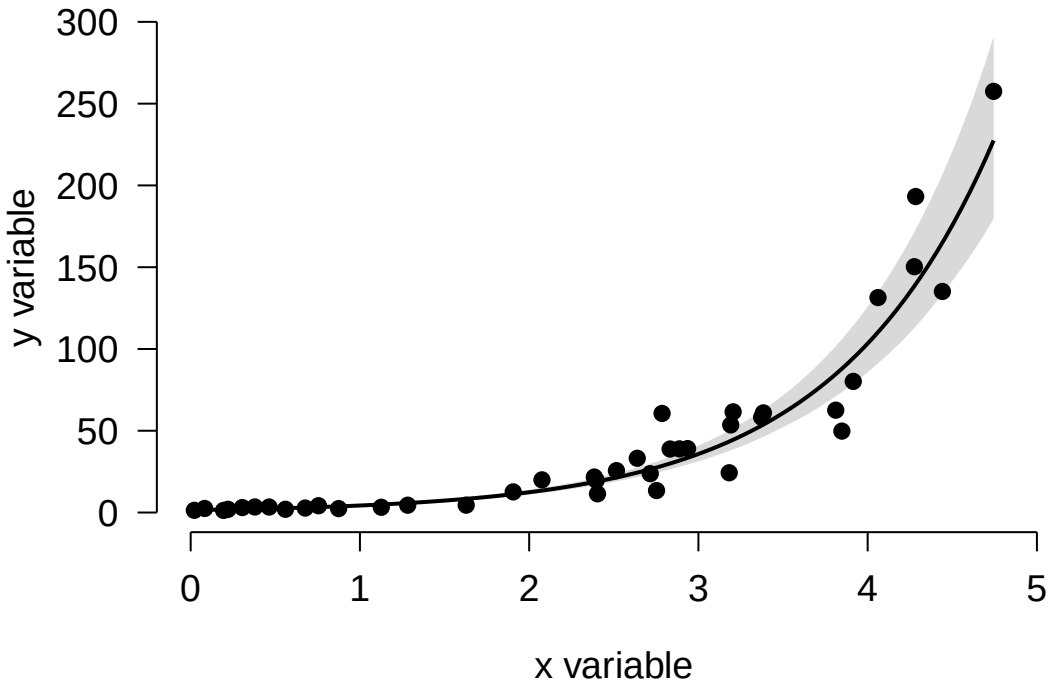
For this plot, we're going to use a shaded area to show the 95% CRI instead of using dashed lines. In R base graphics, this is done with polygon(). The syntax for polygon() takes some getting used to. Basically, the coordinates are used to draw the vertices of the shape either clockwise or anti-clockwise, depending on the values. If there are many vertices (100s), then the shape can appear curved.

```
# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
# make the plot and set basic options
plot(dx$x, dx$y, type="n",
    ylim=c(0, 300), xlim=c(0, 5),
    xlab="x variable", ylab="y variable")
# make a polygon for the CRI
polygon(x=c(px, rev(px)), y=c(plotdf$lo, rev(plotdf$up)),
    border=NA, col="grey85")
# predicted mean
points(plotdf$x, plotdf$mn, type="l", lwd=2)
```

Figure 4.4: Illustration of how R's `polygon()` function constructs a filled polygon. The left panel shows that the polygon is traced by following the upper boundary from left to right, then returning along the lower boundary in reverse order to close the shape. The right panel provides a simple four-vertex example with coordinates and the corresponding R code needed to draw the polygon.




```
# put original data on top, so not covered by
# lines and etc.
points(dx$x, dx$y, pch=16, cex=1.2)
```

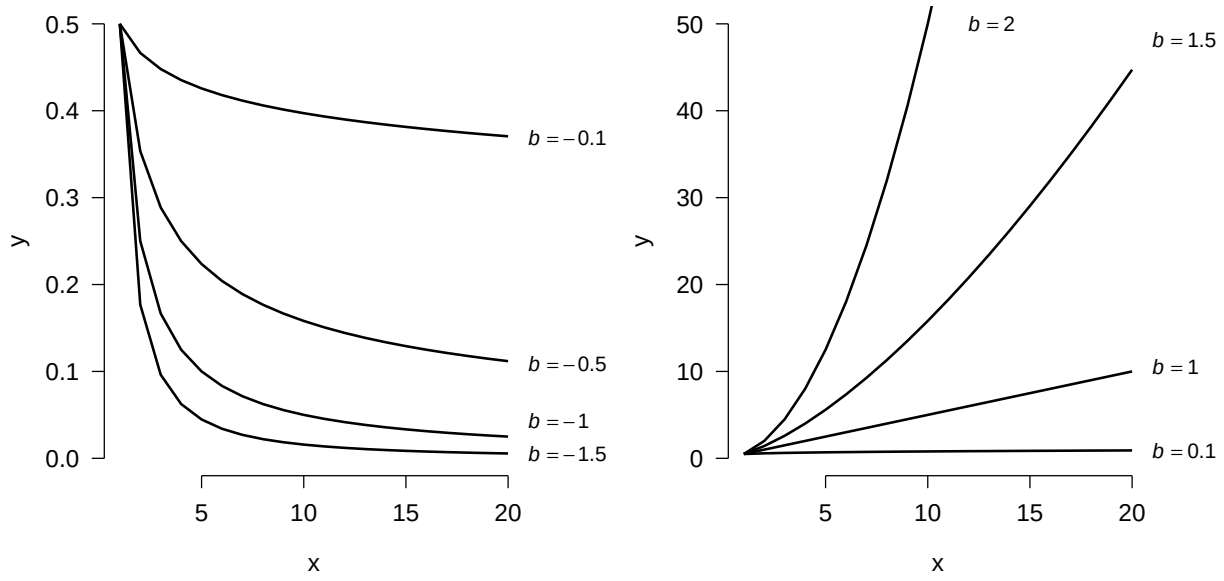


4.5.2 Power laws (log-log) models

The last model we'll examine is a model that appears nonlinear at first glance, but can actually be fit as a linear model. This is power law or allometric equation:

$$y = ax^b$$

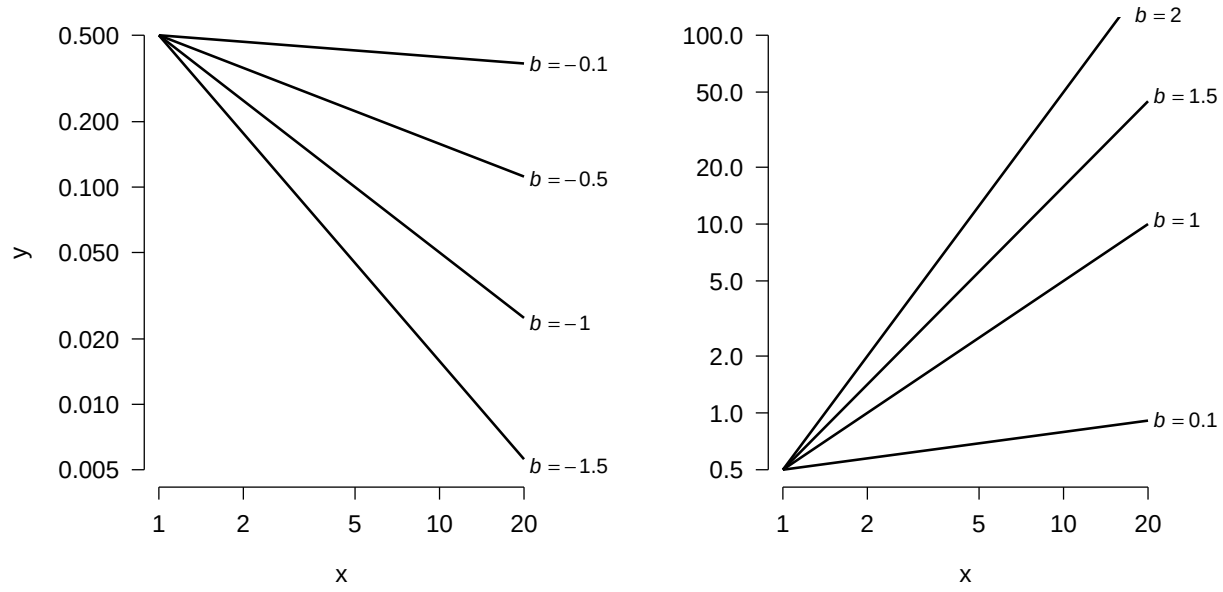
The terms in this equation are a, the value when$x = 1$, and b, a growth constant. The sign of b determines the sign of the slope of the curve; the magnitude of b determines how fast the function increases or decreases.



While this model looks nonlinear, it can be analyzed as a linear equation by taking advantage of some properties of logarithms. Taking the logarithm of both sides:

$$log(y) = log(a) + blog(x)$$

we get an equation were $log(y)$ is linear with respect to $log(x)$. Here is the previous figure on the new log-log scale. Notice how the values on the axes are spaced, and how the lines are now straight.



What the power law and its log-transform mean is that when x is scaled (multiplied) by some factor z, y is scaled by a factor of z^b. In other words, the power law describes how proportional changes in x relate to

proportional changes in y. This is because the power law with a proportional change z to x

$$y = a\left(zx\right)^b$$

can be rearranged to:

$$y = az^b x^b$$

Taking the logarithm of both sides we can see how the linearization works:

$$\log\left(y\right) = \log\left(a\right) + b\log\left(z\right) + b\log\left(x\right)$$

In its transformed state, we see that $log(y)$ is linear with respect to both $log(z)$ and $log(x)$. This can be simplified by factoring out b out of the terms that contain it, to get the linearized power law with slope b.

$$\log\left(y\right) = \log\left(a\right) + b\left(\log\left(z\right) + \log\left(x\right)\right)$$

This shows that multiplying x times a factor z is equivalent to adding $log(z)$ to $log(x)$. This is one of the definitions of the logarithm. This is also why the effect of x on y is linear on the log-log scale but curved on the linear-linear scale.

Transforming the power law model to linearize it will get you the same parameter estimates, but with the benefit of being able to use the ease and convenience of linear models. The hardest part of fitting this model in JAGS is keeping track of which values are on the log scale and which values are not. Complicating this is the fact that JAGS and BUGS handle data transformations differently (Lunn et al. 2000, Plummer 2003).

BUGS vs. JAGS

BUGS (WinBUGS and OpenBUGS) was a predecessor to JAGS. Most published textbooks, and online help examples, are old enough to have been written in BUGS. There are very few differences between BUGS and JAGS, and this issue with data transformation is probably one of the most important. The model script here should work in BUGS as well as JAGS, but the old-style BUGS equivalent might not work in JAGS. This took me a long time to figure out when I was writing this module.

As with the log-linear model, we have two options for how to code the linearized power law. Method 1 is to model y on its original scale with a lognormal distribution. Method 2 is to transform y in a data block and use a normal distribution. Both methods will get you the same answer.

4.5.2.1 Power law option 1: lognormal y

```
# simulate some data
set.seed(42)
n <- 30
```

```
dx <- data.frame(x=runif(n, 0, 20))

A <- 2.1
logA <- log(A)
B <- 1.5
sigma.res <- 0.3
dx$y <- exp(logA + B*log(dx$x) + rnorm(n, 0, sigma.res))

# MUST be TRUE:
all(dx$y > 0)
```

```
[1] TRUE
```

```
# power law (allometric equation)
mod.name <- "jags_power_law.txt"
sink(mod.name)
cat("
    model{
        # priors
        ## intercept
        logA ~ dnorm(0, 0.001)

        ## growth constant
        B ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            # expected value on LOG scale
            eta[i] <- logA +  B * log(x[i])
            y[i] ~ dlnorm(eta[i], tau)
        }#i

        # derived quantity: original scale A
        A <- exp(logA)

    }#model

", fill=TRUE)
```

```
sink()

# parameters to monitor
params <- c("A", "B", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# package data for JAGS
in.data <- list(y=dx$y,
    n=n, x=dx$x)

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
        res[[i]] <- list(logA=rnorm(1, 0, 100),
                         B=rnorm(1, 0, 100),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod13 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 30
   Unobserved stochastic nodes: 3
   Total graph size: 162
```

```
Initializing model
```

```
mod13
```

```
Inference for Bugs model at "jags_power_law.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 7.7 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
A           2.248   0.558  1.325  1.852  2.169  2.580  3.473 1.000  1200
B           1.439   0.100  1.243  1.374  1.439  1.509  1.641 1.000  1200
sigma       0.367   0.051  0.283  0.331  0.362  0.395  0.479 1.002   930
deviance  272.060   2.577 269.198 270.185 271.395 273.214 278.616 1.001  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 3.3 and DIC = 275.4
DIC is an estimate of expected predictive error (lower deviance is better).
```

4.5.2.2 Power law option 2: linearized

The equivalent approach is to handle the transformation of y in a separate data block. Either way is fine; I would probably use Method 1 in my own work because it's a little simpler.

Here's the alternative method. Only the model file has changed; everything else is the same. The alternative method uses a data block to transform the y values in the input, and uses a normal distribution to relate the expected value `eta` to the observed value `z`. The method above uses the lognormal distribution, which effectively undoes the log transform of y which was implied by the fact that the deterministic part of the model was on the log scale.

```
# power law (allometric equation) with
# transformation in data block
mod.name <- "jags_power_law_v2.txt"
sink(mod.name)
cat("
    data{
        for(i in 1:n){z[i] <- log(y[i])}#i
    }

    model{
        # priors
        ## intercept
        logA ~ dnorm(0, 0.001)
```

```
        ## growth constant
        B ~ dnorm(0, 0.001)

        ## residual SD and precision
        sigma ~ dlnorm(1, 0.25)
        tau <- 1 / (sigma^2)

        # likelihood
        for(i in 1:n){
            # expected value on LOG scale
            eta[i] <- logA +  B * log(x[i])
            # NOTE: likelihood uses transformed z, not y
            # and dnorm(), not dlnorm()
            z[i] ~ dnorm(eta[i], tau)
        }#i

        # derived quantity: original scale A
        A <- exp(logA)

    }#model

", fill=TRUE)
sink()

# parameters to monitor
params <- c("A", "B", "sigma")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3

# package data for JAGS
in.data <- list(y=dx$y,
    n=n, x=dx$x)

# function to define initial values for MCMC chains
init.fun <- function(nc){
    res <- vector("list", length=nc)
    for(i in 1:nc){
```

```
        res[[i]] <- list(logA=rnorm(1, 0, 100),
                         B=rnorm(1, 0, 100),
                         sigma=rlnorm(1, 1, 2))
    }
    return(res)
}
inits <- init.fun(nchains)

# fit model
mod13v2 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling data graph
   Resolving undeclared variables
   Allocating nodes
   Initializing
   Reading data back into data table
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 30
   Unobserved stochastic nodes: 3
   Total graph size: 192

Initializing model
```

```
mod13v2
```

```
Inference for Bugs model at "jags_power_law_v2.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 2.7 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
A          2.263   0.544  1.380  1.885  2.213  2.551  3.543 1.001  1200
B          1.437   0.096  1.244  1.374  1.436  1.500  1.618 1.000  1200
sigma      0.369   0.052  0.289  0.332  0.360  0.399  0.485 1.001  1200
deviance  24.015   2.580 21.140 22.168 23.355 25.144 30.715 1.002  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).
```

```
DIC info (using the rule: pV = var(deviance)/2)
pV = 3.3 and DIC = 27.3
DIC is an estimate of expected predictive error (lower deviance is better).
```

4.5.2.3 Plotting power laws

Getting predicted values for the Bayesian linearized power law fit is very similar to the process for the log-linear model. We just have to make sure to use the correct equation, and keep track of what is on the log scale and what is on the linear scale. Below we see two methods: Method 1 calculates y directly on the linear (untransformed) scale using the model output A (which we included in the model as a derived quantity). Method 2 calculates log(y) using log(A), then exponentiates to get the linear scale predictions. Both methods work just fine. In my own work I'd probably use Method 1 unless I had a compelling reason not to.

4.5.2.3.1 Method 1: predict y on linear scale

```
# matrix of coefficients
prx <- mod13$BUGSoutput$sims.matrix
## don't need deviance or sigma
prx <- prx[,c("A", "B")]
post.n <- nrow(prx)

# x values at which to predict
pred.n <- 100
px <- seq(min(dx$x), max(dx$x), length=pred.n)

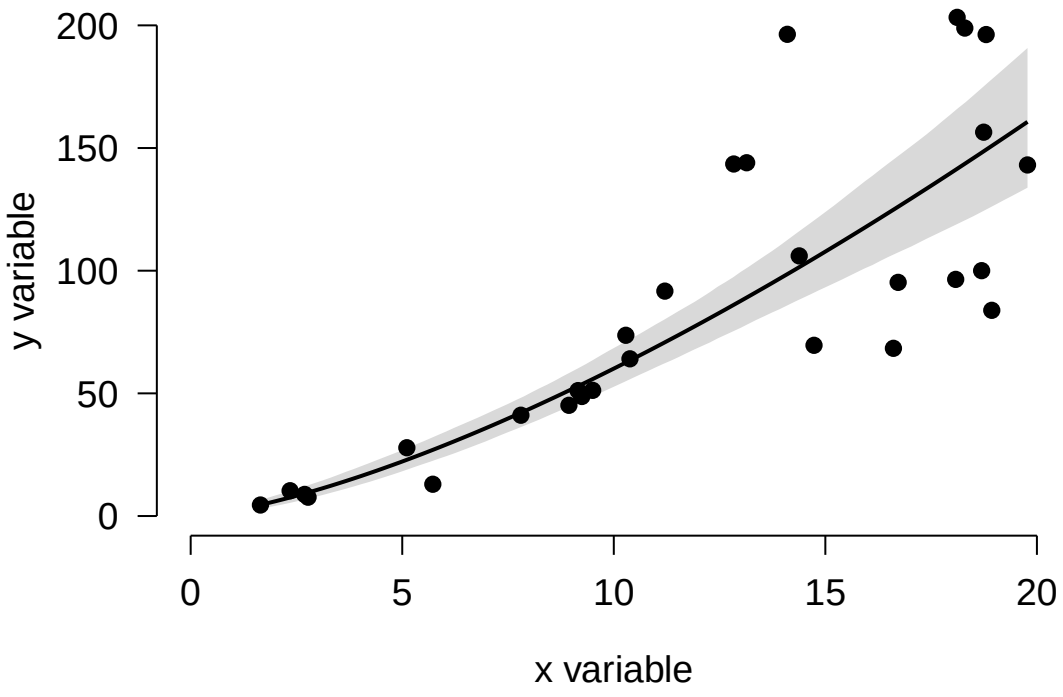
# set up matrix
pred.mat <- matrix(NA, nrow=pred.n, ncol=post.n)

# fill in a loop
for(i in 1:post.n){
    pred.mat[,i] <- prx[i,"A"] * px^prx[i,"B"]
}#i

# set up data frame with mean and CRI
# NOTE: don't need to exponentiate, because we
#       have linear-scale A and can thus
#       calculate linear-scale y directly.
plotdf <- data.frame(x=px,
    mn=apply(pred.mat, 1, mean),
    lo=apply(pred.mat, 1, quantile, 0.025),
```

```
    up=apply(pred.mat, 1, quantile, 0.975))

# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
# make the plot and set basic options
plot(dx$x, dx$y, type="n",
    ylim=c(0, 200), xlim=c(0, 20),
    xlab="x variable", ylab="y variable")
# make a polygon for the CRI
polygon(x=c(px, rev(px)), y=c(plotdf$lo, rev(plotdf$up)),
    border=NA, col="grey85")
# predicted mean
points(plotdf$x, plotdf$mn, type="l", lwd=2)
# put original data on top, so not covered by
# lines and etc.
points(dx$x, dx$y, pch=16, cex=1.2)
```



4.5.2.3.2 Method 2: predict y on log scale

Oops, we forgot to calculate the linear scale A in our model and don't feel like going back to re-run the model. That's fine, we can just generate predictions on the log-log scale and backtransform . Very little in the code

below is changed from Method 1: the prediction equation for `pred.mat[,i]` is linear, and the results are exponentiated with `exp()`.

```
# matrix of coefficients
prx <- mod13$BUGSoutput$sims.matrix
## don't need deviance or sigma
prx <- prx[,c("A", "B")]

## OOPS, we forgot to get linear-scale A
## in the model, and don't feel like
## re-running it. This line will get us
## the log-scale A posteriors.
prx[,"A"] <- log(prx[,"A"])

post.n <- nrow(prx)

# x values at which to predict
pred.n <- 100
px <- seq(min(dx$x), max(dx$x), length=pred.n)

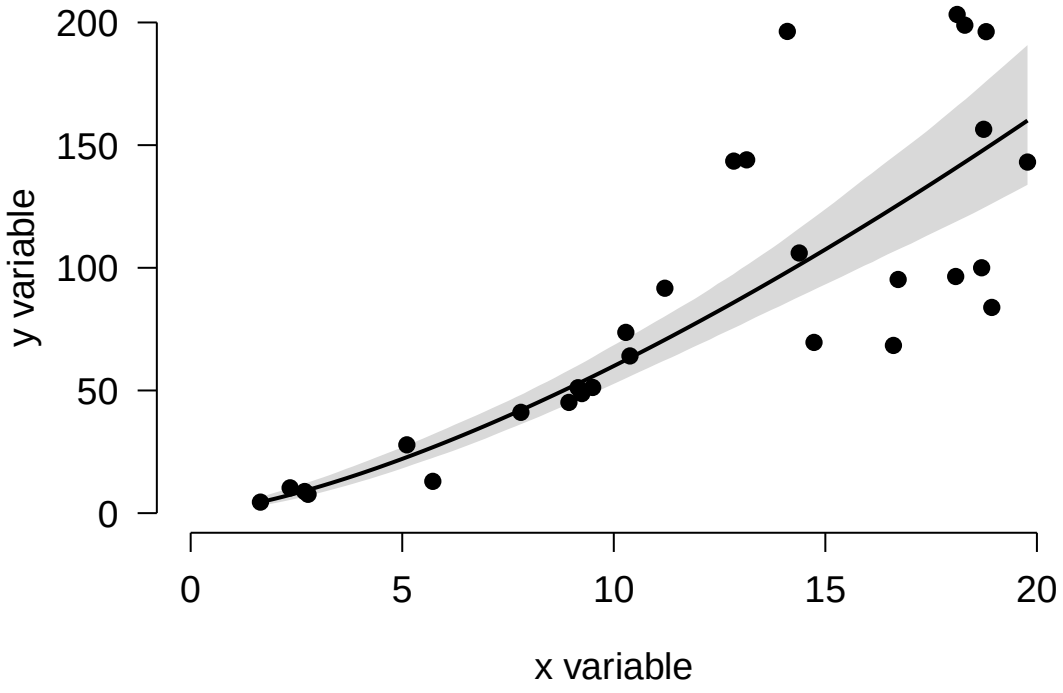
# set up matrix
pred.mat <- matrix(NA, nrow=pred.n, ncol=post.n)

# fill in a loop
# NOTE: we are predicting on log-log scale,
#       so don't forget to log-transform x!
for(i in 1:post.n){
    pred.mat[,i] <- prx[i,"A"] + log(px) * prx[i,"B"]
}#i

# set up data frame with mean and CRI
# NOTE: need to exponentiate, because
#       predictions are on log scale.
plotdf <- data.frame(x=px,
    mn=exp(apply(pred.mat, 1, mean)),
    lo=exp(apply(pred.mat, 1, quantile, 0.025)),
    up=exp(apply(pred.mat, 1, quantile, 0.975)))

# fancy plot options with par()
par(mfrow=c(1,1), mar=c(5.1, 5.1, 1.1, 1.1),
    bty="n", lend=1,
    las=1, cex.axis=1.2, cex.lab=1.2)
```

```
# make the plot and set basic options
plot(dx$x, dx$y, type="n",
    ylim=c(0, 200), xlim=c(0, 20),
    xlab="x variable", ylab="y variable")
# make a polygon for the CRI
polygon(x=c(px, rev(px)), y=c(plotdf$lo, rev(plotdf$up)),
    border=NA, col="grey85")
# predicted mean
points(plotdf$x, plotdf$mn, type="l", lwd=2)
# put original data on top, so not covered by
# lines and etc.
points(dx$x, dx$y, pch=16, cex=1.2)
```



The plot from Method 2 is exactly the same as the plot for Method 1. This illustrates the freedom that comes with understanding your model and the various ways it can be expressed.

4.6 Conclusions

Now that you have seen the ways we fit linear models in JAGS, you should have both a better understanding of linear models and Bayesian model fitting. In particular, wrapping your head around the ANCOVA models should set you up to understand one of the more advanced topics in this course, mixed models. It only takes one subtle change to convert an ANCOVA into a mixed model.

Likewise, understanding how the linear regression model was put together will help you understand the generalized linear model (GLM). The GLM generalized the linear model by allowing the response distribution to be something other than the normal, and allowing a **link** function that relates the scale of the response variable and the scale of the deterministic part of the model. Both of these changes are very straightforward in JAGS…much more so than in R!

Chapter 5

GLM for count data

Over the next two modules we will extend our JAGS language knowledge and our statistical skills to another important class of statistical methods: **generalized linear models (GLM)**. GLMs are an indispensable tool in the modern biologist's toolbox. The GLM framework includes all linear models, as well as a wide variety of other techniques:

- Gamma GLM for overdispersed continuous data
- Poisson and negative binomial models for counts
- Logistic regression for binary outcomes
- Binomial regression for proportional data
- And many more.

These models may appear different on the surface, and take different kinds of data as inputs, but are actually the same kind of model. This is much more apparent when fitting models in JAGS than in R.

This module explores GLMs for **count** data. Count data are exactly what they sound like: values resulting from counting things. Count data take the form of non-negative integers. Non-negative integers, however, are not necessarily count data. For example, summer temperatures in Georgia can be rounded to the nearest integer, but they will not be counts.

> **i** About this module
>
> For this module you will need a recent versions of R, JAGS, and the R packages rjags and R2jags (Plummer (2003), Plummer (2021), R Core Team (2025), Su and Masanao Yajima (2021)). This module and all code examples were developed and tested using R version 4.5.1, JAGS 4.3.1, rjags version 4-17, and R2jags version 0.8-9.

5.1 GLM refresher

Generalized linear models (GLM) extend the linear model by relaxing two assumptions:

1. GLM allows response distributions other than the normal distribution
2. GLM allows the linear predictor to be on a scale other than the scale that the data are on; the data and linear predictor scales are related by a link function.

So, while a linear model is written as:

$$\text{Stochastic part}: y \sim \text{Normal}\left(E\left(y\right), \sigma_{res}\right)$$

$$\text{Linear predictor}: E\left(y\right) = \beta\mathbf{x}$$

Here we use matrix shorthand for the linear predictor to indicate that β is a vector of coefficients (β_0, β_1, β_2, and so on), and **x** is the design matrix that contains the predictor variables (x_1, x_2, and so on; see **?@sec-linear-matrix**).

A GLM is written as:

$$\text{Stochastic part}: y \sim \text{SomeDistribution}\left(E\left(y\right), \theta\right)$$

$$\text{Linear predictor}: \eta = \beta\mathbf{x}$$

where θ stands in for other parameters of the response distribution (if there are any), and η ("eta") is the expected value on the scale of the linear predictor–which may or may not be the same scale as y and $E\left(y\right)$. We relate $E\left(y\right)$ and η via a **link function**:

$$\text{Link function}: f\left(E\left(y\right)\right) = \eta$$

> **i** GLM response distributions
>
> Technically, the response distribution in a GLM must be one of the exponential family. This family includes a huge variety: the normal, exponential, gamma, beta, Poisson, and geometric. Under certain conditions, the binomial, negative binomial, and multinomial can be included as well.

This means that linear regression itself is a special case of GLM, with a normal response distribution and the identity link function (the identity function of any value is that value itself).

$$\text{Stochastic part}: y \sim \text{Normal}\left(E\left(y\right), \sigma_{res}\right)$$

$$\text{Link function}: E\left(y\right) = \eta$$

$$\text{Linear predictor}: \eta = \beta\mathbf{x}$$

If you are unfamiliar with GLM, check out the notes to my data anlysis course for a more in-depth explanation.

5.2 Poisson GLM for counts

5.2.1 Definition

The Poisson GLM models the logarithm of expected count λ as a linear function of predictors $\beta\mathbf{x}$. Here we use matrix shorthand for the linear predictor to indicate that β is a vector of coefficients (β_0, β_1, β_2, and so on), and **x** is the design matrix that contains the predictor variables (x_1, x_2, and so on). Instead of saying that the Poisson GLM "models the logarithm of the expected count", we could also say that it uses a log link function.

$$\text{Stochastic part}: y \sim Poisson\left(\lambda\right)$$

$$\text{Link function}: \log\left(\lambda\right) = \eta$$

$$\text{Linear predictor}: \eta = \beta\mathbf{x}$$

Notice that the Poisson distribution has only one parameter, λ ("lambda"), which is both the mean and variance of the distribution. This means that we should not be surprised or concerned if a scatterplot suggests heteroscedasticity with variance increasing with the expected value. This property of the Poisson distribution and the log link function means that a "cone" of points on a scatterplot does not necessarily imply heteroscedasticity or overdispersion. **Overdispersion** means that the data contain more variance than would be expected under a Poisson distribution. If overdispersion is suspected, you can try a negative binomial GLM instead of a Poisson GLM (Section 5.3).

5.2.2 Example

5.2.2.1 Data simulation

Let's simulate a dataset and analyze it.

```
# random number seed for reproducibility
set.seed(123)

# sample size
n <- 100

# coefficients
beta0 <- 1.5
beta1 <- 0.2
```

```r
# x values
x <- runif(n, 0, 10)

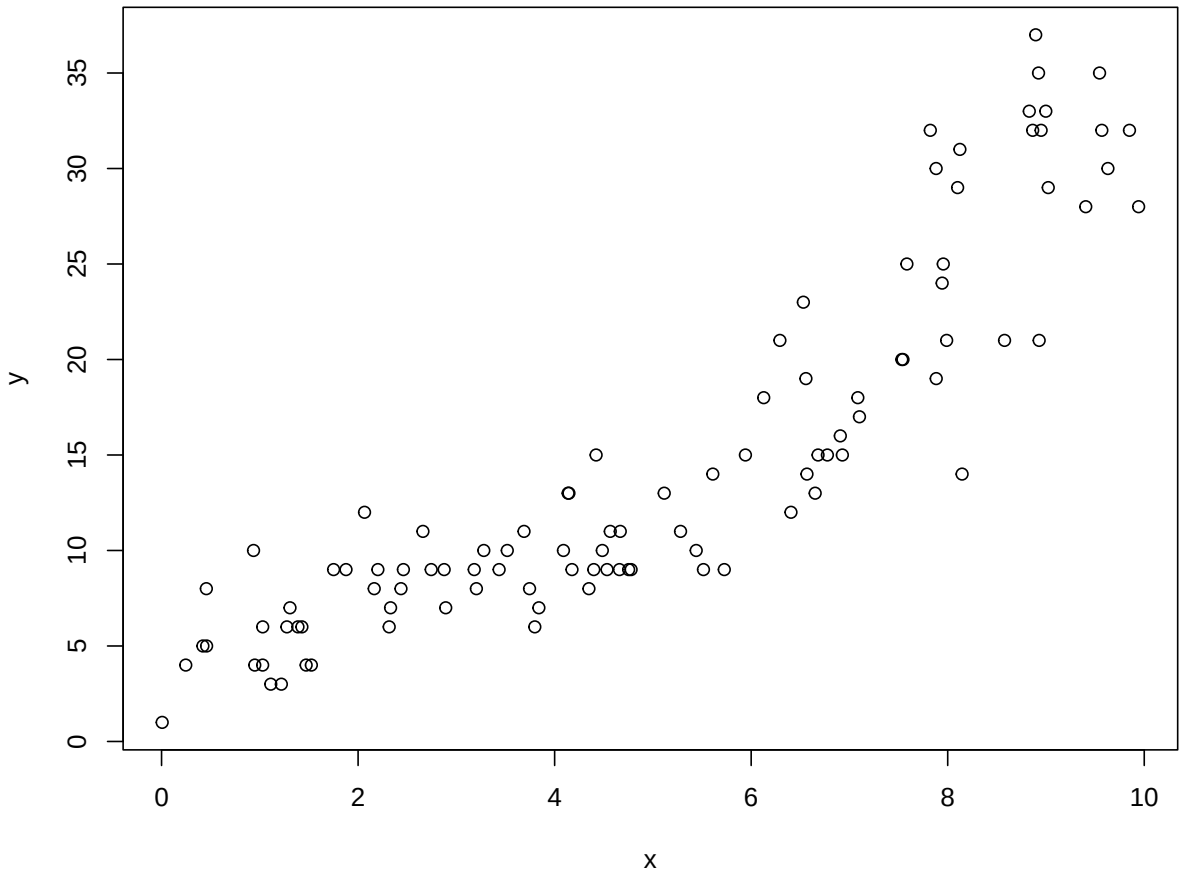
# linear predictor
eta <- beta0 + beta1 * x

# inverse link function
lambda <- exp(eta)

# response variable
y <- rpois(n, lambda)

# put everything in a data frame
dx <- data.frame(x=x,y=y)

# make the plot
par(mar=c(5.1, 5.1, 1.1, 1.1))
plot(x,y)
```



The relationship between y and x looks almost linear, but because we are clever biologists, we know that linear models are not appropriate for two reasons. First, linear models predict continuous responses, and counts are not continuous (they are discrete). Second, linear models can predict negative responses, and counts cannot be negative. So, we will try Poisson regression.

5.2.2.2 Frequentist analysis

We could fit a Poisson GLM of y vs. x in R with this command:

```
mod01 <- glm(y~x, data=dx, family=poisson)
# model summary:
summary(mod01)
```

```
Call:
glm(formula = y ~ x, family = poisson, data = dx)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.46480    0.07144    20.5   <2e-16 ***
x            0.21025    0.01011    20.8   <2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 552.212  on 99  degrees of freedom
Residual deviance:  75.704  on 98  degrees of freedom
AIC: 513.17

Number of Fisher Scoring iterations: 4
```

The frequentist analysis gets pretty close to the real coefficients, and the model reduced the deviance by an impressive 86%.

```
1-mod01$deviance/mod01$null.deviance
```

```
[1] 0.8629071
```

5.2.2.3 Bayesian analysis with JAGS

Now let's repeat the analysis in JAGS. The analysis takes a little more work in JAGS, but it might be worth it.

First, define a model file for the Poisson GLM.

```
# Poisson GLM with one predictor x
mod.name <- "jags-glm-poisson.txt"
sink(mod.name)
cat("
    model{
        # priors
        # reasonable ranges...extreme values may cause
        # numerical overflow. Use some number sense!
        beta0 ~ dnorm(0, 1)
        beta1 ~ dnorm(0, 1)

        # likelihood
        for(i in 1:n){
            y[i] ~ dpois(lambda[i])
            log(lambda[i]) <- beta0 + beta1 * x[i]
        }#i
    }#model
", fill=TRUE)
sink()
```

Then, set up the usual MCMC parameters for JAGS and tell JAGS what parameters to monitor.

```
# parameters to monitor
params <- c("beta0", "beta1")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3
```

As always, we need a function to define initial values for the MCMC chains. Remember that EVERY parameter in the model defined as a stochastic node needs an initial value. This means any variable that is ever on the left side of a ~ symbol.

```
# function to define initial values for MCMC chains
init.fun <- function(nc){
  # output will be a list with one element per chain
  res <- vector("list", length=nc)

  # fill in a for loop over chains
  for(i in 1:nc){
    # note that
    res[[i]] <- list(beta0=rnorm(1, 0, 1),
                     beta1=rnorm(1, 0, 1))
  }#i
  return(res)
}

# generate the intial values
inits <- init.fun(nchains)
```

Package up the data for JAGS

```
# package data for JAGS
in.data <- list(x=dx$x, y=dx$y, n=n)
```

And run the model.

```
library(R2jags)
library(rjags)
```

```
mod02 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
```

```
                n.chains=nchains, n.iter=n.iter,
                n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 100
   Unobserved stochastic nodes: 2
   Total graph size: 505

Initializing model
```

And inspect the results:

```
mod02
```

```
Inference for Bugs model at "jags-glm-poisson.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 41.21 secs
          mu.vect sd.vect    2.5%     25%     50%     75%   97.5%  Rhat n.eff
beta0       1.456   0.07   1.327   1.410   1.455   1.500   1.596 1.001  1200
beta1       0.211   0.01   0.192   0.205   0.212   0.217   0.230 1.000  1200
deviance  511.122   2.01 509.212 509.699 510.528 511.821 516.818 1.000  1200

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 2.0 and DIC = 513.1
DIC is an estimate of expected predictive error (lower deviance is better).
```

The model converged, and the estimated values for `beta0` and `beta1` are pretty close to the truth…and, to the estimates from the frequentist version of the model.

Just as with linear models, the best way to present the results of your Poisson GLM is to plot the model predictions against the data. Plotting the expected value λ against the data is easy. What's tricky is plotting the uncertainty. We will plot uncertainty as the 95% credible interval (CRI) of the predicted value λ. Unless you think carefully about what you're doing, it's easy to confuse plotting uncertainty about the expected values with plotting uncertainty about the observable values. In frequentist terms, the former is a confidence interval (CI) and the latter is a prediction interval (PI).

```
# set up x values at which to predict
pn <- 100
px <- seq(min(dx$x), max(dx$x), length=pn)
```

```
prx <- data.frame(x=px)

# extract permuted posteriors for
# model parameters
mx <- mod02$BUGSoutput$sims.matrix
npost <- nrow(mx)

# make matrix to hold model predictions
pred <- matrix(NA, nrow=nrow(prx), ncol=nrow(mx))

# fill matrix in a loop
for(i in 1:npost){
    pred[,i] <- mx[i,"beta0"] + mx[i,"beta1"] * prx$x
}#i

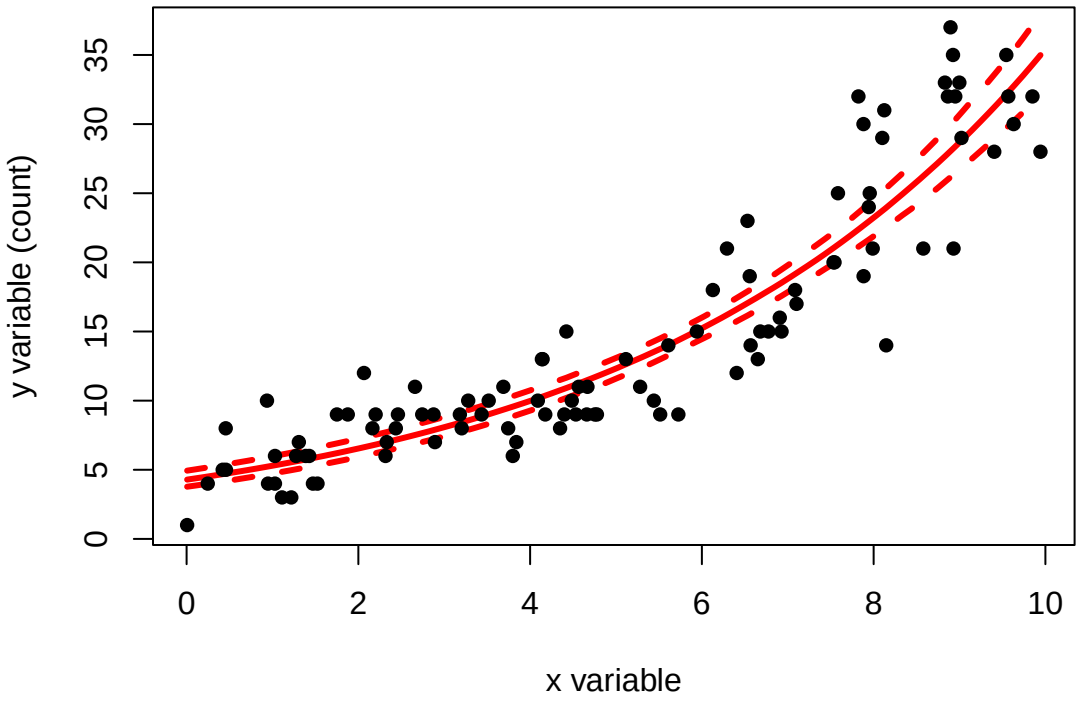
# summarize across columns of pred to
# get predicted values and 95% credible interval
prx$lo <- apply(pred, 1, quantile, 0.025)
prx$mn <- apply(pred, 1, quantile, 0.5)
prx$up <- apply(pred, 1, quantile, 0.975)
```

The values in `prx` are still on the link scale, where the linear predictor lives. So, we need to apply the inverse link function to get values on the data scale.

```
prx$lo <- exp(prx$lo)
prx$mn <- exp(prx$mn)
prx$up <- exp(prx$up)
```

Now we can construct our plot:

```
par(mar=c(5,5,1,1))
plot(dx$x, dx$y, type="n",
    xlab="x variable",
    ylab="y variable (count)")
points(prx$x, prx$mn, type="l", lwd=3, col="red")
points(prx$x, prx$lo, type="l", lwd=3, col="red", lty=2)
points(prx$x, prx$up, type="l", lwd=3, col="red", lty=2)
points(dx$x, dx$y, pch=16)
```



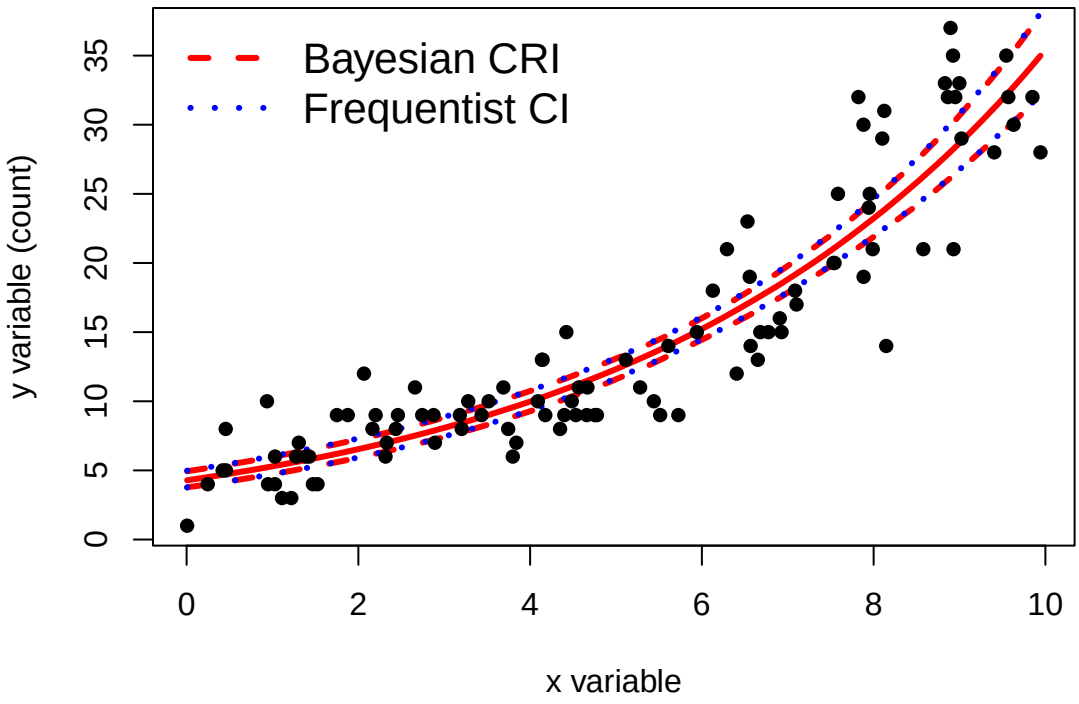
The interval plotted is the 95% **credible interval (CRI)**, not a confidence interval. Remember that a credible interval is calculated from samples of the posterior distribution, not estimated from a fitted function like a frequentist confidence interval (CI) (see Section 1.4.6). That means we can say, "we are 95% certain that the true expected value lies within the interval shown by the dashed lines.

A CRI is often similar in width to a CI, but they are not the same.

```
# compare to CI
predf <- predict(mod01, newdata=prx,
                 type="link", se.fit=TRUE)
prx$lof <- exp(qnorm(0.025, predf$fit, predf$se.fit))
prx$mnf <- exp(qnorm(0.5, predf$fit, predf$se.fit))
prx$upf <- exp(qnorm(0.975, predf$fit, predf$se.fit))


par(mar=c(5,5,1,1))
plot(dx$x, dx$y, type="n",
    xlab="x variable",
    ylab="y variable (count)")
points(prx$x, prx$mn, type="l", lwd=3, col="red")
points(prx$x, prx$lo, type="l", lwd=3, col="red", lty=2)
points(prx$x, prx$up, type="l", lwd=3, col="red", lty=2)
points(prx$x, prx$lof, type="l", lwd=3, col="blue", lty=3)
```

```
points(prx$x, prx$upf, type="l", lwd=3, col="blue", lty=3)
points(dx$x, dx$y, pch=16)
legend("topleft",
       legend=c("Bayesian CRI", "Frequentist CI"),
       lty=c(2,3), lwd=3,
       col=c("red", "blue"),
       bty="n", cex=1.3)
```



5.2.2.4 Bayesian analysis with brms

JAGS is a powerful engine for fitting models, and I love JAGS (and its spiritual predecessor, WinBUGS), but it is not the only option. Let's use a newer package, `brms`. It has the advantage of being able to fit models using a formula interface similar to the standard R `lm()` or `glm()` functions. This has the side effect of being able to fit models without a lot of the programming overhead needed for JAGS.

You will need the package `brms`. Load it and set the backend to use stan as its engine. Using the data frame `dx` from earlier:

```
library(brms)
options(brms.backend = "cmdstanr")
```

```
# random number seed for reproducibility
set.seed(123)
```

```r
# sample size
n <- 100

# coefficients
beta0 <- 1.5
beta1 <- 0.2

# x values
x <- runif(n, 0, 10)

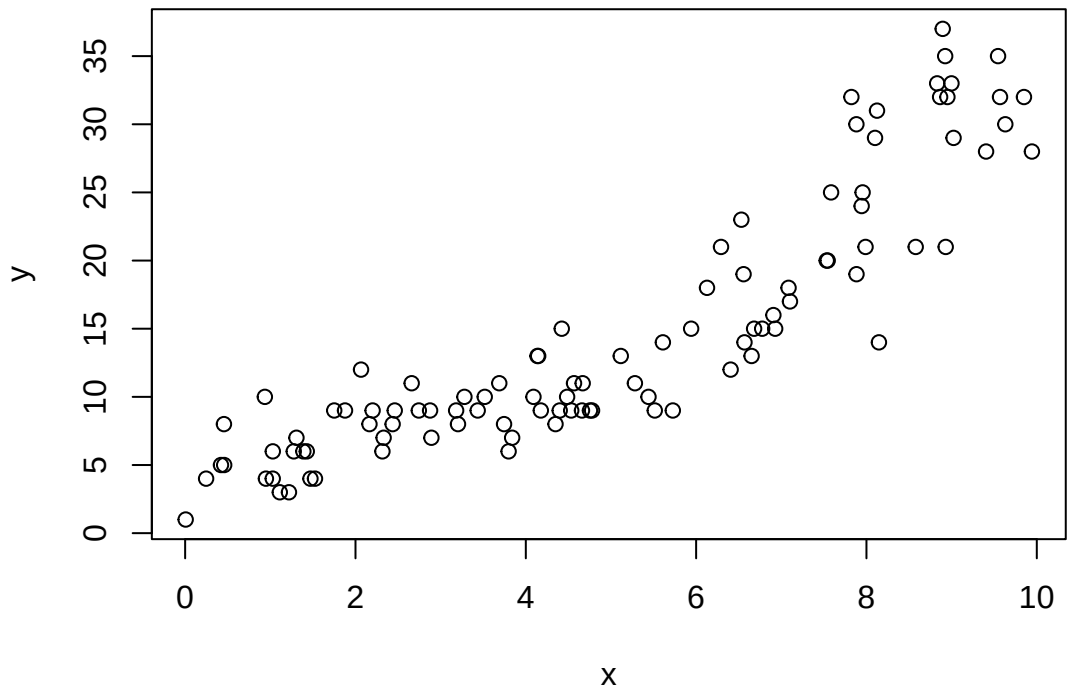
# linear predictor
eta <- beta0 + beta1 * x

# inverse link function
lambda <- exp(eta)

# response variable
y <- rpois(n, lambda)

# put everything in a data frame
dx <- data.frame(x=x,y=y)

# make the plot
par(mar=c(5.1, 5.1, 1.1, 1.1))
plot(x,y)
```



Now let's fit the model. This step might take a minute or two as the stan model compiles in the background…but to be fair, it would have taken longer than that to write all the JAGS code!

```
# Fit Bayesian Poisson GLM
mod03 <- brm(
  y ~ x,
  data = dx,
  family = poisson(link = "log"),
  prior = c(
    prior(normal(0, 2), class = "Intercept"),
    prior(normal(0, 1), class = "b")
  ),
  chains = 3,
  iter = 2000,
  warmup = 1000,
  seed = 123
)
```

```
Running MCMC with 3 sequential chains...

Chain 1 Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1 Iteration:  100 / 2000 [  5%]  (Warmup)
Chain 1 Iteration:  200 / 2000 [ 10%]  (Warmup)
```

```
Chain 1 Iteration:  300 / 2000 [ 15%]  (Warmup)
Chain 1 Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1 Iteration:  500 / 2000 [ 25%]  (Warmup)
Chain 1 Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1 Iteration:  700 / 2000 [ 35%]  (Warmup)
Chain 1 Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1 Iteration:  900 / 2000 [ 45%]  (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%]  (Sampling)
Chain 1 Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1 Iteration: 1300 / 2000 [ 65%]  (Sampling)
Chain 1 Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1 Iteration: 1500 / 2000 [ 75%]  (Sampling)
Chain 1 Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1 Iteration: 1700 / 2000 [ 85%]  (Sampling)
Chain 1 Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%]  (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1 finished in 0.2 seconds.
Chain 2 Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2 Iteration:  100 / 2000 [  5%]  (Warmup)
Chain 2 Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2 Iteration:  300 / 2000 [ 15%]  (Warmup)
Chain 2 Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2 Iteration:  500 / 2000 [ 25%]  (Warmup)
Chain 2 Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2 Iteration:  700 / 2000 [ 35%]  (Warmup)
Chain 2 Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2 Iteration:  900 / 2000 [ 45%]  (Warmup)
Chain 2 Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2 Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 2 Iteration: 1100 / 2000 [ 55%]  (Sampling)
Chain 2 Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2 Iteration: 1300 / 2000 [ 65%]  (Sampling)
Chain 2 Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%]  (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%]  (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%]  (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%]  (Sampling)
```

```
Chain 2 finished in 0.2 seconds.
Chain 3 Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%]  (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3 Iteration:  300 / 2000 [ 15%]  (Warmup)
Chain 3 Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3 Iteration:  500 / 2000 [ 25%]  (Warmup)
Chain 3 Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3 Iteration:  700 / 2000 [ 35%]  (Warmup)
Chain 3 Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3 Iteration:  900 / 2000 [ 45%]  (Warmup)
Chain 3 Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3 Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3 Iteration: 1100 / 2000 [ 55%]  (Sampling)
Chain 3 Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3 Iteration: 1300 / 2000 [ 65%]  (Sampling)
Chain 3 Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3 Iteration: 1500 / 2000 [ 75%]  (Sampling)
Chain 3 Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3 Iteration: 1700 / 2000 [ 85%]  (Sampling)
Chain 3 Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%]  (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3 finished in 0.2 seconds.

All 3 chains finished successfully.
Mean chain execution time: 0.2 seconds.
Total execution time: 1.0 seconds.
```

The first time you run the model, R will have to compile the model into stan. This might take a few minutes. After compilation, the MCMC sampling begins. If everything works, the model object will be returned and you can inspect the output.

```
# see the posterior summary
summary(mod03)
```

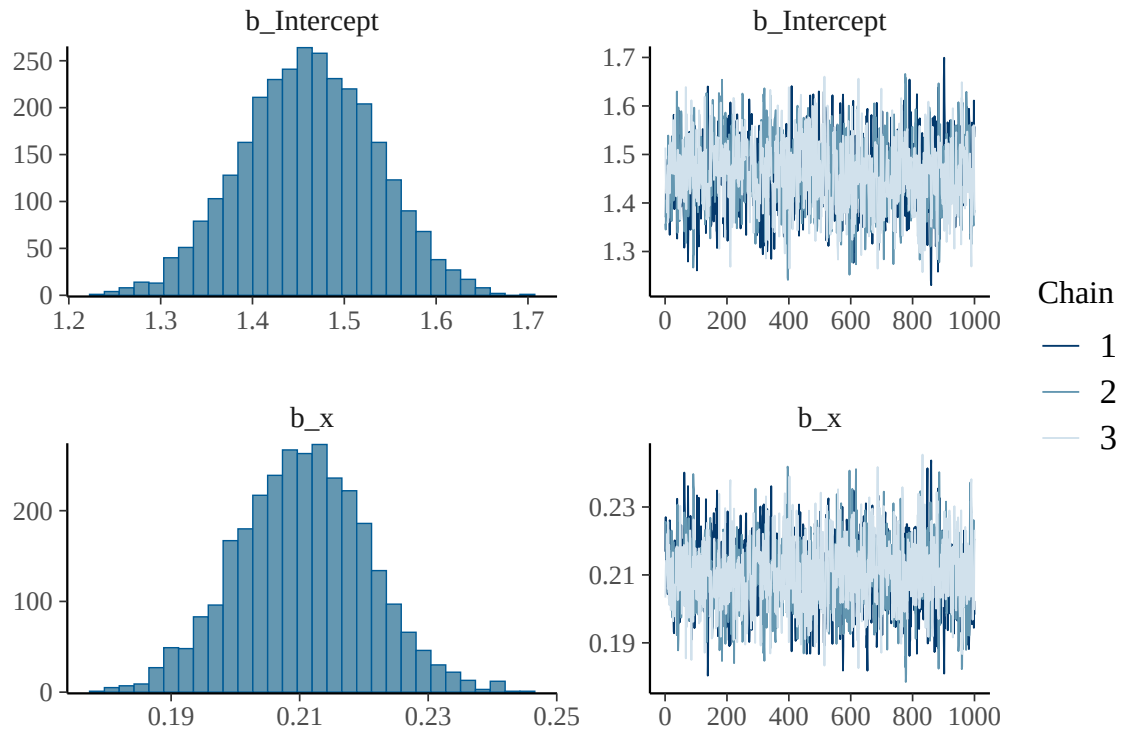
```
 Family: poisson
  Links: mu = log
Formula: y ~ x
   Data: dx (Number of observations: 100)
  Draws: 3 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 3000
```

```
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     1.46      0.07     1.32     1.60 1.00     1071     1612
x             0.21      0.01     0.19     0.23 1.00     1305     1787

Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

And plot the results. The default `plot` method for a `brm` output will include a histogram of the posteriors and a trace plot for each parameter.

```
plot(mod03)
```



Getting predicted values in `brms` is easier than in JAGS and more like a `lm` or `glm` fit.

```
# make a data frame with a range of
# x values covering the observed x
px <- seq(min(dx$x), max(dx$x), length=100)
prx <- data.frame(x=px)

# obtain fitted values with fitted
pred <- fitted(mod03, newdata=prx)
```

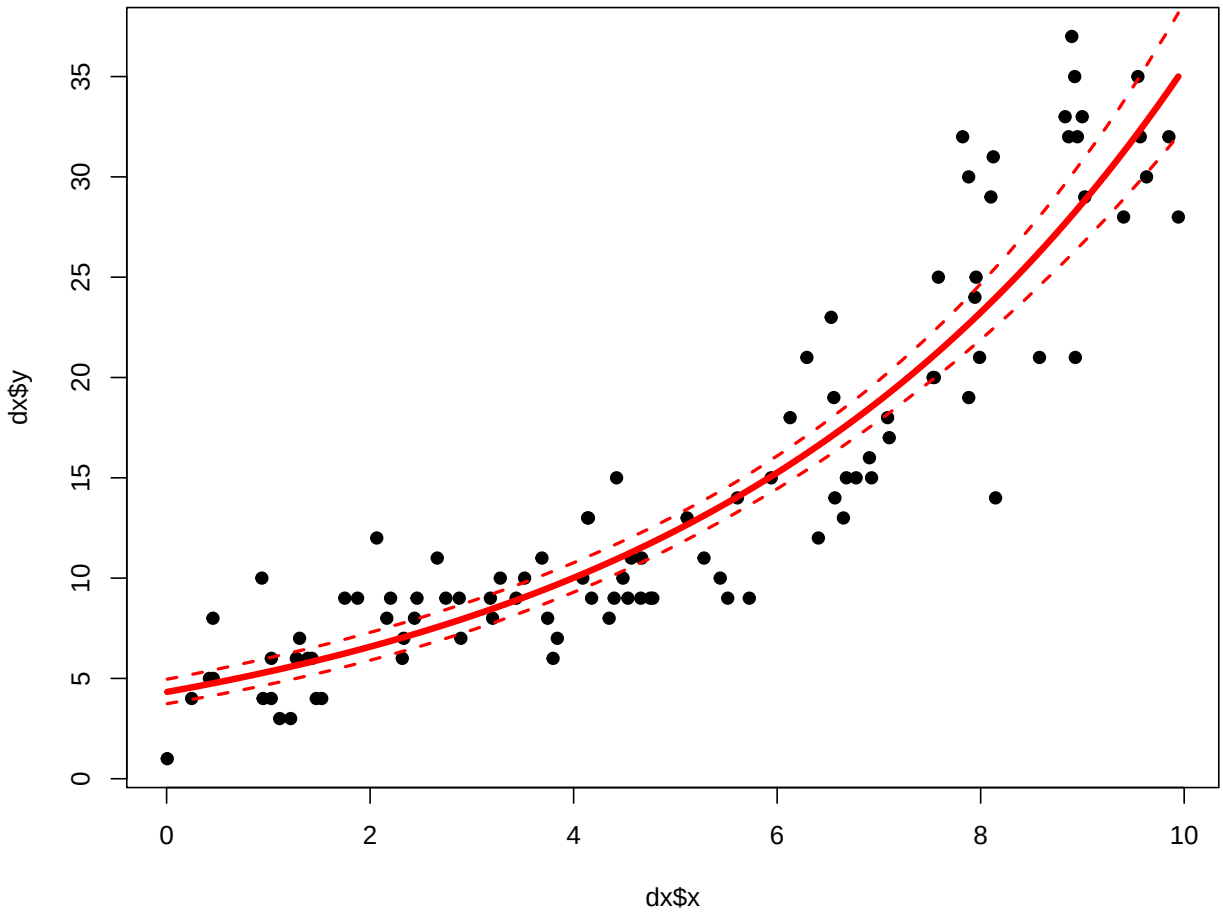
This returns object `pred`, a matrix with four columns: the posterior mean, posterior SD, and the 95%

credible interval limits. Combine these with the new data frame.

```
prx <- cbind(prx, pred)
```

Now we can use base R graphics to construct our figure:

```
par(mar=c(4,4,1,1))
plot(dx$x, dx$y, pch=19)
lines(prx$x, prx$Estimate, lwd=4, col="red")
lines(prx$x, prx$Q2.5, lwd=2, lty=2, col="red")
lines(prx$x, prx$Q97.5, lwd=2, lty=2, col="red")
```



The difference here can be a bit confusing. In `brms`, `fitted()` returns the posterior expected response and its 95% credible interval. This reflects uncertainty in the estimated mean response due to uncertainty in the model parameters. This is what we usually get from `predict.lm()` or `predict.glm()` with `se.fit=TRUE, interval="confidence"`.

In `brms`, `predict()` returns the posterior predictive distribution of a new observation and its 95% posterior predictive interval. This includes both parameter uncertainty and residual/sampling variation. This is equivalent to `predict.lm()` or `predict.glm()` with `se.fit=TRUE, interval="prediction"`.

5.2.3 A note on numeracy (number sense)

It may have been tempting to put a very wide prior on the intercept and slope, like we did for coefficients in a linear model. But, this can be a bad idea when working on the log scale. In the linear models in the last module, we usually used priors like $Normal\,(0,\,100)$ for `beta0` and `beta1` terms. This was fine for linear models, where the response variable and linear predictor were on the same scale. For our Poisson GLM, however, the parameters are on the logarithmic scale because of the log link function. Because the linear predictor ($\beta_0 + \beta_1 x$) is exponentiated in the model likelihood, calculations like the following might occur:

$$\log\left(\lambda\right) = 200 + 1500x$$

which really means

$$\lambda = e^{200+1500x}$$

which even for a small value like $x = 1$ would evaluate to an absurdly large number, too large to be represented in R or JAGS. R can represent values with an absolute value up to about 1.79×10^{308}. Anything larger returns `Inf` (or `-Inf`, for negative numbers).

```
.Machine$double.xmax
```

```
[1] 1.797693e+308
```

which is approximately 10^{228} times greater than the number of atoms in the universe. Similarly, R can only represent numbers as small as about 2.2×10^{-308}. So, even if R and JAGS could represent those numbers, would they really be relevant to your analysis? Probably not.

Any of these could cause inappropriately large or small values of □_i within JAGS. Values that are too large cannot be represented in JAGS and may be treated as infinity…which will cause an error. Values that are too small will be arbitrarily close to 0, and may be treated as 0…which will also cause an error. When you try to feed a JAGS distribution function with an illegitimate input, such as a negative or 0 value for `dpois()`, you will get the dreaded "Node inconsistent with parents" error. This error can be very hard to track down, because it is not possible to see current parameter values within JAGS. This means you have to examine each node and work out if extreme values of any of its inputs could lead to illegitimate values.

In JAGS, the expected value of a Poisson distribution must be finite and strictly positive. So, we need to be careful to set up the model in such a way as to avoid these kinds of numerical issues.

There are two takeaways from this digression. First, you need to be smart about what numbers you are implying in your JAGS models. Your model might have the correct relationships, but if the numbers lead to illegitimate values, it is all for naught. Try rewriting the model above with the extremely broad priors and see what happens.

Second, sometimes you need to think like a programmer instead of a biologist or statistician. There's an old saying: "If it's stupid and its works, then it's not stupid." Many people might opt to relate `x` and `y` with a

construction like this in their JAGS code:

```
# stochastic part as usual
y[i] ~ dpois(lambda[i])

# limit lambda to [0.0001, 9999]
lambda[i] <- max(0.0001, min(9999, exp(eta[i])))

# calculate log scale expected value eta
eta[i] <- exp(beta0 + beta1 * x[i])
```

What does this do? By separating the value of the linear predictor `eta[i]` from the expected value of the Poisson distribution `lambda[i]`, this code allows us to limit the values that go into `dpois()`. This practice is less elegant than choosing appropriate prior values, but it's not wrong. In fact, this practice is explicitly recommended for some GLMs by Kéry (2010). We'll use it below when we get to logistic regression (see Section 6.1).

Some might argue that putting this kind of limit in your JAGS model might bias the parameter estimates. In my experience, however, this is not the case. As long as you don't make the acceptable interval too narrow, the MCMC algorithm will probably still find the right solution. The only issue I've ever had is when the "true" coefficient value might be very close to one of the limits. In that case, the parameter estimates can be biased high if the correct value is just above a limit, or biased low if the correct value is just below a limit.

Another option might be to simply add a small constant to λ. This would solve the problem of λ that were too small, but not the problem of λ that were too large. Personally, I prefer the first solution, because the estimated `beta0` and `beta1` are for the expected value, not the expected value minus a small constant.[1]

```
y[i] ~ dpois(lambda[i])
lambda[i] <- eta[i] + 0.00001
eta[i] <- exp(beta0 + beta1 * x[i])
```

5.3 Negative binomial GLM for overdispersed counts

Poisson GLMs assume that the single parameter λ of the Poisson distribution is both the mean and the variance of the response distribution (i.e., that $\lambda = \mu = \sigma^2$). This is not always the case in real datasets. Many biological processes that produce counts have a variance much greater than their mean. This phenomenon is called **overdispersion**. The "over" in "overdispersion" means "more variance than expected for a Poisson distribution with the same mean".

One way to deal with overdispersion is to use a distribution other than the Poisson. Most textbooks recommend either the **quasi-Poisson** or the **negative binomial**. The negative binomial is easier to work with in

[1] I once needed to model a population of deer, which could occur in groups on a landscape. We had a parameter for mean group size, but there was a problem. Using the Poisson distribution for group size would frequently predict groups of 0 deer…in other words, no group! Our solution was to reparametrize the model to work with "group size minus 1". This meant that 0s returned from any Poisson draws would have 1 added to them, so the minimum group size in the model was a biologically reasonable 1. If it's stupid but it works, then it isn't stupid.

most software packages because it has a full likelihood function.

5.3.1 Definition

There are several definitions of the **negative binomial distribution**. The definition biologists usually use has a mean μ or λ and an overdispersion parameter called θ ("theta") or k. Smaller θ leads to more variance; at larger θ the distribution becomes indistinguishable from the Poisson.

$$\sigma^2 \left(\text{NegBin}\right) = \lambda + \frac{\lambda^2}{\theta}$$

In the R functions that deal with the negative binomial, θ is referred to as the `size` parameter. As θ gets smaller, the fraction $\frac{\lambda^2}{\theta}$ gets larger, and the distribution picks up more and more variance compared to the Poisson distribution (because for a Poisson, $\sigma^2 = \lambda$). Thus, as θ increases to positive infinity, the fraction gets smaller and the variance approaches that of a Poisson.

Other people define the negative binomial as a gamma-Poisson mixture . That is, a Poisson distribution with expected value λ, where λ itself varies as a gamma distribution. Both can be obtained in JAGS, which uses an altogether different definition of the negative binomial. In JAGS, the expected value (linear predictor) needs to be converted to a probability parameter π. The `r` parameter here is the overdispersion parameter which is also fit as part of the model.

$$y \sim \text{NegBin}\left(\pi, r\right)$$

$$\pi = \frac{r}{r + \mu}$$

$$\log \mu = \beta \mathbf{x}$$

This multi-step process if fraught with opportunities for numerical errors, including the dreaded "Node inconsistent with parents". Here be dragons.

5.3.2 Example

5.3.2.1 Data simulation

Let's simulate a dataset and analyze it.

```
# random number seed for reproducibility
set.seed(42)

# sample size
n <- 100
```

```
# coefficients
beta0 <- 4      # intercept
beta1 <- 0.15   # slope
theta <- 3      # overdispersion

# x values
x <- runif(n, 0, 10)

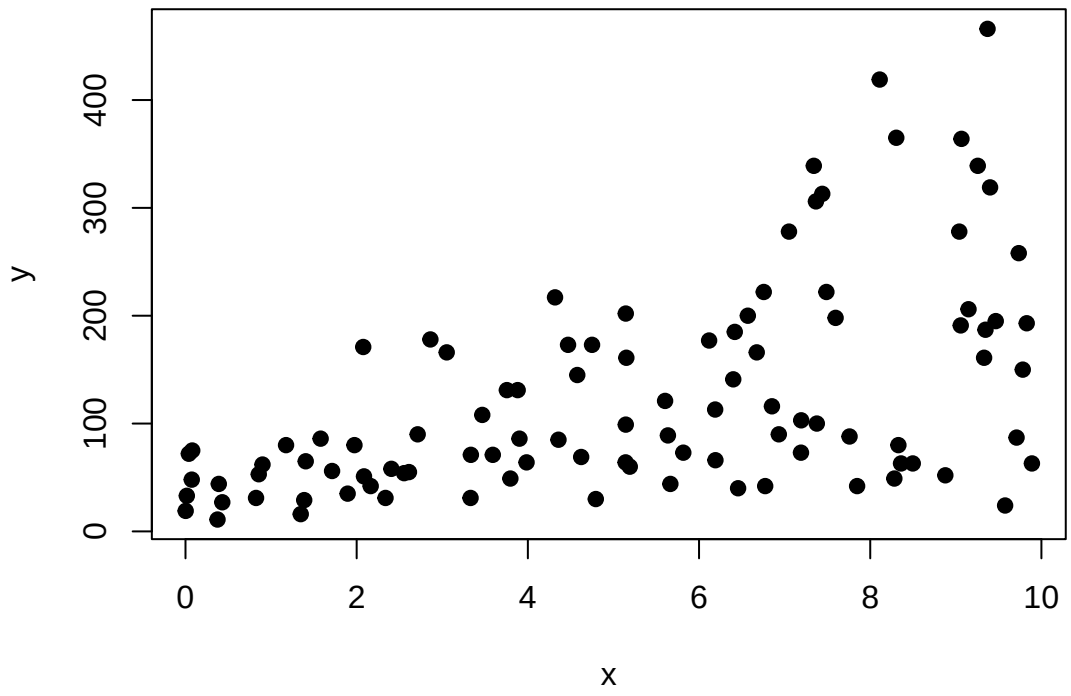
# linear predictor
eta <- beta0 + beta1 * x

# inverse link function
lambda <- exp(eta)

# response variable
y <- rnbinom(n, mu=lambda, size=theta)

# assemble into data frame
dn <- data.frame(x=x, y=y)

# make the plot
par(mar=c(5.1, 5.1, 1.1, 1.1))
plot(x,y, pch=19)
```



5.3.2.2 Frequentist analysis

There is no base R function for a negative binomial GLM, so we need to find an add-on package for it. The most common package for this is probably `MASS`.

```
library(MASS)
```

Now let's fit the negative binomial GLM.

```
mod3 <- glm.nb(y~x, data=dx)
summary(mod3)
```

```
Call:
glm.nb(formula = y ~ x, data = dx, init.theta = 201722.9736,
    link = log)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.46480    0.07145    20.5   <2e-16 ***
x            0.21025    0.01011    20.8   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(201723) family taken to be 1)
```

```
    Null deviance: 552.169  on 99  degrees of freedom
Residual deviance:  75.698  on 98  degrees of freedom
AIC: 515.17

Number of Fisher Scoring iterations: 1

              Theta:  201723
          Std. Err.:  2992532
Warning while fitting theta: iteration limit reached

 2 x log-likelihood:  -509.173
```

The model estimates for the parameters are pretty close to reality.

5.3.2.3 Bayesian analysis with JAGS

The Bayesian version is a little more involved. First, write a model file for the model. Notice the difference from the Poisson model.

```
# Negative binomial GLM with one predictor x
mod.name <- "jags-glm-negbin-v1.txt"
sink(mod.name)
cat("
    model{
        # priors
        beta0 ~ dnorm(0, 1)
        beta1 ~ dnorm(0, 1)
        r ~ dgamma(0.1, 0.1)

        # likelihood
        for(i in 1:n){
            y[i] ~ dnegbin(pi[i],r)
            pi[i] <- r/(r+mu[i])
            log(mu[i]) <- beta0 + beta1 * x[i]

        }#i
    }#model
", fill=TRUE)
sink()
```

Then, set up the usual MCMC parameters for JAGS and tell JAGS what parameters to monitor.

```
# parameters to monitor
params <- c("beta0", "beta1", "r")

# MCMC parameters
n.iter <- 5e4
n.burnin <- 1e4
n.thin <- 100
nchains <- 3
```

As always, we need a function to define initial values for the MCMC chains. Remember that EVERY parameter in the model defined as a stochastic node needs an initial value. This means any variable that is ever on the left side of a ~ symbol.

```
# function to define initial values for MCMC chains
init.fun <- function(nc){
  # output will be a list with one element per chain
  res <- vector("list", length=nc)

  # fill in a for loop over chains
  for(i in 1:nc){
    # note that
    res[[i]] <- list(beta0=rnorm(1, 0, 1),
                     beta1=rnorm(1, 0, 1),
                     r=rgamma(1,
                              shape=0.1,
                              rate=0.1))
  }#i
  return(res)
}

# generate the intial values
inits <- init.fun(nchains)
```

Package up the data for JAGS

```
# package data for JAGS
in.data <- list(x=dx$x, y=dx$y, n=n)
```

And run the model.

```
library(R2jags)
library(rjags)
```

```
mod03 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 100
   Unobserved stochastic nodes: 3
   Total graph size: 707

Initializing model
```

And inspect the results:

```
mod03
```

```
Inference for Bugs model at "jags-glm-negbin-v1.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 91.39 secs
          mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
beta0      1.463   0.079  1.310  1.405  1.463  1.517  1.611 1.004   490
beta1      0.210   0.012  0.188  0.202  0.210  0.218  0.234 1.003   680
r         49.875  16.328 25.530 38.044 47.255 58.720 89.167 1.001  1200
deviance 519.960   3.819 514.307 517.279 519.421 522.102 528.482 1.002   860

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 7.3 and DIC = 527.2
DIC is an estimate of expected predictive error (lower deviance is better).
```

The parameters are pretty close to the truth.

We can also use the Poisson-Gamma mixture definition of the negative binomial, if you prefer that. The model file is very similar, and the initial values structure is the same.

```
# Negative binomial GLM with one predictor x
mod.name <- "jags-glm-negbin-v2.txt"
sink(mod.name)
cat("
```

```
    model{
        # priors
        beta0 ~ dnorm(0, 1)
        beta1 ~ dnorm(0, 1)
        r ~ dgamma(0.1, 0.1)

        # likelihood
        for(i in 1:n){
            y[i] ~ dpois(lambda[i])
            lambda[i] ~ dgamma(r, r/mu[i])
            log(mu[i]) <- beta0 + beta1 * x[i]

        }#i
    }#model
", fill=TRUE)
sink()
```

```
mod04 <- jags(data=in.data, inits=inits,
              parameters.to.save=params,
              model.file=mod.name,
              n.chains=nchains, n.iter=n.iter,
              n.burnin=n.burnin, n.thin=n.thin)#jags
```

```
Compiling model graph
   Resolving undeclared variables
   Allocating nodes
Graph information:
   Observed stochastic nodes: 100
   Unobserved stochastic nodes: 103
   Total graph size: 707

Initializing model
```

And we see essentially the same estimates for `beta0`, `beta1`, and `r`:

```
mod04
```

```
Inference for Bugs model at "jags-glm-negbin-v2.txt", fit using jags,
 3 chains, each with 50000 iterations (first 10000 discarded), n.thin = 100
 n.sims = 1200 iterations saved. Running time = 55.48 secs
      mu.vect sd.vect   2.5%    25%    50%    75%  97.5%  Rhat n.eff
beta0    1.456   0.078  1.307  1.404  1.455  1.510  1.607 1.001  1200
beta1    0.211   0.011  0.189  0.204  0.211  0.219  0.234 1.001  1100
r       48.984  15.641 25.861 38.008 46.465 57.515 86.794 1.001  1200
```

```
deviance 502.220   7.444 487.522 497.356 502.270 507.038 517.458 1.011   180

For each parameter, n.eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor (at convergence, Rhat=1).

DIC info (using the rule: pV = var(deviance)/2)
pV = 27.4 and DIC = 529.7
DIC is an estimate of expected predictive error (lower deviance is better).
```

Producing the figure with model predictions and 95% CRI for the negative binomial model is similar to the procedure for the Poisson model.

```
# set up x values at which to predict
pn <- 100
px <- seq(min(dx$x), max(dx$x), length=pn)

# get posterior draws
mx <- mod04$BUGSoutput$sims.matrix

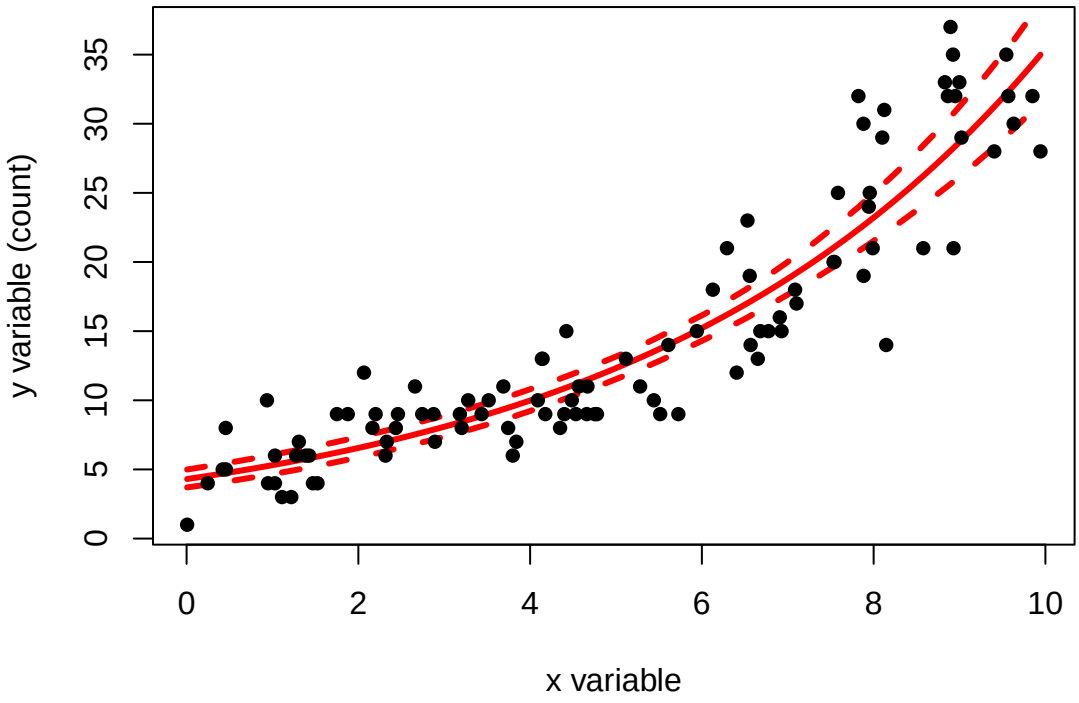
# Matrices: rows = posterior draws, columns = x values
mu <- exp(outer(mx[, "beta1"], px, "*") + mx[, "beta0"])

# Simulate posterior predictive observations
yp <- matrix(
    rnbinom(
        length(mu),
        mu = as.vector(mu),
        size = rep(mx[, "r"], times = length(px))
    ),
    nrow = nrow(mx)
)

# Expected count and 95% credible interval
pred <- data.frame(
    x = px,
    fit = colMeans(mu),
    lwr = apply(mu, 2, quantile, 0.025),
    upr = apply(mu, 2, quantile, 0.975),
    pred.lwr = apply(yp, 2, quantile, 0.025),
    pred.upr = apply(yp, 2, quantile, 0.975)
)
```

Now we can construct our plot:

```
par(mar=c(5,5,1,1))
plot(dx$x, dx$y, type="n",
    xlab="x variable",
    ylab="y variable (count)")
points(pred$x, pred$fit, type="l", lwd=3, col="red")
points(pred$x, pred$lwr, type="l", lwd=3, col="red", lty=2)
points(pred$x, pred$upr, type="l", lwd=3, col="red", lty=2)
points(dx$x, dx$y, pch=16)
```



5.4 Counts with zero-inflation

5.4.1 Definition

5.4.2 Example

5.4.2.1 Data simulation

5.4.2.2 Bayesian analysis with JAGS

Chapter 6

GLM for binomial data

6.1 Logistic GLM for binary outcomes

6.1.1 Definition

6.1.2 Example

6.1.2.1 Data generation

6.1.2.2 Frequentist analysis

6.1.2.3 Bayesian analysis

6.2 Binomial GLM for proportions

6.2.1 Definition

6.2.2 Example

6.2.2.1 Data generation

6.2.2.2 Frequentist analysis

6.2.2.3 Bayesian analysis

Chapter 7

GLM for continuous data

Chapter 8

Nonlinear models in JAGS

Chapter 9

Mixed models and random effects

Part III

Special applications of Bayesian inference

State space state space state space…

Chapter 10

Mark-recapture analysis

Chapter 11

State space models

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